# LARGE DEVIATION PRINCIPLES

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## 1. Introduction

The goal of this lecture note is to give a very brief introduction to the theory of large deviations. In probability theory, a recurrent theme is the convergence of random variables, and the law of large numbers (LLN) is among the fundamental results concerning their "typical" behaviors. But beyond the scope, mathematicians yearn for more a refined picture of the "atypical configurations" or "rare events" that deviates from the expected outcome either by a small or a large amount. In this pursuit, the former is addressed in the central limit theorem (CLT) while the latter give rise to the large deviation principle (LDP). To demonstrate these, we will like to begin with the following example.

**Example 1.1** (Bernoulli trial)**.** A Bernoulli trial is a series of identical and independent experiments with exactly two possible outcomes, say 0 and 1, which may be modeled as a sequence of i.i.d. random variables $(X_i)_{i=1}^{\infty}$ with the Bernoulli distribution:

$$\mathbb{P}(X_i = 1) = p \quad \text{and} \quad \mathbb{P}(X_i = 0) = 1 - p \quad (p \in (0,1)).$$

The core questions regarding the trials revolve around the number of 1′s in an $n$ -trial, which is mathematically phrased as a random variable $S_n = \sum_{i=1}^{n} X_i$. It is not hard to determine the distribution of $S_n$:

$$\mathbb{P}(S_n = \lfloor nx \rfloor) = \binom{n}{\lfloor nx \rfloor} p^{\lfloor ns \rfloor}(1-p)^{n-\lfloor nx \rfloor} \quad \text{if } s \in [0,1],$$

for which, here and throughout study, we intend to examine the logarithm of the probability the associated events. Specifically, by Stirling's approximation ($\log n! = n \log n - n + O(\log n)$),

$$\begin{aligned} & n^{-1} \log \mathbb{P}(S_n = \lfloor nx \rfloor) \\ = & \frac{\lfloor nx \rfloor}{n} \log\left(\frac{p}{\frac{\lfloor nx \rfloor}{n}}\right) + \left(1 - \frac{\lfloor nx \rfloor}{n}\right) \log\left(\frac{1-p}{1-\frac{\lfloor nx \rfloor}{n}}\right) + O\left(\frac{\log n}{n}\right). \end{aligned}$$

Asymptotically, given that $x \mapsto -x \log x$ is uniformly continuous on $[0,1]$,

$$\lim_{n\to\infty} n^{-1} \log \mathbb{P}(S_n = \lfloor ns \rfloor) = -I(s),$$

where

$$I(s) = x \log\left(\frac{x}{p}\right) + (1-x) \log\left(\frac{1-x}{1-p}\right)$$

is plotted in Figure 1. In the language of large deviation theory, the sequence $n^{-1}S_n$ is said to satisfy the large deviation principle (which will be defined rigorously later) with rate function $I$.

Some comments are in order.

- $I(x)$ is a convex function with

  $$I'(x) = \log\left(\frac{x}{1-x}\right) - \log\left(\frac{p}{1-p}\right) \quad \text{and} \quad I''(x) = \frac{1}{x} + \frac{1}{1-x}.$$

  In particular, its minimum is attained by a unique point $x = p = \mathbb{E}(X_1)$. This implies the law of large numbers by the Borel-Cantelli lemma.

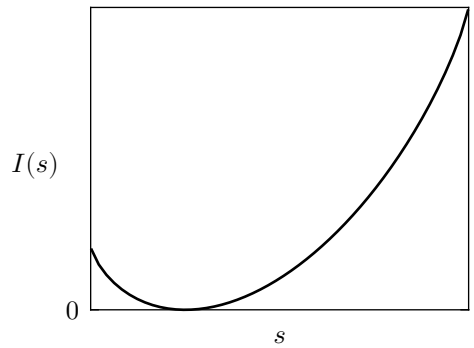


FIGURE 1. rate function $I(s)$

- Observe that the second-order Taylor expansion

$$I(x) = \frac{(x-p)^2}{2p(1-p)} + O(|x-p|^2).$$

When considering a small deviation $x = p + y/\sqrt{n}$, this is "roughly" consistent with the central limit theorem:

$$\mathbb{P}\left(\frac{\sqrt{n}(n^{-1}S_n - p)}{\sqrt{p(1-p)}} = \frac{y}{\sqrt{p(1-p)}}\right) \approx \frac{1}{\sqrt{2\pi p(1-p)}} e^{\frac{y^2}{2p(1-p)}}.$$

### 1.1. **The large deviation principle.**

Throughout the note, we denote by $\{\mu_\varepsilon\}$ be the collection of probability measures on a common measurable space $(\mathcal{X}, \mathcal{B})$ whose limiting behavior as $\varepsilon \to 0$ is of interest. Within the scope, $\mu_\varepsilon$ is always complete and $\mathcal{X}$ is always a Hausdorff topological space associated with the completed Borel $\sigma$-algebra $\mathcal{B}_\mathcal{X}$.

**Definition 1.2** (rate function)**.** Let $I : \mathcal{X} \to [0, \infty]$ be a function with *sublevel sets* $\Psi_I(\alpha) := \{x : I(x) \le \alpha\}$ and *effective domain* $\mathcal{D}_I := \{x : I(x) < \infty\}$.

- $I$ is called a *rate function* if it is lower semicontinuous.
- A rate function $I$ is said to be *good* if $\Psi_I(\alpha)$ is compact for all $\alpha < \infty$.

**Definition 1.3** (large deviation principle)**.** $\{\mu_\varepsilon\}_\varepsilon$ is said to satisfy the *large deviation principle with rate* $I : \mathcal{X} \to [0, \infty]$ if for every $\Gamma \in \mathcal{B}$,

$$-\inf_{x\in\hat{\Gamma}} I(x) \le \liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(\Gamma) \le \limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(\Gamma) \le -\inf_{x\in\bar{\Gamma}} I(x). \tag{1.1}$$

**Remark 1.4.** *Suppose $\mathcal{B}_\mathcal{X} \subseteq \mathcal{B}$. Then, $\{\mu_\varepsilon\}_\varepsilon$ satisfies the large deviation principle with rate $I$ if and only if the following hold.*

- *(upper bound) For any closed set $F \subset \mathcal{X}$,*

$$\limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(F) \le -\inf_{x\in F} I(x). \tag{1.2}$$

- *(lower bound) For any open set $G \subset \mathcal{X}$,*

$$\liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(G) \ge -\inf_{x\in G} I(x). \tag{1.3}$$

**Definition 1.5.** Suppose $\mathcal{B}$ contains all compact subsets of $\mathcal{X}$. Then, $\{\mu_\varepsilon\}$ is said to be *exponentially tight* if for every $\alpha < \infty$, there is a compact set $K$ such that

$$\limsup_{\varepsilon \to 0} \varepsilon \log \mu_\varepsilon(K^c) < -\alpha.$$

**Remark 1.6.** *Suppose $\mathcal{B}_\mathcal{X} \subset \mathcal{B}$. If $\{\mu_\varepsilon\}$ is exponentially tight, then the following hold.*

- *(upper bound) If (1.2) holds for every compact sets, so does it for every closed set*
- *(lower bound) If (1.3) holds for every open set, then $I$ is good.*

*Notably, any $\{\mu_\varepsilon\}$ admitting a rate function satisfying the relaxed conditions is said to satisfy the weak LDP.*

The *large deviation principle* (LDP) characterizes the limiting behavior of a sequence of measures.

## 2. Method of types

The *method of types* refers to a combinatorial argument addressing random variables taking values in a finite alphabet $\Sigma = \left\{a_1, a_2, ..., a_{\{|\Sigma|\}}\right\}$, which made its debut in Example 1.1 and will be revisited in this section in a systematic and more purely combinatorial fashion.

To demonstrate the argument, we consider the case with i.i.d. random variables $(Y_i)_{i=1}^{\infty}$ taking values in $\Sigma$. By writing $M_1(\Sigma)$ (which can be identified as a subset of $\mathbb{R}^{|\Sigma|}$) to mean the space of all probability measures (laws) on $\Sigma$ and assuming the law of for each $Y_i$ is $\mu \in M_1(\Sigma)$, we denote by $P_\mu$ the joint law of $(Y_i)_{i=1}^{\infty}$, i.e., the probability measure on $\mathbb{R}^{\mathbb{N}}$.

**Definition 2.1.** The *empirical measure (law)* $L_n^{\mathbf{y}}$ of a finite sequence $\mathbf{y} = (y_1, y_2, ..., y_n) \in \Sigma^n$ is the measure in $M_1(\Sigma)$ that catalogs the proportion of occurrences of $a_i$'s; namely, $L_n^{\mathbf{y}} = (L_n^{\mathbf{y}}(a_1), ..., L_n^{\mathbf{y}}(a_N))$ with

$$L^{\mathbf{y}}(a_i) = \frac{1}{n}\sum_{j=1}^{n} \mathbb{1}_{a_i}(y_j).$$

Below are some related notions.

- (support of measure $\nu$) $\Sigma_\nu = \{a_i \in \Sigma : \nu(a_i) > 0\}$.
- (empirical distribution associated with $\mathbf{y}$) $\mathcal{L}_n = \{L_n^{\mathbf{y}} : \mathbf{y} \in \Sigma^n\}$.
- (sequences associated law $\nu$) $T_n(\nu) = \{\mathbf{y} : L_n^{\mathbf{y}} = \nu\}$ for $\nu \in M_1(\Sigma)$.
- (variational distance) For $\nu, \nu' \in M_1(\Sigma)$, $d_V(\nu, \nu') := \frac{1}{2}\sum_{a\in\Sigma}|\nu(a) - \nu(a')|$.

We have the following fundamental estimates of the size of $T_n(\nu)$ and $\mathcal{L}_n$.

**Lemma 2.2.**
*(1)* $|T_n(\nu)| = \binom{n}{n\nu(a_1), n\nu(a_2), ..., n\nu(a_N)}$.
*(2)* $1 \le \mathcal{L}_n \le (n+1)^{|\Sigma|}$.

*Proof.* (1) It follows from a combinatorial argument. (2) Note that

$$n \cdot \mathcal{L}_n = \left\{(k_a)_{a\in\Sigma} : k_a \in \mathbb{Z}_+, \sum_{a\in\Sigma} k_a = n\right\},$$

where the latter set has cardinality

$$\binom{n + |\Sigma| - 1}{n} = \prod_{i=1}^{|\Sigma|-1} \frac{n+i}{i} \le (n+1)^{|\Sigma|}$$

as desired. □

In the following, we estimate the large deviation probabilities.

**Definition 2.3** (entropy)**.** Let $\nu, \eta \in M_1(\Sigma)$.

- The *entropy* of $\nu \in M_1(\Sigma)$ is defined as

$$H(\nu) = -\sum_{a\in\Sigma} \nu(a)\log\nu(a).$$

- The *relative entropy* of $\nu$ with respect to $\eta$ is defined as

$$H(\nu|\eta) = \sum_{a\in\Sigma} \nu(a) \log \frac{\nu(a)}{\eta(a)}.$$

We adopt the convention that $0 \log 0 = 0$ and $0 \log \frac{0}{0} = 0$.

**Remark 2.4.** *Let $M_1(\Sigma)$ be endowed with the variational distance. Then,*
*(1) $M_1(\mu)$ is compact.*
*(2) $H(\cdot)$ is continuous on $M_1(\Sigma)$*
*(3) $H(\cdot\,|\mu)$ is continuous on the compact set $\{\nu \in M_1(\Sigma) : \Sigma_\nu \subset \Sigma_\mu\}$.*

**Lemma 2.5.** *If $\mathbf{y} \in T_n(\nu)$, then the following hold.*
*(1) $P_\mu((Y_1, ..., Y_n) = \mathbf{y}) = e^{-n(H(\nu)+H(\nu|\mu))}$.*
*(2) $(n+1)^{-|\Sigma|} e^{nH(\nu)} \le T_n(\nu) \le e^{nH(\nu)}$.*
*(3) $(n+1)^{-|\Sigma|} e^{-nH(\nu|\mu)} \le P_\mu(L_n^{\mathbf{Y}} = \nu) \le e^{-nH(\nu|\mu)}$.*

*Proof.* (1) The case $\Sigma_\nu \subseteq \Sigma_\mu$ is trivial, as $P_\mu((Y_1, ..., Y_n) = \mathbf{y}) = 0$ and $H(\nu|\mu) = -\infty$. Assuming $\Sigma_\nu \subseteq \Sigma_\mu$, we have by definition that

$$P_\mu((Y_1, ..., Y_n) = \mathbf{y}) = \prod_{a\in\Sigma} \mu(a)^{n\nu(a)} = e^{-n(H(\nu)+H(\nu|\mu))}.$$

(2) The second inequality follows immediately from the following:

$$1 \ge P_\nu((Y_1, ..., Y_n) \in T_n(\nu)) = e^{-nH(\nu)}\ |T_n(\nu)|.$$

To prove the first, we claim that if $\Sigma_{\nu'} \subset \Sigma_\nu$, then

$$\frac{P_\nu(L_n^{\mathbf{Y}} \in T_n(\nu))}{P_\nu(L_n^{\mathbf{Y}} \in T_n(\nu'))} = \prod_{a\in\Sigma} \frac{|T_n(\nu)|\nu(a)^{n\nu(a)}}{|T_n(\nu')|\nu(a)^{n\nu'(a)}} = \prod_{a\in\Sigma} \frac{(n\nu'(a))!}{(n\nu(a))!} \nu(a)^{n\nu(a)-n\nu'(a)} \ge 1.$$

Indeed, the second last expression is of the form $\frac{m!}{\ell!}\left(\frac{\ell}{n}\right)^{\ell-m}$ and the inequality is validated using the fact $\frac{m!}{\ell!} \ge \ell^{m-\ell}$:

$$\frac{P_\nu(L_n^{\mathbf{Y}} \in T_n(\nu))}{P_{\nu'}(L_n^{\mathbf{Y}} \in T_n(\nu))} \ge n^{n\left[\sum_{a\in\Sigma} \nu'(a) - \sum_{a\in\Sigma} \nu(a)\right]} = 1.$$

To prove the the desired inequality, observe that

$$1 = \sum_{\nu'\in\mathcal{L}_n} P_\nu(L_n^{\mathbf{Y}} = \nu') \le |\mathcal{L}_n|\ P_\nu(L_n^{\mathbf{Y}} = \nu) = |\mathcal{L}_n|\ e^{-nH(\nu)}\ |T_n(\nu)|.$$

The proof is finished by applying Lemma 2.2.

(3) Combining (1) and (2), we arrive at the estimate

$$P_\mu(L_n^{\mathbf{Y}} = \nu)) = |T_n(\nu)|\ P_\mu((Y_1, ..., Y_n) = \mathbf{y}) \le |T_n(\nu)|\ e^{-n(H(\nu)+H(\nu|\mu))} = e^{-n(H(\nu|\mu))},$$

whereas the other inequality follows naturally. □

**Theorem 2.6** (Sanov)**.** *For every set of probability vectors in $M_1(\Sigma)$,*

$$\begin{aligned} -\inf_{\nu\in\mathring{\Gamma}} H(\nu|\mu) &\le \liminf_{n\to\infty} \frac{1}{n} \log P_\mu(L_n^{\mathbf{Y}} \in \Gamma) \\ &\le \limsup_{n\to\infty} \frac{1}{n} \log P_\mu(L_n^{\mathbf{Y}} \in \Gamma) \le -\inf_{\nu\in\Gamma} H(\nu|\mu). \end{aligned}$$

*Proof.* By Lemma 2.5, we obtain the upper bound for the probability in question:

$$P_\mu(L_n^{\mathbf{Y}} \in \Gamma) = \sum_{\nu \in \Gamma \cap \mathcal{L}_n} P_\nu(L_n^{\mathbf{Y}} = \nu) \le |\Gamma \cap \mathcal{L}_n| \; e^{-n \inf_{\nu \in \Gamma \cap \mathcal{L}_n} H(\nu|\mu)}$$
$$\le (n+1)^{|\Sigma|} e^{-n \inf_{\nu \in \Gamma \cap \mathcal{L}_n} H(\nu|\mu)},$$

yielding immediately the proposed upper bound:

$$\limsup_{n\to\infty} \frac{1}{n} \log P_\nu(L_n^{\mathbf{Y}} \in \Gamma) \le -\liminf_{n\to\infty} \inf_{\nu \in \Gamma \cap \mathcal{L}_n} H(\nu|\mu) \le -\inf_{\nu \in \overline{\Gamma}} H(\nu|\mu).$$

To prove the lower bound, we begin with an analogous estimate

$$P_\nu(L_n^{\mathbf{Y}} \in \Gamma) = \sum_{\nu \in \Gamma \cap \mathcal{L}_n} P_\nu(L_n^{\mathbf{Y}} = \nu) \ge (n+1)^{-|\Sigma|} e^{-n \inf_{\nu \in \Gamma \cap \mathcal{L}_n} H(\nu|\mu)}.$$

Since $\sup_{\nu \in M_1(\Sigma)} d_V(\nu, \mathcal{L}_n) \le \frac{|\Sigma|}{2n}$, for every $\nu \in \mathring{\Gamma}$ with $\Sigma_\nu \subset \Sigma_\mu$, one can find for every sufficiently large $n$ a $\nu_n \in \mathring{\Gamma} \cap \mathcal{L}_n$ such that $\nu_n \to \nu$ and that $\Sigma_{\nu_n} \subset \Sigma_\mu$. Hence, by continuity of the relative entropy,

$$\liminf_{n\to\infty} \frac{1}{n} \log P_\nu(L_n^{\mathbf{Y}} \in \Gamma) \ge -\limsup_{n\to\infty} H(\nu_n|\mu) \ge -H(\nu|\mu).$$

This proves the theorem, for $\nu \in \mathring{\Gamma}$ is arbitrary as long as $\Sigma_\nu \subset \Sigma_\mu$ and $H(\nu|\mu) = \infty$ whenever $\Sigma_\nu \not\subset \Sigma_\mu$. □

As an application of Sanov's theorem, a version of Cramér's theorem concerning the large deviations of the empirical mean of i.i.d. random variables is proved. Let $f : \Sigma \to \mathbb{R}$ be a function and consider the large deviation principle of the empirical average $\hat{S}_n = \sum_{i=1}^n f(Y_i)$. For convenience, we identify $f$ as $\mathbf{f} \in \mathbb{R}^\Sigma$ so we may write $\hat{S}_n = \langle L_n^{\mathbf{Y}}, \mathbf{f} \rangle$ as an inner product. Sanov's theorem then imply the following theorem.

**Theorem 2.7** (Cramér)**.** *For any set $A \subseteq \mathbb{R}$,*

$$-\inf_{\nu \in \mathring{A}} I(x) \le \liminf_{n\to\infty} \frac{1}{n} \log P_\mu\big(\hat{S}_n \in A\big)$$
$$\le \limsup_{n\to\infty} \frac{1}{n} \log P_\mu\big(\hat{S}_n \in A\big) \le -\inf_{\nu \in \Gamma} I(x),$$

*where $I(x) = \inf_{\nu : \langle \nu, \mathbf{f} \rangle = x} H(\nu|\mu)$. The rate function $I(x)$ is continuous on the interval $[\min_{a \in \Sigma} f(a), \max_{a \in \Sigma} f(a)]$ and satisfies there*

$$I(x) = \sup_{\lambda \in \mathbb{R}} [\lambda x - \Lambda(\lambda)],$$

*where*

$$\Lambda(\lambda) = \log \sum_{a \in \Sigma} \mu(a) e^{\lambda f(a)}.$$

## 3. Cramér's theorem in $\mathbb{R}^d$

The aim of this section is to establish the large deviation principle for i.i.d. random vectors in $\mathbb{R}^d$. Let $(X_i)_{i=1}^{\infty}$ be i.i.d. $d$-dimensional random vectors each of which is distributed with respect to the law $\mu \in M_1(\mathbb{R}^d)$, and let $\mu_n$ be the law of $\hat{S}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Write $\overline{x} := \mathbb{E}X_1$.

**Definition 3.1.** The *logarithmic moment generating function* associated with the law $\mu$ is defined as

$$\Lambda(\lambda) := \log \mathbb{E}\big(e^{\langle \lambda, X_1\rangle}\big). \tag{3.1}$$

**Definition 3.2.** The *Fenchel-Legendre transform* of $\Lambda(\lambda)$ is

$$\Lambda^*(x) := \sup_{\lambda\in\mathbb{R}^d} [\langle \lambda, x\rangle - \Lambda(\lambda)]. \tag{3.2}$$

### 3.1. Cramér's theorem in $\mathbb{R}$.

The subsection is devoted to the proof of the following theorem.

**Theorem 3.3** (Cramér)**.** *Let $X_i$, $\mu_n$, $\Lambda$, and $\Lambda^*$ be as defined on $\mathbb{R}$. Then, $\{\mu_n\}$ satisfies the LDP with the convex rate function $\Lambda^*$, namely,*

*(1) For any closed set $F$, $\limsup_{n\to\infty} \frac{1}{n}\log \mu_n(F) \le -\inf_{x\in F} \Lambda^*(x)$.*

*(2) For any open set $G$, $\limsup_{n\to\infty} \frac{1}{n}\log \mu_n(G) \ge -\inf_{x\in G} \Lambda^*(x)$.*

*Furthermore,*

*(3) If $0 \in \mathring{\mathcal{D}}_\Lambda$, then $\Lambda^*$ is good.*

Before delving into the exposition, we should first demonstrate the role played by the logarithmic moment generating function. For real-valued random variables, the key inequality in need to establish the upper bound of the LDP (i.e., (1.2)) is Markov's inequality: For all $\lambda \ge 0$,

$$\mu_n[x,\infty) \le e^{-\lambda x}\mathbb{E}\Big(e^{\lambda \hat{S}_n}\Big) = e^{-\lambda x}\big(\mathbb{E}\big(e^{\lambda X_1}\big)\big)^n$$

$$\Rightarrow \limsup_{n\to\infty} \frac{1}{n}\log \mu_n[x,\infty) \le \inf_{\lambda\ge 0} -\lambda x + \Lambda(\lambda).$$

Similarly,

$$\limsup_{n\to\infty} \frac{1}{n}\log \mu_n(-\infty, x] \le \inf_{\lambda\le 0} -\lambda x + \Lambda(\lambda).$$

These observations hinted at a candidate of rate function and urges a closer inspection on them.

The following lemma provides clear a picture of $\Lambda$ and $\Lambda^*$.

**Lemma 3.4.** *Let $\Lambda$ and $\Lambda^*$ be as defined. Then,*

*(1) $\Lambda$ is a convex function and $\Lambda^*$ is a convex rate function.*

*(2) Either of following holds.*

- *If $\Lambda(\lambda) < \infty$ only when $\lambda = 0$, then $\Lambda^*$ is identically zero.*
- *If $\Lambda(\lambda) < \infty$ for some $\lambda > 0$ (respectively, $\lambda < 0$), then $\overline{x} < \infty$ (respectively, $\overline{x} > -\infty$) is well-defined. Under the circumstances,*

$$\Lambda^*(x) = \begin{cases} \sup_{\lambda\geq 0}[\lambda x - \Lambda(\lambda)] \text{ if } x \geq \overline{x}, \\ \sup_{\lambda\leq 0}[\lambda x - \Lambda(\lambda)] \text{ if } x \leq \overline{x}, \end{cases} \tag{3.3}$$

*which satisfies*

- $\Lambda^*$ *is decreasing on* $(-\infty, \overline{x}]$ *and increasing on* $[\overline{x}, \infty)$*, and*
- $\inf_{x\in\mathbb{R}} \Lambda^*(x) = 0$ *and if* $\overline{x} \in \mathbb{R}$*, then* $\Lambda^*(\overline{x}) = 0$.

*(3)* $\Lambda$ *is differentiable in* $\mathring{\mathcal{D}}_\Lambda$ *with* $\Lambda'(\lambda) = \big(\mathbb{E}\big(e^{\lambda X_1}\big)\big)^{-1}\mathbb{E}\big(X_1 e^{\lambda X_1}\big)$ *and*

$$\Lambda'(\lambda) = x \Rightarrow \Lambda^*(x) = \lambda x - \Lambda(\lambda).$$

*(4) If* $0 \in \mathring{\mathcal{D}}_\Lambda$*, then* $\Lambda^*$ *is a good rate function. Moreover, if* $\mathcal{D}_\Lambda = \mathbb{R}$*, then* $\lim_{|x\mapsto\infty} \frac{\Lambda^*(x)}{|x|} = \infty$.

*Proof.* (1) By Hölder's inequality, given any $\lambda, \lambda' \in \mathbb{R}$ and any $t, t' \in [0,1]$ satisfying $t + t' = 1$,

$$\log \mathbb{E}\Big(e^{\langle t\lambda + t'\lambda', X_1\rangle}\Big) \leq t \log \mathbb{E}\big(e^{\langle \lambda, X_1\rangle}\big) + t' \log \mathbb{E}\Big(e^{\langle \lambda', X_1\rangle}\Big),$$

proving the convexity of $\Lambda$. The convexity of $\Lambda^*$ follows from definition:

$$\sup_{\lambda\in\mathbb{R}}[\langle \lambda, tx + t'x\rangle - \Lambda(\lambda)] \leq t \sup_{\lambda\in\mathbb{R}}[\langle \lambda, x\rangle - \Lambda(\lambda)] + t' \sup_{\lambda\in\mathbb{R}}[\langle \lambda, x'\rangle - \Lambda(\lambda)].$$

To prove the lower semicontinuity, observe that for any $\lambda \in \mathbb{R}$ and any $x \in \mathbb{R}$,

$$\liminf_{y\to x} \langle \lambda, y\rangle - \Lambda(\lambda) = \langle \lambda, x\rangle - \Lambda(\lambda),$$

implying $\liminf_{y\to x} \Lambda^*(y) \geq \Lambda^*(x)$. The non-negativity of $\Lambda^*$ follows from the fact $\Lambda^*(x) \geq \langle 0, x\rangle - \Lambda(0) = 0$.

(2) The case $\mathcal{D}_\Lambda = \{0\}$ is automatic. If $\Lambda(\lambda) < \infty$ for some $\lambda > 0$, then, due to the fact that $e^x \geq x$ for all $x \geq 0$,

$$\mathbb{E}\Big[X_1 \mathbb{1}_{\{X_1\geq 0\}}\Big] \leq \lambda^{-1}\mathbb{E}\Big[e^{\langle \lambda, X_1\rangle}\mathbb{1}_{\{X_1\geq 0\}}\Big] \leq \lambda^{-1}\exp(\Lambda(\lambda)) < \infty,$$

meaning both $\mathbb{E}(\langle \lambda, X_1\rangle)$ and $\Lambda(\lambda)$ are well-defined. Hence, by Jensen's inequality,

$$\overline{x} = \lambda^{-1} \log \circ \exp(\mathbb{E}(\langle \lambda, X_1\rangle)) \leq \lambda^{-1}\Lambda(\lambda) < \infty.$$

The arguments proceeds similarly for $\Lambda(\lambda) > -\infty$ whenever $\lambda < 0$.

We then proceed to prove (3.3) and its related properties. Since now $\overline{x}$ is well-defined, by Jensen's inequality, for all $\eta < 0$ and $x \geq \overline{x}$ (similar for $\eta > 0$ and $x \leq \overline{x}$),

$$\langle \eta, x\rangle - \Lambda(\eta) = \log \mathbb{E}\big(e^{\langle \eta, x - X_1\rangle}\big) \leq \langle \eta, x - \overline{x}\rangle \leq 0 = \langle 0, x\rangle - \Lambda(0), \tag{3.4}$$

from which (3.3) follows. Moreover, (3.3) implies the monotonicity on $(\overline{x}, \infty)$ and $(\infty, \overline{x})$. Finally, if $\overline{x} \in \mathbb{R}$, then by (3.4), $\inf_{x\in\mathbb{R}} \Lambda^*(x) = \Lambda^*(\overline{x}) = 0$. If $\overline{x} = -\infty$ (similar for $\overline{x} = -\infty$), we deduce from Markov's inequality that for all $\lambda \geq 0$

$$\log \mu[x, \infty) \leq \log e^{-\langle \lambda, x\rangle}\mathbb{E}\big(e^{\langle \lambda, X_1\rangle}\big),$$

from which it follows that $0 \leq \inf_{x\in\mathbb{R}} \Lambda^*(x) \leq \lim_{x\to\overline{x}} \Lambda^*(x) \leq 0$ as desired.

(3) The differentiability follows from the dominated convergence theorem. Let $f_\varepsilon(x) = \frac{e^{(\lambda+\varepsilon)x} - e^{\lambda x}}{\varepsilon}$ so that $f_\varepsilon(x) \to xe^{\lambda x}$ as $\varepsilon \to 0$ and $|f_\varepsilon(x)| \leq \frac{e^{\lambda x}\left(e^{\delta\,|x|} - 1\right)}{\delta} =: h_\delta(x)$ whenever $|\varepsilon| < \delta$. Since $\mathbb{E}(h_\delta(X_1)) < \infty$ for all sufficiently small $\delta$, we may apply the dominated convergence theorem to derive the derivative of $\Lambda$. Finally, the function

$g(\eta) := \langle \eta, x\rangle - \Lambda(\eta)$ is concave and thus $g'(\lambda) = 0$ implies $g(\lambda) = \sup_{\eta\in\mathbb{R}} g(\eta)$, proving the proposed property.

(4) Suppose $[\lambda_-, \lambda_+] \subset \mathcal{D}_\Lambda$ is a non-degenerate interval containing 0. Then, for $\lambda \in [\lambda_-, \lambda_+]$

$$\liminf_{|x|\mapsto\infty} \frac{\Lambda^*(x)}{|x|} \geq \liminf_{|x|\mapsto\infty} \left[\lambda \operatorname{sign}(x) - \frac{\Lambda(\lambda)}{|x|}\right] \geq \min\{-\lambda_-, \lambda_+\} > 0,$$

implying $\Lambda^*(x) \to \infty$ as $|x| \to \infty$. Hence, the sublevel set $\Psi_{\Lambda^*}(\alpha)$ is closed and bounded for all $\alpha < \infty$, and $\Lambda^*$ is good. When $\mathcal{D}_\Lambda = \mathbb{R}$, then the result follows by letting $\lambda_+ = -\lambda_- \to \infty$. □

*Proof of Theorem 3.3.* (1) For economy, let $I_F = \inf_{x\in F}\langle \lambda, x\rangle - \Lambda(\lambda)$. If $I_F = 0$, then the inequality is trivial. Hence, we assume $I_F > 0$. Under the circumstances, the numbers $x_+ = \inf[\overline{x}, \infty) \cap F$ and $x_- = \sup(-\infty, \overline{x}] \cap F$ are different from $\overline{x}$. By Markov's inequality, for all $\lambda > 0$ and $\lambda' < 0$,

$$\mu_n(F) \leq \mu_n[x_+, \infty) + \mu_n(-\infty, x_-] \leq e^{-n(\langle\lambda, x_+\rangle - \Lambda(\lambda))} + e^{-n(\langle\lambda', x_-\rangle - \Lambda(\lambda'))}.$$

By Lemma 3.4,

$$\mu_n(F) \leq e^{-n\Lambda^*(x_+)} + e^{-n\Lambda^*(x_-)} \leq 2e^{I_F}.$$

Taking the normalized logarithm and letting $n \to \infty$ yields the desired inequality.

(2) It suffices to show that

$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(-\delta, \delta) \geq \inf_{\lambda\in\mathbb{R}} \Lambda(\lambda) = -\Lambda^*(0), \tag{3.5}$$

for it is once proved, one may simply consider the translation $Y = X - x$ to deduce that the logarithmic moment generating function $\Lambda_Y(\lambda) = \Lambda(\lambda) - \langle\lambda, x\rangle$ and that $\Lambda_Y^*(z) = \Lambda^*(z + x)$, which in turn imply

$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(x - \delta, x + \delta) \geq -\Lambda^*(x).$$

This proves the desired lower bound.

To prove (3.5), assume first that (a) $\mu(-\infty, 0) > 0$, (b) $\mu(0, \infty) > 0$, and (c) $\mu$ is boundedly supported. Under the circumstances, (a) and (b) imply $\Lambda(\lambda) \to \infty$ as $|\lambda| \to \infty$, and (c) implies $\Lambda$ is finite on $\mathbb{R}$. Consequently, by Lemma 3.4, there exists $\eta \in \mathbb{R}$ such that

$$\Lambda'(\lambda) = 0 \quad \text{and} \quad \Lambda(\lambda) = \inf_{\eta\in\mathbb{R}} \Lambda(\eta).$$

define a probability measure $\mu$ by

$$\frac{d\tilde{\mu}}{d\mu}(x) = e^{\langle\lambda, x\rangle - \Lambda(\lambda)}.$$

Observe that

$$\begin{aligned}\mu_n(-\varepsilon,\varepsilon) &= \int_{|\sum_{i=1}^n x_i|<n\varepsilon} d\mu(x_1)\cdots d\mu(x_n)\\ &= \int_{|\sum_{i=1}^n x_i|<n\varepsilon} e^{\sum_{i=1}^n[-\langle\lambda,x_i\rangle+\Lambda(\lambda)]} d\tilde{\mu}(x_1)\cdots d\tilde{\mu}(x_n).\\ &\geq e^{n\Lambda(\lambda)-n\varepsilon|\lambda|}\tilde{\mu}_n(-\varepsilon,\varepsilon).\end{aligned}$$

By Lemma 3.4,

$$\mathbb{E}_{\tilde{X}\sim\tilde{\mu}}(\tilde{X}) = \int x e^{\langle\lambda,x\rangle-\Lambda(\lambda)} d\mu(x) = \big(\mathbb{E}\big(e^{\langle\lambda,x\rangle}\big)\big)^{-1}\mathbb{E}\big(X_1 e^{\langle\lambda,X_1\rangle}\big) = \Lambda'(\lambda) = 0.$$

Hence, by the law of large numbers,

$$\lim_{n\to\infty}\tilde{\mu}_n(-\varepsilon,\varepsilon) = 1.$$

Hence, for all $0<\varepsilon<\delta$,

$$\liminf_{n\to\infty}\frac{1}{n}\log\mu_n(-\delta,\delta) \geq \liminf_{n\to\infty}\frac{1}{n}\log\mu_n(-\varepsilon,\varepsilon) \geq \Lambda(\lambda)-\varepsilon|\lambda|,$$

proving (3.5) by letting $\varepsilon\to 0$.

Now if support of $\mu$ is not bounded, fix $M>0$ such that (a) $\mu[-M,0)>0$, (b) $\mu(0,M]>0$. Hence, by letting $\nu$ be the normalized law of $X_1$ on $\{|X_1|\leq M\}$, we have that

$$\begin{aligned}\mu_n(-\delta,\delta) &\geq \int_{\frac{1}{n}\sum_{i=1}^n x_i\in(-\delta,\delta)} \prod_{i=1}^n \mathbb{1}_{[-M,M]}(x_i) d\mu(x_1)\cdots d\mu(x_n)\\ &= \nu_n(-\delta,\delta)\cdot\mu[-M,M]^n\end{aligned}$$

and that the logarithmic moment generating function associated with $\nu$ is

$$\log\int_{-M}^{M} e^{\langle\lambda,x\rangle} d\mu(x) - \log\mu[-M,M] =: \Lambda_M(\lambda) - \log\mu[-M,M].$$

Hence, by the case of bounded support,

$$\liminf_{n\to\infty}\frac{1}{n}\log\mu_n(-\varepsilon,\varepsilon) \geq \log\mu[-M,M] + \liminf_{n\to\infty}\frac{1}{n}\log\nu_n(-\varepsilon,\varepsilon) \geq \inf_{\lambda\in\mathbb{R}}\Lambda_M(\lambda),$$

and therefore, by writing $I_M = -\inf_{\lambda\in\mathbb{R}}\Lambda_M(\lambda)$ and $I^* = \limsup_{M\to\infty} I_M$,

$$\liminf_{n\to\infty}\frac{1}{n}\log\mu_n(-\varepsilon,\varepsilon) \geq -I^*. \tag{3.6}$$

Since $\Lambda_M$ is increasing in $M$, so is $-I_M$. Therefore, $I^*>-\infty$ and the level sets $\{\lambda:\Lambda_M(\lambda)\leq -I^*\}$ are decreasing compacted sets, admitting some $\lambda_0$ in their intersection. By monotone convergence theorem,

$$\Lambda(\lambda_0) = \lim_{M\to\infty}\Lambda_M(\lambda_0) \leq -I^*. \tag{3.7}$$

Combining (3.6) and (3.7) proves (3.5).

Finally, if $\mu(-\infty,0)=0$ or $\mu(0,\infty)=0$, then $\Lambda$ is monotone and $\inf_{\lambda\in\mathbb{R}}\Lambda(\lambda) = \log\mu(\{0\})$. Therefore, $\mu_n(-\delta,\delta)\geq\mu_n(\{0\})=\mu(\{0\})^n$, implying (3.5).

(3) It follows from Lemma 3.4(4). □

### 3.2. Cramér's theorem in $\mathbb{R}^d$.

In this section, we prove the Cramér's theorem in higher dimensions. In contrast with its one-dimensional counterpart, we need additional assumptions for the full LDP.

**Theorem 3.5** (Cramér)**.** *Let $X_i$, $\mu_n$, $\Lambda$, and $\Lambda^*$ be as defined on $\mathbb{R}^d$ with $d \geq 2$. Assume further that $\mathcal{D}_\Lambda = \mathbb{R}^d$. Then, $\{\mu_n\}$ satisfies the LDP with the convex rate function $\Lambda^*$, namely,*

*(1) For any closed set $F$, $\limsup_{n\to\infty} \frac{1}{n} \log \mu_n(F) \leq -\inf_{x\in F} \Lambda^*(x)$.*

*(2) For any open set $G$, $\limsup_{n\to\infty} \frac{1}{n} \log \mu_n(G) \geq -\inf_{x\in G} \Lambda^*(x)$.*

*Furthermore,*

*(3) If $0 \in \mathring{\mathcal{D}}_\Lambda$, then $\Lambda^*$ is good.*

**Remark 3.6.** *The theorem will be revisited, if time permits, later with the following generalizations.*

*(1) Even without the assumption $\mathcal{D}_\Lambda = \mathbb{R}^d$, the weak LDP holds.*

*(2) If $0 \in \mathring{\mathcal{D}}_\Lambda$, then the full LDP holds.*

Our starting point is a number of properties regarding logarithmic generating function.

**Lemma 3.7.** *Let $\Lambda$ and $\Lambda^*$ be as defined. Then,*

*(1) $\Lambda$ is a convex function and $\Lambda^*$ is a convex rate function.*

*(2) $\Lambda$ is differentiable in $\mathring{\mathcal{D}}_\Lambda$ with $\nabla\Lambda(\lambda) = \big(\mathbb{E}\big(e^{\langle\lambda, X_1\rangle}\big)\big)^{-1}\mathbb{E}\big(X_1 e^{\langle\lambda, X_1\rangle}\big)$ and*

$$\nabla\Lambda(\lambda) = x \Rightarrow \Lambda^*(x) = \langle\lambda, x\rangle - \Lambda(\lambda).$$

*(3) If $0 \in \mathring{\mathcal{D}}_\Lambda$, then $\Lambda^*$ is a good rate function. Moreover, if $\mathcal{D}_\Lambda = \mathbb{R}^d$, then $\lim_{|x\mapsto\infty} \frac{\Lambda^*(x)}{|x|} = \infty$.*

*Proof.* (1) Same as Lemma 3.4(1).

(2) The gradient can be obtained through a similar calculation as in Lemma 3.4(3), from which the differentiability follows naturally as it is continuous $\mathring{\mathcal{D}}_\Lambda$. To prove the remaining property, define

$$g_\eta(t) = \langle t\eta + (1-t)\lambda, x\rangle - \Lambda(t\eta + (1-t)\lambda)$$

to paraphrase the property as $g_\eta(1) \leq g_\eta(0)$ for all $\eta \in \mathcal{D}_\Lambda$ and all $t \in [0,1]$. To this end, observe that $g_\eta(t)$ is concave in $t$ and that

$$g'_\eta(0) = \langle\eta - \lambda, x\rangle - \langle\eta - \lambda, \nabla\Lambda(\lambda)\rangle = 0,$$

proving the proposed inequality.

(3) Note that if $0 \in \mathring{\mathcal{D}}_\Lambda$, then there exists $\rho > 0$ such that the $\rho$-ball centered at the origin is contained in $\rho(\mathcal{D}_\Lambda)$ and

$$\Lambda^*(x) \geq \rho|x| - \sup_{|\lambda|=\rho} \Lambda(\lambda) \to \infty \text{ as } |x| \to \infty.$$

The case $\mathcal{D}_\Lambda = \mathbb{R}^d$ follows similarly. □

**Definition 3.8.** The $\delta$-rate function associated with $I$ is

$$I^\delta(x) = \min\Big\{I(x) - \delta, \frac{1}{\delta}\Big\}. \tag{3.8}$$

*Proof of Theorem 3.5 for* $\mathbb{R}^d$. (1) To prove the upper bound, it suffices to show that for every closed set $F$ and every $\delta > 0$,

$$\limsup_{n\to\infty} \frac{1}{n} \log \mu_n(F) \le -\inf_{x\in F} I^\delta(x), \tag{3.9}$$

where $I^\delta$ is the $\delta$-rate function associated with $\Lambda^*$ defined as in (3.8).

We first prove (3.9) for any compact set $F$. Choose for each $x \in \Gamma$ an $\lambda_x \in \mathbb{R}^d$ and a open neighborhood $A_x$ of $x$ such that

$$\langle \lambda_x, x\rangle - \Lambda(x) \ge I^\delta(x),$$
$$\inf_{y\in A_x} \{\langle \lambda_x, y\rangle - \langle \lambda_x, x\rangle\} \ge -\delta,$$

Applying the Markov's inequality yields

$$\mu_n(A_x) \le \mathbb{E}\Big(e^{n\cdot(\langle \lambda_x, \cdot \hat{S}_n\rangle - \langle \lambda_x, x\rangle)}\Big) \exp\Big(-n \cdot \inf_{y\in A_x} \langle \lambda_x, y\rangle - \langle \lambda_x, x\rangle]\Big),$$

which in turn implies

$$\frac{1}{n} \log \mu_n(A_x) \le \delta - [\langle \lambda_x, x\rangle - \Lambda(\lambda_x)].$$

Now that $F$ is a compact set, one can find a finite open cover $\{A_{x_i}\}_{i=1}^N$ with $x_i \in F$ such that

$$\limsup_{\varepsilon\to 0} \frac{1}{n} \log \mu_n(F) \le \delta - \min_{1\le i\le N} I^\delta(x_i) \le \delta - \inf_{x\in F} I^\delta(x),$$

proving (3.9) by letting $\delta \to 0$.

It remains to show that $\mu_n$ is an exponentially tight family (See Remark 1.6). Let $H_\rho = [-\rho, \rho]^d$ and $\mu^j$ denote the marginal law of $\mu$ on the $j$-th coordinate, so that $\mu_n\big(H_\rho^c\big) \le \sum_{j=1}^d \mu_n^j[-\rho,\rho]^c$. Since $0 \in \mathring{\mathcal{D}}_\Lambda$, Lemma 3.7 then implies

$$\lim_{\rho\to\infty} \limsup_{n\to\infty} \frac{1}{n} \log \mu_n\big(H_\rho^c\big) \le -\infty.$$

(2) For the lower bound, we need to prove that for every $x \in \mathcal{D}_{\Lambda^*}$ and every $\delta > 0$,

$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(B_\delta(x)) \ge -\Lambda^*(y). \tag{3.10}$$

To prove (3.10), assume first that $x = \nabla\Lambda(\lambda)$ for some $\lambda \in \mathbb{R}^d$ and define a probability measure $\mu$ by

$$\frac{d\tilde{\mu}}{d\mu}(y) = e^{\langle \lambda, y\rangle - \Lambda(\lambda)}.$$

Observe that

$$\begin{aligned}\frac{1}{n} \log \mu_n(B_\delta(x)) &= \Lambda(\lambda) - \langle \lambda, x\rangle + \frac{1}{n} \log \int_{B_\delta(x)} e^{n\cdot\langle \lambda, x-y\rangle} d\tilde{\mu}_n(y) \\ &\ge \Lambda(\lambda) - \langle \lambda, x\rangle - \delta|\lambda| + \frac{1}{n} \log \tilde{\mu}_n(B_\delta(x)).\end{aligned}$$

By the dominated convergence theorem and Lemma 3.7,

$$\mathbb{E}_{\tilde{X}_1\sim\tilde{\mu}}(\tilde{X}_1) = \int xe^{\langle\lambda,x\rangle-\Lambda(\lambda)}d\mu(x) = \big(\mathbb{E}\big(e^{\langle\lambda,x\rangle}\big)\big)^{-1}\mathbb{E}\big(X_1e^{\langle\lambda,X_1\rangle}\big) = \nabla\Lambda(\lambda) = y,$$

and thus, by the weak law of large numbers, $\lim_{n\to\infty}\log\tilde{\mu}_n(B_\delta(y)) = 1$. Consequently,

$$\liminf_{n\to\infty}\frac{1}{n}\log\mu_n(B_\delta(y)) \geq \Lambda(\lambda) - \langle\lambda,x\rangle - \delta|\lambda| = -\Lambda^*(x) - \delta|\lambda|,$$

which proves (3.10) as

$$\liminf_{n\to\infty}\frac{1}{n}\log\mu_n(B_\delta(y)) \geq \lim_{\delta\to 0}\liminf_{n\to\infty}\frac{1}{n}\log\mu_n(B_\delta(y)) \geq -\Lambda^*(x).$$

To extend the lower bound (3.10) to cover any $x \in \mathcal{D}_{\Lambda^*}$, we introduce i.i.d. multivariate normal random variables $V_i$ that are independent of $X_i$ and define $Y_i = X_i + \frac{V_i}{\sqrt{M}}$, $M > 0$. Denote by $\Lambda_M$ the logarithmic moment generating function of $Y_1$ and $\nu_n$ the law governing $\hat{S}^{(M)} = \frac{1}{n}\sum_{j=1}^n Y_j$. Clearly,

$$\Lambda_M(\eta) = \Lambda(\eta) + \frac{1}{2M}\ |\eta|^2,$$

and

$$\Lambda^*(x) \geq \sup_{\eta\in\mathbb{R}^d}[\langle\eta,x\rangle - \Lambda_M(x)].$$

Since $0 \in \mathring{\mathcal{D}}_\Lambda$ implies $\overline{x}$ is finite, we have, by Jensen's inequality, $\langle\eta,x\rangle - \Lambda(x) \leq 0$ and

$$\limsup_{\rho\to\infty}\sup_{|\eta|>\rho}\langle\eta,x\rangle - \Lambda_M(\eta) \leq \limsup_{\rho\to\infty}\sup_{|\eta|>\rho} -\frac{1}{2M}\ |\eta|^2 = -\infty$$

Hence, the supremum of $g(\lambda) := \langle\lambda,x\rangle - \Lambda(\lambda)$ is attained at some finite $\lambda \in \mathbb{R}^d$, and, by differentiability of $\Lambda_M$,

$$0 = \nabla g(\lambda) = x - \nabla\Lambda_M(\lambda).$$

Thus, by the discussion above, for every $\delta > 0$,

$$\liminf_{n\to\infty}\frac{1}{n}\log\nu_n(B_\delta(x)) \geq -\Lambda^*(x) > -\infty. \tag{3.11}$$

Now that $\hat{S}_n^{(M)}$ has the same distribution as $\hat{S}_n + \frac{V}{\sqrt{Mn}}$, where $V \sim \text{Normal}(0, I)$ is independent of $\hat{S}_n$, we derive

$$\mu_n(B_{2\delta}(x)) \geq \nu_n(B_\delta(x)) - \mathbb{P}\Big(|V| \geq \sqrt{Mn}\delta\Big) \tag{3.12}$$

with

$$\limsup_{n\to\infty}\frac{1}{n}\log\mathbb{P}\Big(|V| \geq \sqrt{Mn}\delta\Big) \leq -\frac{M\delta^2}{2}. \tag{3.13}$$

Combining (3.11)-(3.13) proves (3.10). □

## 4. Some backgrounds

The general setting for the rest of this note is that $\mathcal{X}$ is, unless specified otherwise, a regular Hausdorff topological space with $\mathcal{B}_{\mathcal{X}} \subset \mathcal{B}$, as defined below.

**Definition 4.1** (Hausdorff topological space)**.** A topological space $\mathcal{X}$ is said to be *regular* if for every point $x$ and closed set $y$ there exists disjoint open sets $U$, $V$ satisfying $x \in U$ and $y \in V$.

**Definition 4.2** (regular topological space)**.** A topological space $\mathcal{X}$ is said to be *regular* if for every point $x$ and closed set $F \not\supseteq \{x\}$ there exists disjoint open sets $U$, $V$ satisfying $x \in U$ and $F \subset V$.

### 4.1. Existence and uniqueness of LDP.

**Proposition 4.3** (uniqueness)**.** *A family of probability measures $\{\mu_\varepsilon\}$ on a regular topological space can have at most one rate function associated with its LDP.*

*Proof.* Suppose $I_1$ and $I_2$ are two rate functions associated with the LDP of $\mu_\varepsilon$, with a point $x_0$ satisfying $I_1(x_0) < I_2(x_0)$. Then, by lower semicontinuity, there exists an open neighborhood $A$ of $x_0$ such that $\inf_{x\in A} I_2(x) \geq I_2^\delta(x_0)$, where $I_2^\delta$ is the $\delta$-rate function as defined in Definition 3.8. By regularity, we may assume $\inf_{x\in \overline{A}} I_2(x) \geq I_2^\delta(x_0)$ (by choosing a subset if necessary). Hence,

$$-I_2^\delta(x_0) \geq \limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(A) \geq \liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(A) \geq -I_1(x_0),$$

contradicting $I_1(x_0) < I_2(x_0)$ if letting $\delta \to 0$. □

**Remark 4.4.**

(1) *Regular spaces include all metric spaces.*
(2) *If $\mathcal{X}$ is regular and locally compact, then the LDP Proposition 4.3 can be relaxed by weak LDP.*

**Proposition 4.5** (existence)**.** *Let $\mathcal{A}$ be a base of the topology of the topological space $\mathcal{X}$. For every $A \in \mathcal{A}$, define*

$$\underline{\mathcal{L}}_A := \liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(A) \quad \text{and} \quad \overline{\mathcal{L}}_A := \limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(A),$$

*and*

$$\underline{I}(x) := -\inf_{A\in\mathcal{A}: x\in A} \underline{\mathcal{L}}_A \quad \text{and} \quad \overline{I}(x) := -\inf_{A\in\mathcal{A}: x\in A} \overline{\mathcal{L}}_A.$$

*If $\underline{I} = \overline{I}$, then $\mu_\varepsilon$ satisfies the weak LDP with the rate function $\underline{I}$.*

*Proof.* We begin by noting the fact that $I$ is indeed a rate function, which follows naturally from definition.

For the lower bound in LDP, observe that for every open set $G$ and any $x \in G$, there exists $A \in \mathcal{A}$ such that $x \in A \subset G$ such that

$$\underline{\mathcal{L}}_G \geq \underline{\mathcal{L}}_A \geq -\underline{I}(x),$$

from which the desired lower bound follows naturally.

For upper bound, given any compact set $F$, we can find for every point $x$ a neighborhood $A_x$ satisfying $\overline{\mathcal{L}}_{A_x} \leq -\overline{I}^\delta(x)$. By compactness, it admits some finite open subcover $\{A_{x_i}\}_{i=1}^m$, and thus

$$\overline{\mathcal{L}}_F \leq \max_{1\leq i\leq m} \overline{\mathcal{L}}_{A_i} \leq \max_{1\leq i\leq m} -\overline{I}^{\delta}(x_i) \leq -\inf_{x\in F} \overline{I}^{\delta}(x).$$

This finishes the proof. □

**Remark 4.6.** *Proposition 4.5 holds for parametrized family* $\{\mu_{\varepsilon,\sigma}\}$ *for any given* $\sigma \in \Sigma$ *if*

$$\underline{\mathcal{L}}_A := \liminf_{\varepsilon\to 0} \varepsilon \log\Big[\inf_{\sigma\in\Sigma} \mu_{\varepsilon,\sigma}(A)\Big] \quad \text{and} \quad \overline{\mathcal{L}}_A := \limsup_{\varepsilon\to 0} \varepsilon \log\Big[\sup_{\sigma\in\Sigma} \mu_{\varepsilon,\sigma}(A)\Big].$$

*This may be useful in describing certain processes, e.g., Markov chains conditioned on initial state* $\sigma$*.*

We also have a partial converse of Proposition 4.5.

**Proposition 4.7.** *Suppose that* $\{\mu_\varepsilon\}$ *satisfies the LDP in a regular topological space* $\mathcal{X}$ *with rate function* $I$*. Then, for any base* $\mathcal{A}$ *of the topology of* $\mathcal{X}$*,* $I = \underline{I} = \overline{I}$ *with both functions defined in Proposition 4.5.*

*Proof.* Exercise. □

## 5. General principles

### 5.1. Contraction principles.

**Theorem 5.1** (Contraction principle)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be Hausdorff topological spaces and $f : \mathcal{X} \to \mathcal{Y}$ a continuous function. Consider a good rate function $I : X \to [0, \infty]$.*

*(1) For each $y \in \mathcal{Y}$, define*

$$I'(y) := \inf\{I(x) : x \in \mathcal{X}, y = f(x)\}.$$

*Then, $I'$ is a good rate function on $\mathcal{Y}$*

*(2) If $I$ controls the LDP associated with a family of probability measures $\{\mu_\varepsilon\}$ on $\mathcal{X}$ , then $I'$ controls the LDP associated with the family of probability measures $\{\mu_\varepsilon \circ f^{-1}\}$ on $\mathcal{Y}$.*

*Proof.* (1) $I'$ is non-negative by definition. To prove the lower semicontinuity, observe that by compactness of $\Psi_I(\alpha)$,

$$\Psi_{I'}(\alpha) = \left\{ y : \inf_{x: f(x)=y} I(x) \le \alpha \right\} = \{f(x) : I(x) \le \alpha\} = f(\Psi_I(\alpha))$$

is again compact as desired.

(2) Note that for each $A \subset \mathcal{Y}$, $\inf_{y \in A} I'(y) = \inf_{x \in f^{-1}(A)} I(x)$, where $f^{-1}(A)$ is open (respectively, closed) if $A$ is open (respectively, closed). With this LDP is established naturally. □

**Theorem 5.2** (Inverse contraction principle)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be Hausdorff topological spaces. Suppose that $g : \mathcal{Y} \to \mathcal{X}$ is a continuous bijection, and that $\{\nu_\varepsilon\}$ is an exponentially tight family of probability measures on $\mathcal{Y}$. If $\{\nu_\varepsilon \circ g^{-1}\}$ satisfies the LDP with the rate function $I : \mathcal{X} \to [0, \infty]$, then $\{\nu_\varepsilon\}$ satisfies the LDP with the good rate function $I' := I \circ g$.*

*Proof.* The non-negativity of $I'$ is clear from definition, and, for every $\alpha < \infty$,

$$\Psi_{I'}(\alpha) = \{y : I(g(y)) \le \alpha\} = g^{-1}(\Psi_\alpha).$$

is closed, implying the lower semicontinuity of $I'$. Therefore, $I'$ is a rate function. Now since $\{\nu_\varepsilon\}$ is exponentially tight, for every $\alpha$, it remains to show that $\{\nu_\varepsilon\}$ satisfies the weak LDP.

For the upper bound, fix any compact set $K \subset \mathcal{Y}$ and apply the large deviation upper bound for $\nu_\varepsilon \circ g^{-1}$ to obtain

$$\limsup_{\varepsilon \to 0} \varepsilon \log \nu_\varepsilon(K) = \limsup_{\varepsilon \to 0} \varepsilon \log \nu_\varepsilon(g^{-1} \circ g(K)) \le -\inf_{y \in K} I'(y).$$

To demonstrate the lower bound, fix any $y \in \mathcal{Y}$ with $I(y) = \alpha < \infty$ and any open neighborhood $G \subset \mathcal{Y}$ of $y$. In addition, choose, thanks to the exponential tightness, a compact set $K$ such that

$$\limsup_{\varepsilon \to 0} \varepsilon \log \nu_\varepsilon(K) < -\alpha.$$

Hence, since $g(K^c) = g(K)^c$ is open (a consequence of Hausdorff topology), it follows from the LDP for $\nu_\varepsilon \circ g^{-1}$ that

$$-\alpha > \limsup_{\varepsilon \to 0} \varepsilon \log \nu_\varepsilon(g^{-1} \circ g(K^c)) \ge -\inf_{x \in g(K^c)} I(x). \tag{5.1}$$

In addition, observe that $g^{-1}|_{g(K)}$ is continuous, there exists an open neighborhood $G'$ of $g(y)$ such that

$$G' \subset g(G \cap K) \cup g(K^c) = g(G \cup K^c),$$

which in turn gives

$$\begin{aligned} &\max\Big\{\limsup_{\varepsilon\to 0} \varepsilon \log \nu_\varepsilon(G \cap K), \limsup_{\varepsilon\to 0} \varepsilon \log \nu_\varepsilon(K^c)\Big\} \\ &\geq \liminf_{\varepsilon} \varepsilon \log \nu_\varepsilon(g^{-1}(G')) \geq -I'(y). \end{aligned}$$

Inequality (5.1) then rules out $\limsup_{\varepsilon\to 0} \varepsilon \log \nu_\varepsilon(K^c) \geq -I'(y)$ and asserts

$$\limsup_{\varepsilon\to 0} \varepsilon \log \nu_\varepsilon(G) \geq \limsup_{\varepsilon\to 0} \varepsilon \log \nu_\varepsilon(G \cap K) \geq -I'(y)$$

Since the neighborhood $G$ of $y$ is arbitrary, the proof of the lower bound is complete. □

**Corollary 5.3.** *Let $\{\mu_\varepsilon\}$ be an exponentially tight family of probability measures on $X$ equipped with the topology $\tau_1$. If $\{\mu_\varepsilon\}$ satisfies an LDP with respect to a Hausdorff topology $\tau_2$ on $X$ that is coarser than $\tau_1$, then the same LDP holds with respect to the topology $\tau_1$.*

*Proof.* It follows straightforwardly from Theorem 5.2. □

5.2. **Varadhan's integral lemma.**

In this subsection we assume $\mathcal{X}$ to be a regular topological space.

**Theorem 5.4** (Varadhan)**.** *Suppose that $\{\mu_\varepsilon\}$ satisfies the LDP with a good rate function $I : \mathcal{X} \to [0, \infty]$, and let $\varphi : \mathcal{X} \to \mathbb{R}$ be any continuous function. Assume further either the tail condition*

$$\lim_{M\to\infty} \limsup_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{\{\varphi(Z_\varepsilon)\geq M\}}\Big] = -\infty. \tag{5.2}$$

*or the following moment condition for some $\gamma > 1$,*

$$\limsup_{\rho\to 0} \varepsilon \log \mathbb{E}\Big[e^{\frac{\gamma\varphi(Z_\varepsilon)}{\varepsilon}}\Big] < \infty. \tag{5.3}$$

*Then,*

$$\lim_{\rho\to 0} \varepsilon \log \mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}}\Big] = \sup_{x\in\mathcal{X}} [\varphi(x) - I(x)].$$

The theorem is a consequence of the following three lemmas.

**Lemma 5.5.** *If $\varphi : \mathcal{X} \to \mathbb{R}$ is lower semicontinuous and the large deviations lower bound holds with $I : \mathcal{X} \to [0, \infty]$, then*

$$\liminf_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big(e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}}\Big) \geq \sup_{x\in\mathcal{X}} [\varphi(x) - I(x)].$$

**Lemma 5.6.** *If $\varphi : \mathcal{X} \to \mathbb{R}$ is an upper semicontinuous function for which the tail condition (5.2) holds, and the large deviations upper bound holds with the good rate function $I : \mathcal{X} \to [0, \infty]$, then*

$$\limsup_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big(e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}}\Big) \leq \sup_{x\in\mathcal{X}} [\varphi(x) - I(x)].$$

**Lemma 5.7.** *Condition (5.3) implies the tail condition (5.2).*

*Proof of Lemma 5.5.* Fix $x \in \mathcal{X}$. Since $\varphi$ is lower semicontinuous, for every $\delta > 0$ there exists open neighborhood $G$ of $x$ such that

$$-\varphi(x) \geq -\inf_{y\in G} \varphi(y) - \delta.$$

Hence, by Markov's inequality and the lower bound of LDP,

$$\begin{aligned}
&\liminf_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big(e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}}\Big) \geq \liminf_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big(e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{Z_\varepsilon \in G}\Big) \\
&\geq \inf_{y\in G} \varphi(y) + \liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(G) \\
&\geq \inf_{y\in G} \varphi(y) - \inf_{y\in G} I(y) \geq \varphi(x) - I(x) - \delta,
\end{aligned}$$

proving the inequality since $x$ and $\delta$ are arbitrary. □

*Proof of Lemma 5.6.* To begin, note that by semicontinuity of $\varphi$, $I$ and regularity of $\mathcal{X}$, for each $x \in \mathcal{X}$ and each $\delta > 0$ there is an open neighborhood $A_x$ such that

$$\inf_{y\in \overline{A}_x} I(y) \geq I(x) - \delta \quad \text{and} \quad \sup_{y\in A_x} \varphi(y) \leq \varphi(x) + \delta,$$

which gives a local estimate $\mathbb{E}\Big(e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{Z_\varepsilon \in A_x}\Big) \leq e^{\frac{\varphi(x)+\delta}{\varepsilon}} \mu(\varphi^{-1}(A_x))$. Hence, for each $\alpha < \infty$, we may find a finite cover $\Big\{A_{x_i}\Big\}_{i=1}^N$ of the compact set $\Psi_I(\alpha)$ such that

$$\begin{aligned}
&\limsup_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{\{\varphi(Z_\varepsilon)\leq M\}}\Big] \\
&\leq \limsup_{\varepsilon\to 0} \max\Big\{M - \alpha, \max_{1\leq i\leq N} \Big[\varphi(x_i) + \delta + \varepsilon \log \mu_\varepsilon\Big(A_{x_i}\Big)\Big]\Big\} \\
&\leq \max\Bigg\{M - \alpha, \max_{1\leq i\leq N} \Bigg[\varphi(x_i) + \delta - \inf_{x\in \overline{A}_{x_i}} I(x)\Bigg]\Bigg\} \\
&\leq \max\Big\{M - \alpha, \max_{x\in \Psi_I(\alpha)} [\varphi(x) - I(x) + 2\delta]\Big\}.
\end{aligned}$$

First letting $\delta \to 0$ and then letting $\alpha \to 0$ yields

$$\mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{\{\varphi(Z_\varepsilon)\leq M\}}\Big] \leq \max_{x\in\mathcal{X}} [\varphi(x) - I(x)].$$

The proof is concluded, by assumption (5.2) that

$$\begin{aligned}
&\limsup_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big(e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}}\Big) \\
&= \lim_{M\to\infty} \limsup_{\varepsilon\to 0} \max\Big\{\varepsilon \log \mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{\{\varphi(Z_\varepsilon)\leq M\}}\Big], \varepsilon \log \mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{\{\varphi(Z_\varepsilon)> M\}}\Big]\Big\} \\
&= \lim_{M\to\infty} \limsup_{\varepsilon\to 0} \varepsilon \log \mathbb{E}\Big[e^{\frac{\varphi(Z_\varepsilon)}{\varepsilon}} \mathbb{1}_{\{\varphi(Z_\varepsilon)\leq M\}}\Big].
\end{aligned}$$

□

*Proof of Lemma 5.7.* The lemma follows from the fact that for any non-negative random variable $X$ and any $\gamma > 1$,

$$\mathbb{E}(X^\gamma \mathbb{1}_{X\geq\alpha}) = \alpha^\gamma \mathbb{P}(X \geq \alpha) + \int_\alpha^\infty x^{\gamma-1}\mathbb{P}(X \geq x)\, dx.$$

Hence, $\mathbb{E}(X^\gamma \mathbb{1}_{X\geq\alpha}) \geq \alpha^{1-\gamma}\mathbb{E}(X\mathbb{1}_{X\geq\alpha})$, which proves the theorem by taking $X = \varphi(Z_\varepsilon)/\varepsilon$ and $\alpha = \varphi(M)/\varepsilon$. □

5.3. **Bryc's inverse Varadhan lemma.**

Recall the following definition.

**Definition 5.8.** A topological space $\mathcal{X}$ is said to be completely regular if for any closed set $F$ and any point $x \notin F$, there is continuous function $f : \mathcal{X} \to [0, 1]$ such that $f(x) = 1$ and $f(y) = 0$ for all $y \in F$.

For each Borel measurable function $f : \mathcal{X} \to \mathbb{R}$, define

$$\Lambda_f := \lim_{\varepsilon\to 0} \varepsilon \log \int_{\mathcal{X}} e^{\frac{f(x)}{\varepsilon}} d\mu_\varepsilon(x).$$

**Theorem 5.9** (Bryc)**.** *Let $\mathcal{X}$ be a completely regular topological space and $\mathcal{B}_{\mathcal{X}} \subset \mathcal{B}$. Suppose that the family $\{\mu_\varepsilon\}$ is exponentially tight and that the limit $\Lambda_f$ exists for every $f \in C_b(\mathcal{X})$. Then $\{\mu_\varepsilon\}$ satisfies the LDP with the good rate function*

$$I(x) = \sup_{f\in C_b(\mathcal{X})} \big\{f(x) - \Lambda_f\big\}. \tag{5.4}$$

*Furthermore, for every $f \in C_b(\mathcal{X})$,*

$$\Lambda_f = \sup_{x\in\mathcal{X}} \{f(x) - I(x)\}. \tag{5.5}$$

*Proof.* Since $\Lambda_0 = 0$, it follows that $I \geq 0$. Moreover, it is lower semicontinuous since it is the supremum of continuous functions. According to Remark 1.6, it suffices to show that $\{\mu_\varepsilon\}$ satisfies the weak LDP with rate $I$. Finally, by Varadhan's lemma (Theorem 5.4), (5.4) implies (5.5).

The proof of upper bound of the LDP is essentially Theorem 6.2(3) by, formally, replacing $\langle\lambda, x\rangle = e^{\lambda(x)}$ by $\lambda \in C_b(\mathcal{X})$.

For the lower bound, fix $x \in \mathcal{X}$ and an open neighborhood $G$ of $x$. Since $\mathcal{X}$ is completely regular, choose $f \in C_b(\mathcal{X})$ such that $f(x) = 1$ and $f(y) = 0$ for all $y \in G^c$. For $m > 0$, define $f_m = m(f - 1)$ so that

$$\int e^{\frac{f_m(x)}{\varepsilon}} d\mu_\varepsilon(x) \leq e^{-\frac{m}{\varepsilon}} + \mu_\varepsilon(G) \leq e^{-\frac{m}{\varepsilon}} + \mu_\varepsilon(G).$$

Since $f_m \in C_b(\mathcal{X})$ and $f_m(x) = 0$,

$$\begin{aligned}
&\max\Big\{\liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(G), -m\Big\} \\
&\geq \{\liminf_{\varepsilon\to 0} \varepsilon \log \int e^{\frac{f_m(x)}{\varepsilon}} d\mu_\varepsilon(x) = \Lambda_{f_m} \\
&= -\Big[f_m(x) - \Lambda_{f_m}\Big] \geq -\sup_{f\in C_b(\mathcal{X})} \big\{f(x) - \Lambda_f\big\} = -I(x),
\end{aligned}$$

from which the lower bound follows by letting $m \to \infty$. □

5.4. **Large deviation on projective spaces.**

Let $(J, \leq)$ be a partially ordered, right-filtering set (for every $i, j \in J$, there exists $k \in J$ such that $i, j \leq k$) and $\{\mathcal{Y}_j\}_{j\in J}$. A *projective system* $(\mathcal{Y}_j, p_{ij})_{i\leq j}$ consists of topological vector spaces $\mathcal{Y}_j$ and continuous maps $p_{ij} : \mathcal{Y}_j \to \mathcal{Y}_i$ such that $p_{ij} \circ p_{jk} = p_{ik}$ and that $p_{jj}$ is the identity map. The *projective limit* $\mathcal{X} = \lim_j \mathcal{Y}_j$ is defined as the (topological) subspace $\mathcal{X}$ of the product topological space $\mathcal{Y} = \prod_{j\in J} \mathcal{Y}_j$, defined as

$$\mathcal{X} = \big\{(y_j)_{j\in J} \in \mathcal{Y} : p_{ij}(y_j) = y_i, \forall i \leq j\big\},$$

which is naturally associated continuous map $p_j : \mathcal{X} \to \mathcal{Y}_j$.

**Theorem 5.10** (Dawson-Gärtner)**.** *Let $\{\mu_\varepsilon\}$ be a family of probability measures on $\mathcal{X}$, such that for any $j \in J$ the Borel probability measures $\big\{\mu_\varepsilon \circ p^{-1}\big\}$ on $Y_j$ satisfy the LDP with the good rate function $I_j$. Then $\big\{\mu_j\big\}$ satisfies the LDP with the good rate function*

$$I(\mathbf{x}) = \sup\big\{I_j\big(p_j(\mathbf{x})\big) : j \in J\big\} \text{ for } \mathbf{x} \in \mathcal{X}.$$

*Proof.* We first show that $I$ is a good rate function. Obviously, $I$ is nonnegative. Note that $\Psi_{I_j}(\alpha)$ is compact by goodness assumption of $I_j$. By the contraction principle, $I_i(y_i) = \inf_{y_j\in p_{ij}^{-1}(y_i)} I_j\big(y_j\big)$, which, under the goodness assumption, implies $\Psi_{I_i}(\alpha) = p_{ij} \circ \Psi_{I_j}(\alpha)$. Hence, $\Big(\Psi_{I_j}(\alpha), p_{ij}\Big)_{ij}$ is a projective system and

$$\Psi_I(\alpha) = \mathcal{X} \cap \prod_{j\in J} \Psi_{I_j}(\alpha)\Big(= \lim_j \Psi_{I_j}(\alpha)\Big).$$

The lower semicontinuity and goodness of $\Psi_I(\alpha)$ then follows by Tychonoff's theorem.

For lower bound, given any measurable set $A \subset \mathcal{X}$ and every $\mathbf{x} \in A$, there exists an open set $U_j \subset \mathcal{Y}_j$ such that $\mathbf{x} \in p_j^{-1}\big(U_j\big) \cap \mathring{A}$

$$\liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(A) \geq \liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon\big(p_j^{-1}\big(U_j\big)\big) \geq -I_j\big(p_j(\mathbf{x})\big) \geq -I(\mathbf{x}).$$

For the upper bound, we first claim that given any $A \subset \mathcal{X}$, we have that, with $A_j := p_j\big(\overline{A}\big)$, $\big(\overline{A_j}, p_{ij}\big)$ is a projective system and $\overline{A} = \lim_j \overline{A_j}$. Observe that $p_{ij}\big(\overline{A_j}\big) \subset \overline{A_i}$. Hence, $\overline{A} \subset \lim_j \overline{A_j}$. To prove the other inclusion, observe that for every $\mathbf{x} \in \overline{A}^c$, there exists some open set $U_j \subset \mathcal{Y}_j$ such that $\mathbf{x} \in p_j^{-1}\big(U_j\big) \subset \overline{A}^c$. This implies $p_j(\mathbf{x}) \in U_j \subset A_j^c$ and thus $p_j(\mathbf{x}) \notin \overline{A_j}$, proving $\overline{A} \supset \lim_j \overline{A_j}$. From this claim, we infer that

$$\overline{A} \cap \Psi_I(\alpha) = \lim_j \overline{A_j} \cap \Psi_{I_j}(\alpha) \quad \text{for all} \quad \alpha < \infty.$$

By taking $\alpha < \inf_{\underline{x\in \overline{A}}} I(x)$, we obtained, by the finite intersection property, there exists $j$ such that $\overline{A_j} \cap \Psi_{I_j}(\alpha)$, and consequently,

$$\limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(A) \leq \limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon \circ p_j^{-1}\Big(\overline{A_j}\Big) \leq -\alpha.$$

This finishes the proof as $\alpha$ is arbitrary. □

## 6. LDP on topological vector spaces

In the previous sections, we have obtained from the Markov's inequality a natural candidate of rate function $\Lambda^*$. In fact, this candidate is the rate function for many general processes, and for this reason we will study the function systematically in this section. Throughout the section, $\mathcal{X}$ is a Hausdorff real topological vector space.

**Definition 6.1.** Let $\mathcal{X}$ be a Hausdorff real topological vector space and $\mu_\varepsilon \in M_1(\mathcal{X})$ be the probability measure associated with $Z_\varepsilon$.

- Define *the logarithmic generating function* as $\Lambda_{\mu_\varepsilon} : \mathcal{X}^* \to (-\infty, \infty]$
$$\Lambda_{\mu_\varepsilon}(\lambda) = \log \mathbb{E}\big(e^{\langle \lambda, Z_\varepsilon \rangle}\big)$$
- Let $\overline{\Lambda}(\lambda) = \limsup_{\varepsilon \to 0} \varepsilon \Lambda_{\mu_\varepsilon}\big(\frac{\lambda}{\varepsilon}\big)$ and write $\Lambda(\lambda) = \overline{\Lambda}(\lambda)$ if the limit exists.
- Define the *Fenchel-Legendre transform* $\overline{\Lambda}^*$ of $\overline{\Lambda}$ as
$$\overline{\Lambda}^*(x) = \sup_{\lambda \in \mathcal{X}^*} \Big[\langle \lambda, x \rangle - \overline{\Lambda}(\lambda)\Big].$$

### 6.1. General upper bound.

**Theorem 6.2.**
*(1) $\overline{\Lambda}$ is a convex on $\mathcal{X}^*$*
*(2) $\overline{\Lambda}^*$ is a convex rate function.*
*(3) For any compact set $\Gamma \subset \mathcal{X}$,*
$$\limsup_{n \to \infty} \varepsilon \log \mu_\varepsilon(\Gamma) \le -\inf_{x \in \Gamma} \overline{\Lambda}^*(\lambda)$$

*Proof.* (1) By Hölder's inequality, given any $t, t' \in [0,1]$ satisfying $t + t' = 1$,
$$\log \mathbb{E}\Big(e^{\langle t\lambda + t'\lambda', Z_\varepsilon \rangle}\Big) \le t \log \mathbb{E}\big(e^{\langle \lambda, Z_\varepsilon \rangle}\big) + t' \log \mathbb{E}\Big(e^{\langle \lambda', Z_\varepsilon \rangle}\Big),$$
proving the convexity of $\Lambda_{\mu_\varepsilon}$ and hence $\overline{\Lambda}$.

(2) The convexity of $\overline{\Lambda}^*$ follows immediately from definition. To see $\overline{\Lambda}^*$ is rate function, note that (a) it is non-negative
$$\overline{\Lambda}^*(x) \ge \langle 0, \lambda \rangle - \overline{\Lambda}(0) = 0,$$
and that (b) it is lower semicontinuous, as $x \mapsto \langle x, \lambda \rangle - \Lambda(\lambda)$ is continuous for all $\lambda \in \mathcal{X}^*$.

(3) Choose for each $x \in \Gamma$ an $\lambda_x \in \mathcal{X}^*$ and a open neighborhood $A_x$ of $x$ such that
$$\langle \lambda_x, x \rangle - \overline{\Lambda}(x) \ge I^\delta(x),$$
$$\inf_{y \in A_x} \{\langle \lambda_x, y \rangle - \langle \lambda_x, x \rangle\} \ge -\delta,$$
where $I^\delta$ is the $\delta$-rate function associated with $\overline{\Lambda}^*$ as defined in (3.8). Applying the Markov's inequality yields
$$\mu_\varepsilon(A_x) \le \mathbb{E}\Big(e^{\langle \frac{\lambda_x}{\varepsilon}, Z_\varepsilon \rangle - \langle \frac{\lambda_x}{\varepsilon}, x \rangle}\Big) \exp\bigg(-\inf_{y \in A_x} \Big\langle \frac{\lambda_x}{\varepsilon}, y \Big\rangle - \Big\langle \frac{\lambda_x}{\varepsilon}, x \Big\rangle\bigg),$$
which in turn implies
$$\varepsilon \log \mu_\varepsilon(A_x) \le \delta - \bigg[\langle \lambda_x, x \rangle - \varepsilon \Lambda_{\mu_\varepsilon}\bigg(\frac{\lambda_x}{\varepsilon}\bigg)\bigg].$$

Now that $\Gamma$ is a compact set, one can find a finite cover $\{A_{x_i}\}_{i=1}^N$ with $x_i \in \Gamma$ such that

$$\limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(\Gamma) \le \delta - \min_{1\le i\le N} I^\delta(x_i) \le \delta - \inf_{x\in\Gamma} I^\delta(x),$$

proving the theorem by letting $\delta \to 0$. □

6.2. **Rate function and Fenchel-Legendre transform.**

A large class of topological vector spaces, for example when $\mathcal{X}$ is the space of finite measures on Polish space, is actually locally convex. For this reason, we expect good properties regarding LDP for such spaces.

**Theorem 6.3.** *Let $\mathcal{X}$ be a locally convex Hausdorff topological vector space. Assume that $\mu_\varepsilon$ satisfies the LDP with a good rate function $I$. Suppose in addition that*

$$\overline{\Lambda}(\lambda) = \limsup_{\varepsilon\to 0} \varepsilon \Lambda_{\mu_\varepsilon}\left(\frac{\lambda}{\varepsilon}\right), \quad \forall \lambda \in \mathcal{X}^*. \tag{6.1}$$

*(1) For each $\lambda \in \mathcal{X}^*$, $\lim_{\varepsilon\to 0} \varepsilon \Lambda_{\mu_\varepsilon}\left(\frac{\lambda}{\varepsilon}\right)$ exists, is finite, and*

$$\Lambda(\lambda) = \sup_{x\in\mathcal{X}} [\langle \lambda, x\rangle - I(x)].$$

*(2) If $I$ is convex, then it is the Fenchel–Legendre transform of $\Lambda$, namely,*

$$I(x) = \Lambda^*(x) := \sup_{\lambda\in\mathcal{X}^*} [\langle \lambda, x\rangle - \Lambda(x)].$$

*(3) If $I$ is not convex, then $\Lambda^*$ is the affine regularization of $I$, i.e.,*

$$\Lambda^* = \sup\{f : f \le I, f \text{ is convex}\}.$$

*Proof.* (1) Fix $\lambda \in \mathcal{X}^*$ and $\gamma > 1$. By assumption, Varadhan's lemma (Theorem 5.4) asserts

$$\lim_{\varepsilon\to 0} \varepsilon \Lambda_{\mu_\varepsilon}\left(\frac{\lambda}{\varepsilon}\right) = \sup_{\lambda\in\mathcal{X}} [\langle \lambda, x\rangle - I(x)] < \infty.$$

Since $\Lambda$ is convex and $\Lambda(0) = 0$, it follows that $\Lambda > -\infty$.

(2) It follows from the duality lemma (Lemma D.1).

(3) By (1),

$$\Lambda^*(x) = I^{**}(x) = \sup_{\lambda\in\mathcal{X}^*} \inf_{y\in\mathcal{X}} [\langle \lambda, x-y\rangle + I(y)] \le I(x).$$

By Hahn-Banach separation theorem (Theorem C.2),

$$\sup\{f : f \le I, f \text{ is convex}\} = \sup\{\lambda + c : \lambda + c \le I, \lambda \in \mathcal{X}^*, c \in \mathbb{R}\}.$$

If $I \ge \lambda + c$, then it is necessary $\inf_{y\in\mathcal{X}} I(y) - \langle \lambda, y\rangle \ge c$, or equivalently,

$$\inf_{y\in\mathcal{X}} \langle \lambda, x-y\rangle + I(y) \ge \langle \lambda, x\rangle + c,$$

from which the claim follows. □

**Theorem 6.4.** *Suppose that $\{\mu_\varepsilon\}$ satisfies a weak LDP with a convex rate function $I$, and that $\mathcal{X}$ is a locally convex, Hausdorff topological vector space. Assume that for each $\lambda \in \mathcal{X}$, the limits $\Lambda_\lambda(t) = \lim_{\varepsilon\to 0} \varepsilon \Lambda_{\mu_\varepsilon}(t\lambda/\varepsilon)$ exist as extended real numbers, and that $\Lambda_\lambda(t)$ is a lower semicontinuous function of $t \in \mathbb{R}$. If for every $\lambda \in \mathcal{X}^*$ and every $a \in \mathbb{R}$,*

$$\inf_{x:\langle\lambda,x\rangle>a} I(x) \le \inf_{z>a} \Lambda^*_\lambda(z),$$

*then $I = \Lambda^*$, and consequently, $\Lambda^*$ controls a weak LDP associated with $\{\mu_\varepsilon\}$.*

*Proof.* By the assumption,

$$\begin{aligned}\sup_{x\in\mathcal{X}}\{\langle\lambda,x\rangle - I(x)\} &\ge \sup_{a\in\mathbb{R}} \sup_{x:\langle\lambda,x\rangle>a} \{\langle\lambda,x\rangle - I(x)\}\\ &\ge \sup_{a\in\mathbb{R}}\Big[a - \inf_{x:\langle\lambda,x\rangle>a} I(x)\Big]\\ &\ge \sup_{a\in\mathbb{R}}\Big[a - \inf_{z>a}\Lambda^*_\lambda(z)\Big] = \sup_{z\in\mathbb{R}}[z - \Lambda^*_\lambda(z)].\end{aligned}$$

Note that $\Lambda_\lambda$ is convex with $\Lambda_\lambda(0) = 0$ and is lower semicontinuous. Therefore, it can not attain the value $-\infty$. By the duality lemma (Lemma D.1), $\sup_{z\in\mathbb{R}}[z - \Lambda^*_\lambda(z)] = \Lambda_\lambda(1) = \Lambda(\lambda)$, proving

$$\sup_{x\in\mathcal{X}}\{\langle\lambda,x\rangle - I(x)\} \ge \Lambda(\lambda).$$

By Lemma 5.5, we have

$$\Lambda(\lambda) \ge \sup_{x\in\mathcal{X}}[\langle\lambda,x\rangle - I(x)].$$

Combining the above gives $I^* = \Lambda$. Applying the duality lemma (Lemma D.1) again gives $\Lambda^* = I^{**} = I$. □

6.3. **Convexity consideration.**

**Lemma 6.5.** *Let $\mathcal{A}$ be a base for a Hausdorff topological vector space $\mathcal{X}$, such that (1) $\overline{I} = \underline{I}$ and that (2) for every $A_1, A_2 \in \mathcal{A}$, $\overline{\mathcal{L}}_{A_1+A_2} \ge \frac{1}{2}\big(\underline{\mathcal{L}}_{A_1} + \underline{\mathcal{L}}_{A_2}\big)$. Then the rate function $I$ of the weak LDP for $\{\mu_\varepsilon\}$ is convex.*

*Proof.* It suffices to show that $I\big(\frac{x+y}{2}\big) \le \frac{I(x)+I(x)}{2}$ for all $x, y \in \mathcal{X}$, as the assertion amounts to that $I(tx + (1-t)y) \le tI(x) + (1-t)I(y)$ for all $x, y \in \mathcal{X}$ and all $t = k\cdot 2^{-n}$ with $k = 0, 1, ..., 2^n$ and $n \in \mathbb{N}$. By lower semincontiuity, the inequality further extends to hold for all $t \in [0, 1]$.

To prove the claim, our starting point is the observation that the Hausdorff TVS is regular, which follows from the fact that $+ : \mathcal{X}\times\mathcal{X} \to \mathcal{X}$ is continuous. By virtue of the regularity and lower semicontinuity, we can find for every $x, y \in \mathcal{X}$ a neighborhoods $A \in \mathcal{A}$ of $\frac{x+y}{2}$, $A_1 \in \mathcal{A}$ of $x$ and $A_2 \in \mathcal{A}$ of $y$ such that $A \supset \overline{A_1} + \overline{A_2}$ satisfying

$$-I^\delta\Big(\frac{x+y}{2}\Big) \ge \overline{\mathcal{L}}_A \ge \frac{1}{2}\big(\overline{\mathcal{L}}_{A_1+A_2}\big) \ge \underline{\mathcal{L}}_{A_1} + \underline{\mathcal{L}}_{A_2} \ge \frac{1}{2}I(x) + \frac{1}{2}I(y).$$

This conclude the proof. □

**Corollary 6.6.** *Suppose that $\{\mu_\varepsilon\}$ is exponentially tight, satisfies (6.1) and the assumptions of Lemma 6.5. Then $\{\mu_\varepsilon\}$ satisfies in $\mathcal{X}$ the LDP with the good, convex rate function $\Lambda^*$ .*

*Proof.* By Lemma 6.5, $\{\mu_\varepsilon\}$ satisfies a weak LDP with a convex rate function. As $\{\mu_\varepsilon\}$ is exponentially tight, it is deduced that it satisfies the full LDP with a convex, good rate function. The proof is concluded by Theorem 6.3 (1) and (2). □

## 7. Application - Cramér's theorem in Polish space

For convenience, we write

$$\hat{S}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \text{ and } \hat{S}_m^n = \frac{1}{n-m}\sum_{i=m+1}^{n} X_i.$$

**Theorem 7.1.** *Let $\mathcal{X}$ is a locally convex, Hausdorff, topological real vector space, and $X_i$, $\mu_n$, $\Lambda$, and $\Lambda^*$ be as defined. Assume the following hold.*

*(1) There is a closed, convex subset $\mathcal{E}$ of $\mathcal{X}$ such that $\mu(\mathcal{E}) = 1$ and $\mathcal{E}$ can be made into a Polish space with respect to the topology induced by $\mathcal{X}$.*

*(2) The closed convex hull of each compact $K \subset \mathcal{E}$ is compact.*

*Then $\{\mu_n\}$ satisfies in $\mathcal{X}$ (and $\mathcal{E}$) a weak LDP with rate function $\Lambda^*$. Moreover, for every open, convex subset $A \subset \mathcal{X}$,*

$$\lim_{n\to\infty} \log \mu_n(A) = -\inf \Lambda^*(x).$$

**Remark 7.2.** *The theorem, in particular, applies to separable Banach spaces, or $M_1(\Sigma)$ with $\Sigma$ a Polish space.*

As a corollary of the theorem, we generalize the Cramer's theorem to the following.

**Corollary 7.3** (Cramér)**.** *The sequence $\{\mu_n\}$ of the laws of empirical means of $\mathbb{R}^d$ -valued i.i.d. random variables satisfies a weak LDP with the convex rate function $\Lambda^*$. Moreover, if $0 \in \mathring{\mathcal{D}}_\Lambda$, then $\{\mu_n\}$ satisfies the full LDP with the good, convex rate function $\Lambda^*$.*

*Proof.* The weak LDP is a consequence of Theorem 7.1, from which the full LDP follows due to the exponential tightness. □

### 7.1. Proofs.

**Lemma 7.4.** *Let Theorem 7.1(1) hold true. Then, the sequence $\{\mu_n\}$ satisfies the weak LDP in $\mathcal{X}$ with a convex rate function $I$.*

*Proof.* The lemma follows from Lemma 6.5 with a base $\mathcal{A}$ of topology consisting of convex (open) sets, as argued in the following.

Note that for all convex set $A$, $\mu_n(A) = \mu_n(A \cap \mathcal{E})$ and thus $\overline{\mathcal{L}}_A = \underline{\mathcal{L}}_A$ exists by subadditivity. In addition, we know that for any convex open sets $A_1$ and $A_2$,

$$\begin{aligned}\mu_n(A_1)\mu_n(A_2) &= P_\mu\Big(\hat{S}_n \in A_1, \hat{S}_n^{2n} \in A_2\Big)\\ &\le P_\mu\Big(\hat{S}_{2n} \in \frac{1}{2}(A_1 + A_2)\Big) = \mu_{2n}\Big(\frac{1}{2}(A_1 + A_2)\Big),\end{aligned}$$

from which the weak LDP holds for $\{\mu_n\}$ with rate function

$$I(x) = -\inf_{A\in\mathcal{A}: x\in A} \underline{\mathcal{L}}_A.$$

□

**Lemma 7.5.** *Let Theorem 7.1(1)(2) hold true. Then, for every open, convex subset $A \subset \mathcal{X}$,*

$$\lim_{n\to\infty} \frac{1}{n} \log \mu_n(A) = -\inf_{x\in A} I(x),$$

*where $I$ is the convex rate function in Lemma 7.4.*

*Proof.* By definition,

$$-I(x) \le \underline{\mathcal{L}}_A = \overline{\mathcal{L}}_A \Rightarrow -\inf_{x\in A} I(x) \le \underline{\mathcal{L}}_A = \overline{\mathcal{L}}_A.$$

Hence, it suffices to prove the other inequality. To this end, we first make use of the subadditivity to approximate the limit and tightness to approximate the set $A$. Specifically, for all $\delta > 0$, there exist $N \in \mathbb{N}$ and compact set $C \subset A$ such that

$$\underline{\mathcal{L}}_A \le \frac{1}{N}\log \mu_N(A) + \delta \le \frac{1}{N}\log \mu_N(\overline{\mathrm{co}}(C)) + 2\delta \le \overline{\mathcal{L}}_{\overline{\mathrm{co}}(C)} \le -\inf_{x\in A} I(x),$$

where the third inequality follows from subadditivity. This completes the proof. □

*Proof of Theorem 7.1.* In light of Theorem 6.4, it suffices to show that

$$\frac{1}{n}\log \mathbb{E}\Big[e^{t\langle \lambda, n\hat{S}_n\rangle}\Big] = \log \mathbb{E}\big[e^{t\langle \lambda, X_1\rangle}\big] =: \Lambda_\lambda(t)$$

is lower semicontinuous and

$$\inf_{x:\langle\lambda,x\rangle>a} I(x) \le \inf_{z>a} \Lambda^*_\lambda(z) \text{ for all } a \in \mathbb{R} \text{ and } \lambda \in \mathcal{X}^*.$$

The former follows obviously from Fatou's lemma. As for the latter, we consider the following two cases: (1) $\lambda = 0$ and (2) $\lambda \neq 0$.

(1) We assume $\lambda = 0$, so that

$$\Lambda^*_\lambda(x) = \begin{cases} 0 & \text{if } x = 0, \\ \infty & \text{if } x \neq 0, \end{cases}$$

Given that $\mu_n(\mathcal{X}) = 1$, it follows from Lemma 7.5 that $\inf_{x\in\mathcal{X}} I(x) = 0$. Hence, the proposed inequality holds.

(2) Note that the half-plane $H_a := \{x : \langle\lambda, x\rangle > a\}$ is convex and open. By Lemma 7.5,

$$-\inf_{x\in H_a} I(x) = \overline{\mathcal{L}}_{H_a} \ge \sup_{\delta>0} \overline{\mathcal{L}}_{H_{a+\delta}}.$$

To conclude the proof, define $Y_i = \langle\lambda, X_i\rangle$ and $\hat{Z}_n = \sum_{i=1}^n Y_i$ so that the logarithmic moment generating function of $Y_1$ is exactly $\Lambda_\lambda(t)$. By one-dimensional Cramér's theorem (Theorem 3.3), we have

$$\inf_{\delta>0} \overline{\mathcal{L}}_{H_{a+\delta}} = \sup_{\delta>0}\Big[-\inf_{z>a+\delta} I(x)\Big] = -\inf_{z>a} I(x),$$

This proves the theorem. □

## 8. Gärtner-Ellis theorem.

Throughout this section, the space $\mathcal{X}$ is a Hausdorff topological vector space.

### 8.1. Baldi's theorem.

In this subsection, we consider a generalization of Sanov's theorem on topological vector space.

**Definition 8.1.** A point $x \in \mathcal{X}$ is called a an *exposed point* of $\overline{\Lambda}^*$ if there exists an *exposing hyperplane* $\lambda \in \mathcal{X}^*$ such that for every $z \neq x$,

$$\langle \lambda, x\rangle - \overline{\Lambda}^*(x) > \langle \lambda, z\rangle - \overline{\Lambda}^*(z).$$

**Theorem 8.2** (Baldi)**.** *Suppose $\{\mu_\varepsilon\}$ are exponentially tight family of probability measures on $\mathcal{X}$.*

*(1) For every closed set $F \subset \mathcal{X}$, $\limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(F) \leq -\inf_{x\in F} \overline{\Lambda}^*(x)$.*

*(2) Let $\mathcal{F}$ be the set of exposed points of $\overline{\Lambda}^*$ with an exposing hyperplane $\lambda$ for which*

$$\Lambda(\lambda) = \lim_{\varepsilon\to 0} \varepsilon \Lambda_{\mu_\varepsilon}\left(\frac{\lambda}{\varepsilon}\right) \text{ exists and } \overline{\Lambda}(\gamma\lambda) < \infty \text{ for some } \gamma > 1. \tag{8.1}$$

*Then, for every open set $G \subset \mathcal{X}$,*

$$\liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(G) \geq -\inf_{x\in G\cap\mathcal{F}} \overline{\Lambda}^*(x).$$

*(3) If for every open set $G \subset \mathcal{X}$, $\inf_{x\in G\cap\mathcal{F}} \overline{\Lambda}^*(x) = \inf_{x\in G} \overline{\Lambda}^*(x)$, then $\{\mu_\varepsilon\}$ satisfies the LDP with the good rate function $\overline{\Lambda}^*$.*

*Proof.* (1) It follows from the exponential tightness together with Theorem 6.2.

(2) The case $\overline{\Lambda}(\lambda) = -\infty$ for some $\lambda \in \mathcal{X}^*$ is trivial, for $\overline{\Lambda}^*(\cdot) = \infty$ unanimously. Assuming $\overline{\Lambda}(\lambda) < \infty$ for all $\lambda \in \mathcal{X}^*$, one need to translate $\overline{\Lambda}$ by $\lambda$ in the following manner. Let $G$ be any fixed open set, $y \in G \cap \mathcal{F}$, $\delta > 0$, and $\eta \in \mathcal{X}^*$ be an exposing hyperplane for $\overline{\Lambda}^*$ such that (8.1) holds. By continuity of $\eta$, we choose an open neighborhood $B_\delta \subset G$ of $y$ such that

$$\sup_{z\in B_\delta} \langle \eta, z - y\rangle < \delta.$$

According to (8.1), for every sufficiently small $\varepsilon > 0$, $\Lambda_{\mu_\varepsilon}\left(\frac{\eta}{\varepsilon}\right)$ is well-defined, for each of which we may defined a probability measure $\tilde{\mu}_\varepsilon$ equivalent to $\mu_\varepsilon$ (i.e., $\tilde{\mu}_\varepsilon$ and $\mu_\varepsilon$ are absolutely continuous to each other):

$$\frac{d\tilde{\mu}_\varepsilon}{d\mu_\varepsilon}(z) = \exp\left[\left\langle \frac{\eta}{\varepsilon}, z\right\rangle - \Lambda_{\mu_\varepsilon}\left(\frac{\eta}{\varepsilon}\right)\right]$$

so that

$$\begin{aligned}\varepsilon \log \mu_\varepsilon(B_\delta) &= \varepsilon \log \int_{B_\delta} \exp\left[-\left\langle \frac{\eta}{\varepsilon}, z\right\rangle + \Lambda_{\mu_\varepsilon}\left(\frac{\eta}{\varepsilon}\right)\right] d\tilde{\mu}_\varepsilon(z) \\ &= \varepsilon \Lambda_{\mu_\varepsilon}\left(\frac{\eta}{\varepsilon}\right) - \langle \eta, y\rangle + \varepsilon \log \int_{B_\delta} \exp\left[\left\langle \frac{\eta}{\varepsilon}, y - z\right\rangle\right] d\tilde{\mu}_\varepsilon(z),\end{aligned}$$

yielding

$$\liminf_{\varepsilon\to 0}\varepsilon\log\mu_\varepsilon(B_\delta)\geq\overline{\Lambda}(\eta)-\langle\eta,y\rangle+\lim_{\delta\to 0}\liminf_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon(B_\delta),$$
$$\geq-\overline{\Lambda}^*(y)+\lim_{\delta\to 0}\liminf_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon(B_\delta).$$

It remains to show that $\lim_{\delta\to 0}\liminf_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon(B_\delta)=0$. To this end, we may choose, due to exponential tightness, for each $\alpha<\infty$ a compact sets $K_\alpha$ such that

$$\limsup_{\varepsilon\to 0}\varepsilon\log\mu_\varepsilon(K_\alpha^c)<-\alpha$$

and claim that

$$\limsup_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon(B_\delta^c\cap K_\alpha)<0\text{ for all }\delta>0,\alpha>0,\tag{8.2}$$

$$\limsup_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon(K_\alpha^c)<0\text{ for all large }\alpha>0,\tag{8.3}$$

which together implies the desired estimate.

To prove (8.2), define a function for every $\theta\in\mathcal{X}^*$

$$\tilde{\Lambda}(\theta):=\limsup_{\varepsilon\to 0}\varepsilon\Lambda_{\tilde{\mu}_\varepsilon}\left(\frac{\theta}{\varepsilon}\right)$$

and observe that

$$\tilde{\Lambda}(\theta)=\overline{\Lambda}(\theta+\eta)-\Lambda(\eta),$$
$$\tilde{\Lambda}^*(z)=\overline{\Lambda}^*(z)+\Lambda(\eta)-\langle\eta,z\rangle.$$

Combining these with Theorem 6.2(3), one deduces

$$\limsup_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon(B_\delta^c\cap K_\alpha)\leq-\inf_{z\in B_\delta^c\cap K_\alpha}\tilde{\Lambda}^*(z)<0$$

since $\tilde{\Lambda}^*$ is lower semicontinuous and $\eta$ is an exposing hyperplane.

For (8.3), define open half-spaces

$$H_\rho=\{z\in\mathcal{X}:\langle\eta,z\rangle<\rho\}$$

For every $\beta>0$, applying Markov's inequality to $\tilde{\Lambda}(\beta\eta)$ yields

$$\varepsilon\log\tilde{\mu}_\varepsilon\big(H_\rho^c\big)\leq\varepsilon\log\left[\int_{H_\rho^c}e^{\langle\frac{\beta\eta}{\varepsilon},x\rangle-\beta\rho}d\tilde{\mu}_\varepsilon(x)\right]\leq\varepsilon\Lambda_{\tilde{\mu}_\varepsilon}\left(\frac{\beta\eta}{\varepsilon}\right)-\beta\rho,\tag{8.4}$$

and thus, according to (8.1), for all large $\rho>0$,

$$\limsup_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon\big(H_\rho^c\big)\leq\inf_{\beta>0}\big[\tilde{\Lambda}(\beta\eta)-\beta\rho\big]=\inf_{\beta>0}[\Lambda((1+\beta)\eta)-\Lambda(\eta)-\beta\rho]<0.$$

On the other hand, applying Markov's inequality to $\Lambda(\eta)$ gives that for any $\rho$ and all sufficiently large $\alpha=\alpha(\rho)$,

$$\begin{aligned}&\limsup_{\varepsilon\to 0}\varepsilon\log\tilde{\mu}_\varepsilon\big(K_\alpha^c\cap H_\rho\big)\\&\leq\limsup_{\varepsilon\to 0}\varepsilon\log\int_{K_\alpha^c\cap H_\rho}e^{\langle\frac{\eta}{\varepsilon},z\rangle-\Lambda_{\mu_\varepsilon}(\frac{\eta}{\varepsilon})}d\mu_\varepsilon(z)\\&<\rho-\Lambda(\eta)-\alpha<0.\end{aligned}\tag{8.5}$$

Combining (8.4) and (8.5) proves (8.3).

(3) It follows immediately from (1) and (2). □

In the study of large deviations, a general criterion for the lower bound to hold is the differentiability of $\Lambda$ in the following sense.

**Definition 8.3** (Gateaux differentiable)**.** Let $\mathcal{X}$ be a topological vector space. A function $f : \mathcal{X}^* \to \mathbb{R}$ is said to be *Gateaux differentiable* if for all $\lambda, \theta \in \mathcal{X}^*$, the function $t \in \mathbb{R} \mapsto f(\lambda + t\theta)$ is differentiable at $t = 0$.

**Corollary 8.4.** *Let $\mu_\varepsilon$ be exponentially tight probability measures on the Banach space $\mathcal{X}$. Suppose that $\Lambda(\cdot) = \lim_{\varepsilon\to 0} \Lambda_{\mu_\varepsilon}\left(\frac{\cdot}{\varepsilon}\right)$ is finite valued, Gateaux differentiable, and lower semicontinuous in $\mathcal{X}^*$ with respect to the weak* topology. Then $\{\mu_\varepsilon\}$ satisfies the LDP with the good rate function $\Lambda^*$.*

*Proof.* It suffices to show that for any $x \in \mathcal{D}_{\Lambda^*}$, there exists a sequence of exposed points $x_k$ such that $x_k \to x$ and $\Lambda^*(x_k) \to \Lambda^*(x)$. Define

$$\partial\Lambda^*(x) := \left\{\lambda \in \mathcal{X}^* : \langle\lambda, x\rangle - \Lambda^*(x) = \sup_{z\in\mathcal{X}}[\langle\lambda, z\rangle - \Lambda^*(z)]\right\}.$$

and

$$\partial\Lambda^* = \{x \in \mathcal{X} : \partial\Lambda^*(x) \neq \emptyset\}.$$

Assume that $\mathcal{D}_{\Lambda^*} \neq \emptyset$. Since $\Lambda^*$ is convex and lower semicontinuous, by the Brønsted–Rockafellar theorem (Theorem D.4), for every $x \in \mathcal{D}_{\Lambda^*}$, there exists a sequence $x_k \in \partial\Lambda^*$ such that $x_k \to x$ and $\Lambda^*(x_k) \to \Lambda^*(x)$.

It is enough to prove that when $\Lambda$ is Gateaux differentiable and weak* lower semicontinuous, any point in dom $\partial\Lambda^*$ is also an exposed point. To this end, observe that

$$\Lambda(\lambda) = \sup_{x\in\mathcal{X}}[\langle\lambda, x\rangle - \Lambda^*(x)] = \langle\lambda, x\rangle - \Lambda^*(x).$$

Therefore, by Gateaux differentiability, for every $\theta \in \mathcal{X}^*$,

$$\langle\theta, x\rangle \le \lim_{t\to 0^+} \frac{1}{t}[\Lambda(\lambda + t\theta) - \Lambda(\lambda)] =: D\Lambda(\theta) \quad \text{and} \quad D\Lambda(\theta) = -D\Lambda(-\theta).$$

Consequently, $\langle\theta, x\rangle = D\Lambda(\theta)$ for all $\theta \in \mathcal{X}^*$. Due to this property, if there exists $y \in \mathcal{X}$ satisfying

$$\Lambda(\lambda) = \langle\lambda, y\rangle - \Lambda^*(y) \Rightarrow \langle\lambda, y\rangle = D\Lambda(\lambda),$$

then $\langle\theta, y\rangle = \langle\theta, x\rangle$ for all $\theta \in \mathcal{X}^*$, implying $x = y$. This completes the proof. □

8.2. **Gärtner-Ellis theorem.**

For the finite dimensional vector space, the Baldi's theorem reduces to the Gärtner-Ellis theorem. Let $\Lambda_n$ be the logarithmic moment generating function associated with random variables $Z_n$, which can be defined as

$$\Lambda_n(\lambda) := \log \mathbb{E}\left[e^{\langle\lambda, Z_n\rangle}\right].$$

The Gärtner-Ellis theorem states the following.

**Definition 8.5.** A convex function $\Lambda : \mathbb{R}^d \to (-\infty, \infty]$ is *essentially smooth* if:

(1) $\mathring{\mathcal{D}}_\Lambda$ is nonempty.
(2) $\Lambda$ is differentiable throughout $\mathring{\mathcal{D}}_\Lambda$.
(3) $\Lambda$ is steep, namely, $\lim_{n\to\infty}|\nabla\Lambda(\lambda_n)| = \infty$ whenever $\{\lambda_n\}$ is a sequence in $\mathring{\mathcal{D}}_\Lambda$ converging to a boundary point of $\mathring{\mathcal{D}}_\Lambda$.

**Definition 8.6.** The *relative interior* of a nonempty convex set $C$ is defined as

$$\text{ri } C = \{y \in C : \text{for all } x \in C, y - \varepsilon(x-y) \in C \text{ for some } \varepsilon > 0\}.$$

**Theorem 8.7** (Gärtner-Ellis)**.** *Suppose that* $\Lambda(\lambda) := \lim_{n\to\infty} \frac{1}{n}\Lambda_n(n\lambda)$ *exists for all* $\lambda \in \mathbb{R}^d$ *as extended real numbers and* $0 \in \mathring{\mathcal{D}}_\Lambda$.

*(1) For any closed set* $F$*,*

$$\limsup_{n\to\infty} \frac{1}{n} \log \mu_n(F) \le -\inf_{x\in F} \Lambda^*(x).$$

*(2) For any open set* $G$*,*

$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(G) \ge -\inf_{x\in G\cap\mathcal{F}} \Lambda^*(x),$$

*where* $\mathcal{F}$ *is the set of exposed points of* $\Lambda^*$ *admitting an exposing hyperplane belongs to* $\mathring{\mathcal{D}}_\Lambda$.

*(3) If* $\Lambda$ *is an essentially smooth, lower semicontinuous function, then the LDP holds with the good rate function* $\Lambda^*$.

**Lemma 8.8.** *Under the same assumption as in Theorem 8.7, the following hold.*

*(1)* $\Lambda(\lambda)$ *is a convex function,* $\Lambda(\lambda) > -\infty$ *everywhere, and* $\Lambda^*(x)$ *is a good convex rate function.*

*(2) Suppose that* $y = \nabla\Lambda(\eta)$ *for some* $\eta \in \mathring{\mathcal{D}}_\Lambda$*. Then* $\Lambda^*(y) = \langle \eta, y\rangle - \Lambda(\eta)$*. Moreover* $y \in \mathcal{F}$*, with* $\eta$ *being the exposing hyperplane for* $y$*.*

*Proof.* (1) The convexity follows from Hölder's inequality as in Lemma 3.4.

If $\Lambda(\lambda) = -\infty$ for some $\lambda \in \mathbb{R}^d$, then by convexity $\Lambda(\alpha\lambda) = -\infty$ for all $\alpha \in (0,1]$. Since $\Lambda(0) = 0$, it follows by convexity that $\Lambda(-\alpha\lambda) = \infty$ for all $\alpha \in (0,1]$, contradicting the assumption that $0 \in \mathring{\mathcal{D}}$. Thus, $\Lambda > -\infty$ everywhere.

Since $0 \in \mathring{\mathcal{D}}_\Lambda$, it follows that $B_\delta(0) \subset \mathring{\mathcal{D}}_\Lambda$ for some $\delta > 0$, and $c = \sup_{\lambda\in B_\delta(0)} \Lambda(\lambda) < \infty$ since the convex function $\Lambda$ is continuous in $\mathring{\mathcal{D}}_\Lambda$. Therefore,

$$\Lambda^*(x) \ge \sup_{\lambda\in\overline{B}_\delta(0)} [\langle\lambda, x\rangle - \Lambda(\lambda)] \ge |\delta||x| - \sup_{\lambda\in\overline{B}_\delta(0)} \Lambda(x),$$

implying $\Psi_{\Lambda^*}(\alpha)$ is bounded for every $\alpha < \infty$. The function $\Lambda^*$ is convex and lower semicontinuous by a similar argument as that in Lemma 3.4. Combining these implies that $\Lambda^*$ is a good convex rate function.

(2) Suppose now that for some $x \in \mathbb{R}^d$,

$$\Lambda(\eta) = \langle \eta, y\rangle - \Lambda^*(y) \le \langle \eta, x\rangle - \Lambda^*(x).$$

Then, for every $\theta \in \mathbb{R}^d$,

$$\langle \theta, x\rangle \le \Lambda(\eta+\theta) - \Lambda(\eta).$$

In particular,

$$\langle \theta, x\rangle \le \lim_{\varepsilon\to 0} \frac{1}{\varepsilon}[\Lambda(\eta+\varepsilon\theta) - \Lambda(\eta)] = \langle \theta, \nabla\Lambda(\eta)\rangle.$$

Since the inequality holds for all $\theta \in \mathbb{R}^d$, $x = \nabla\Lambda(\eta) = y$. Hence, $y$ is an exposed point of $\Lambda^*$ with exposing hyperplane $\eta \in \mathring{\mathcal{D}}_\Lambda$. □

**Lemma 8.9** (Rockafellar)**.** *If* $\Lambda : \mathbb{R}^d \to (-\infty, \infty]$ *is an essentially smooth, lower semicontinuous, convex function, then* ri $\mathcal{D}_{\Lambda^*} \subseteq \mathcal{F}$.

*Proof.* Assume without loss of generality that $\mathcal{D}_{\Lambda^*} \neq \emptyset$ for the lemma is automatic otherwise. Under the circumstances, fix henceforth $x \in \text{ri } \mathcal{D}_{\Lambda^*}$ and define a function

$$f(\lambda) := \Lambda(\lambda) - \langle \lambda, x \rangle + \Lambda^*(x).$$

Observe that $f : \mathbb{R}^d \to [0, \infty]$ is convex, lower semincontinuous, and $\inf_{\lambda \in \mathbb{R}^d} f(\lambda) = 0$. Moreover, $f^*(\cdot) = \Lambda^*(\cdot + x) - \Lambda^*(x)$. Therefore, from $x \in \text{ri } \mathcal{D}_{\Lambda^*}$ it follows that $0 \in \text{ri } \mathcal{D}_{f^*}$. By Lemma D.2, there exists $\eta \in \mathcal{D}_\Lambda$ such that $f(\eta) = 0$. Let $\tilde{\Lambda}(\cdot) = \Lambda(\cdot + \eta) - \Lambda(\eta)$, so that $\tilde{\Lambda}$ is an essentially smooth, convex function and $\tilde{\Lambda}(0) = 0$. Consequently, by Lemma D.3, $\tilde{\Lambda}$ is finite in a neighborhood of the origin and thus $\eta \in \mathring{\mathcal{D}}_\Lambda$, at which $f$ is differentiable by the assumption. Hence, $f(\eta) = \inf_{\lambda \in \mathbb{R}^d} f(\lambda)$, implying that $\nabla f(\eta) = 0$, i.e., $x = \nabla \Lambda(\eta)$. It now follows from Lemma 8.8(2) that $x \in \mathcal{F}$. Since $x \in \text{ri } \mathcal{D}_{\Lambda^*}$ is arbitrary, the proof is complete. □

*Proof of Theorem 8.7.* (1) and (2) follow from Theorem 8.2 immediately once exponential tightness is proved. To that end, we apply the Markov's inequality to the marginal $\mu_\varepsilon^j$ on the $j$-th coordinate of $\mu_\varepsilon$. Denote by $\boldsymbol{u}_j \in \mathbb{R}^d$ the $j$-th unit vector. Since $0 \in \mathring{\mathcal{D}}_\Lambda$ we can find $\eta_j, \theta_j > 0$ such that

$$\mu_\varepsilon^j[\rho, \infty) \leq \exp\big(-n\eta_j \rho + \Lambda_n(-n\eta_j \boldsymbol{u}_j)\big),$$
$$\mu_\varepsilon^j(-\infty, -\rho] \leq \exp\big(-n\theta_j \rho + \Lambda_n(-n\theta_j \boldsymbol{u}_j)\big).$$

Therefore,

$$\lim_{\rho \to \infty} \limsup_{\varepsilon \to 0} \varepsilon \log \mu_\varepsilon[\rho, \infty) = -\infty,$$
$$\lim_{\rho \to \infty} \limsup_{\varepsilon \to 0} \varepsilon \log \mu_\varepsilon(-\infty, -\rho] = -\infty.$$

The claim is thus proved.

We now prove (3). To this end, it suffices to show that

$$\inf_{x \in G \cap \text{ri } \mathcal{D}_{\Lambda^*}} \Lambda^*(x) \leq \inf_{x \in G} \Lambda^*(x).$$

We shall further assume, without loss of generality, that $G \cap \mathcal{D}_{\Lambda^*}$ is nonempty, for the inequality is obvious otherwise. Under the assumption, ri $\mathcal{D}_{\Lambda^*}$ is also nonempty, and given any fixed $y \in G \cap \mathcal{D}_{\Lambda^*}$ and $z \in \text{ri } \mathcal{D}_{\Lambda^*}$, we have that for all sufficiently small $\varepsilon > 0$,

$$(1 - \varepsilon)y + \varepsilon z \in G \cap \text{ri } \mathcal{D}_{\Lambda^*},$$

which in turn implies

$$\inf_{x \in G \cap \mathcal{D}_{\Lambda^*}} \Lambda^*(x) \leq \limsup_{\varepsilon \to 0} \Lambda^*((1 - \varepsilon)y + \varepsilon z) \leq \Lambda^*(y).$$

The theorem follows immediately as $y \in G \cap \mathcal{D}_{\Lambda^*}$ is arbitrary. □

## 9. Applications

### 9.1. Large deviations for finite state Markov chains.

In this section, we study the large deviations of Markov chains. Let $\Sigma$ be a finite alphabet and $\mathbf{\Pi} = \{\pi(i,j)\}_{i,j=1}^{|\Sigma|}$ be a stochastic matrix, i.e., a nonnegative matrix with sum of each row 1. Consider a Markov chain $\{Y_k\}_{k\geq 0}$ with the transition probability $\mathbf{\Pi}$, for which we denote by $\mathbb{P}_\sigma^\pi$ the Markov probability measure with initial state $\sigma \in \Sigma$, i.e., the probability measure satisfying

$$\mathbb{P}_\sigma^\pi(Y_1 = y_1, \cdots, Y_n = y_n) = \pi(\sigma, y_1) \prod_{i=1}^{n-1} \pi(y_i, y_{i+1}).$$

In the following, we consider the empirical mean

$$Z_n = \sum_{k=1}^{n} X_k,$$

where $X_k = f(Y_k)$ with $f : \Sigma \to \mathbb{R}^d$ a given function. To express the logarithmic moment generating function, we introduce the nonnegative matrix

$$\pi_\lambda(i,j) = \pi(i,j) e^{\langle \lambda, f(j)\rangle} \quad (i,j \in \Sigma).$$

**Theorem 9.1** (Perron-Frobenius)**.** *Let $\mathbf{B} = \{B(i,j)\}_{i,j=1}^{|\Sigma|}$ be an irreducible matrix. Then $B$ possesses an eigenvalue $\rho$ (called the Perron–Frobenius eigenvalue) such that:*

*(1) $\rho > 0$ is real.*
*(2) For any eigenvalue $\lambda$ of $B$, $|\lambda| \leq \rho$.*
*(3) There exist left and right eigenvectors corresponding to the eigenvalue $\rho$ that have strictly positive coordinates.*
*(4) The left and right eigenvectors $\mu$, $\theta$ corresponding to the eigenvalue $\rho$ are unique up to a constant multiple.*
*(5) For every $i \in \Sigma$ and every $\varphi = \left(\varphi_1, \cdots, \varphi_{|\Sigma|}\right)$ such that $\varphi_j > 0$ for all $j$,*

$$\lim_{n\to\infty} \frac{1}{n} \log \left[ \sum_{j=1}^{|\Sigma|} B^n(i,j)\varphi_j \right] = \lim_{n\to\infty} \frac{1}{n} \log \left[ \sum_{j=1}^{|\Sigma|} \varphi_j B^n(j,i) \right] = \log \rho.$$

*Proof.* See, for example, Wikipedia. □

**Theorem 9.2.** *Let $\{Y_k\}$ be a finite state Markov chain possessing an irreducible transition matrix $\mathbf{\Pi}$. For every $z \in \mathbb{R}^d$ , define*

$$I(z) = \sup_{\lambda \in \mathbb{R}^d} \{\langle \lambda, z\rangle - \log \rho(\mathbf{\Pi}_\lambda)\}$$

*Then the empirical mean $Z_n$ satisfies the LDP with the convex, good rate function $I$. Explicitly, for any set $\Gamma \subseteq \mathbb{R}^d$ , and any initial state $\sigma \in \Sigma$,*

$$-\inf_{z\in\mathring{\Gamma}} I(z) \leq \liminf_{n\to\infty} \frac{1}{n} \log \mathbb{P}_\sigma^\pi(Z_n \in \Gamma) \leq \limsup_{n\to\infty} \frac{1}{n} \log \mathbb{P}_\sigma^\pi(Z_n \in \Gamma) \leq -\inf_{z\in\overline{\Gamma}} I(z).$$

*Proof.* Consider the logarithmic generating function

$$\Lambda_n(\lambda) := \log \mathbb{E}_\sigma^\pi\left[e^{\langle \lambda, Z_n\rangle}\right].$$

By Gärtner–Ellis theorem (Theorem 8.7), it is enough to check that the limit

$$\Lambda(\lambda) := \lim_{n\to\infty} \frac{1}{n}\Lambda_n(n\lambda)$$

exists, is finite and differentiable everywhere in $\mathbb{R}^d$, and satisfies $\Lambda(\lambda) = \log\rho(\mathbf{\Pi}_\lambda)$. To begin, note that

$$\Lambda_n(\lambda) = \log \sum_{y_1,\cdots,y_n} \mathbb{P}^\pi_\sigma(Y_1 = y_1, \cdots, Y_n = y_n) \prod_{i=1}^n e^{\langle\lambda, f(y_k)\rangle} = \log \sum_{y_n} (\mathbf{\Pi}_\lambda)^n(\sigma, y_n).$$

Since $\mathbf{\Pi}_\lambda$ is irreducible, we have

$$\Lambda(\lambda) = \log\rho(\mathbf{\Pi}_\lambda).$$

To show that it is differentiable, we apply the implicit function theorem. Explicitly, consider the functions $F_y : \mathbb{R}\times\mathbb{R}^{|\Sigma|}\times\mathbb{R} \to \mathbb{R}\times\mathbb{R}^{|\Sigma|}$, parametrized by $y \in \mathbb{R}^{|\Sigma|}$, defined by

$$F_y(\rho, x, \lambda) = (\langle y, x\rangle - 1, \mathbf{\Pi}_\lambda x - \rho x).$$

Clearly, $F_y$ is continuously differentiable. Hence, it suffices to show that for every $\lambda_0$ with the associated Perron-Frobenius eigenvalue $\rho_0$ and left and right eigenvectors $y_0$ and $x_0$ of $\mathbf{\Pi}_{\lambda_0}$ satisfying $\langle y_0, x_0\rangle = 1$, we have (1) $F_{y_0}(\rho_0, x_0, \lambda_0) = 0$ and (2) the Jacobian matrix

$$\partial_{\rho,x} F_{y_0}(\rho_0, x_0, \lambda_0) = \begin{pmatrix} 0 & y_0^T \\ -x_0 & \mathbf{\Pi}_{\lambda_0} - \rho_0 I. \end{pmatrix}$$

is invertible. Indeed, (1) is clear and (2) holds for if otherwise, there exists $(u, v) \in \mathbb{R}\times\mathbb{R}^{|\Sigma|}\setminus\{(0,0)\}$ such that

$$\begin{cases} \langle y_0, v\rangle = 0, \\ \left\langle y_0, -x_0 + \mathbf{\Pi}_{\lambda_0} v - \rho_0 v\right\rangle = 0, \end{cases}$$

which is impossible. This implies, by the implicit function theorem, that there exists a continuously differentiable function $\lambda \mapsto (\rho(\lambda), x(\lambda))$ on a neighborhood of $\lambda_0$ such that $F_{y_0}(\rho(\lambda), x(\lambda), \lambda) = 0$. In particular, $\lambda \mapsto \log\rho(\mathbf{\Pi}_\lambda) = \log\rho(\lambda)$ is continuously differentiable on a neighborhood of $\lambda_0$. The proof is finished by noting that $\lambda_0$ is arbitrary. □

As a corollary, we have the following.

**Corollary 9.3.** *The empirical averages $L_n^{\mathbf{Y}}(i) = \frac{1}{n}\sum_{k=1}^n \mathbb{1}_i(Y_k)$, with $i \in \Sigma$, satisfies the LDP with the good rate function*

$$I(q) = \sup_{\lambda\in\mathbb{R}^d}\{\langle\lambda, q\rangle - \log\rho(\mathbf{\Pi}_\lambda)\} = \begin{cases} \sup_{u>0}\sum_{j\in\Sigma} q_j \log\left[\frac{u_j}{(\mathbf{u}\mathbf{\Pi})_j}\right] & q \in M_1(\Sigma), \\ \infty & q \notin M_1(\Sigma) \end{cases}$$

*where $\pi_\lambda(i,j) = \pi(i,j)e^{\lambda_j}$ and the inequality between vectors is compared entrywise.*

*Proof.* The first equality follows immediately from Theorem 9.2 by taking

$$f = \left(\mathbb{1}_1, \mathbb{1}_2, \cdots, \mathbb{1}_{|\Sigma|}\right).$$

To prove the second inequality, we first note the one inequality is more obvious than the other: By taking $\mathbf{u}$ to be the left probability eigenvector of $\mathbf{\Pi}_\lambda$, we see the

inequality "$\leq$". To prove the other inequality, assume $q \in M_1(\Sigma)$ and choose $\lambda_j = \log\big[u_j/(\mathbf{u}\mathbf{\Pi})_j\big]$ so that $\mathbf{u}\mathbf{\Pi}_\lambda = \mathbf{u}$ and that $\rho(\mathbf{\Pi}_\lambda) = 1$. Therefore, by definition,

$$I(q) \geq \sum_{j=1}^{|\Sigma|} q_j \log\left[\frac{u_j}{(\mathbf{u}\mathbf{\Pi})_j}\right],$$

finishing the proof. □

We also consider a derived process $\{(Y_k Y_{k+1})\}_{k\geq 0}$ of consecutive pairs to obtain a Sanov's theorem. Such a process has the transition matrix $\mathbf{\Pi}^{(2)}$ defined by

$$\pi^{(2)}((k \times \ell, i \times j)) = \mathbb{1}_\ell(i)\, \pi(i,j).$$

As is discussed in the following, one can determine the large deviations of the pair empirical measure

$$L^{\mathbf{Y}}_{n,2}(y) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_y(Y_{i-1}Y_i)$$

For $q \in M_1(\Sigma^2)$, we write

$$q_1(i) = \sum_{j=1}^{|\Sigma|} q(i,j) \text{ and } q_2(i) = \sum_{j=1}^{|\Sigma|} q(j,i),$$

and call $q$ shift-invariant if $q_1 = q_2$.

**Theorem 9.4.** *Assume that $\mathbf{\Pi}$ is irreducible. Then for every probability measure $q \in M_1(\{(i,j) : \pi(i,j) > 0\})$,*

$$I_2(q) = \begin{cases} \sum_i q_1(i) H(q(\cdot\,|i) \mid \pi(i,\cdot)) & \text{if } q \text{ is shift-invariant,} \\ \infty & \text{otherwise.} \end{cases}$$

*Proof.* By Corollary 9.3,

$$I_2(q) = \sup_{u>0} \sum_{i,j\in\Sigma} q(i,j) \log\left[\frac{u_{i,j}}{(\mathbf{u}\mathbf{\Pi}^{(2)})_{i,j}}\right] = \sup_{u>0} \sum_{i,j\in\Sigma} q_{i,j} \log\left[\frac{u_{i,j}}{\sum_k u_{k,i}\pi_{i,j}}\right].$$

If $q$ is not invariant, then $q_1(j_0) < q_2(j_0)$ for some $j_0$. For $\mathbf{u}$ such that $u(\cdot,j) = 1$ when $j \neq j_0$ and $u(\cdot,j_0) = e^\alpha$, we have $I_2(q) = \infty$ if we let $\alpha \to \infty$.

If $q$ is invariant,

$$\sum_{i,j\in\Sigma} q(i,j) \log\left[\frac{\sum_k u_{k,i} q_2(j)}{\sum_k u_{k,j} q_1(i)}\right] = 0.$$

Hence,

$$I_2(q) - \sum_i q_1(i) H(q(\cdot\,|i) \mid \pi(i|\,\cdot)) = \sup_{u>0} \sum_{i,j\in\Sigma} q(i,j) \log\left[\frac{u_{i,j} q_1(i)}{\sum_k u_{k,i} q(i,j)}\right]$$

$$= \sup_{u>0}\left\{-\sum_j q_2(j) H(q'(\cdot\,|j) \mid u'(\cdot\,|j))\right\},$$

where $u'(\cdot\,|j) = \frac{u(\cdot,j)}{\sum_i u(i,j)}$ and $q'(\cdot\,|j) = \frac{q(\cdot,j)}{\sum_i q(i,j)}$. This implies

$$I_2(q) \le \sum_{i\in\Sigma} q_1(i) H(q(\cdot\,|i) \mid \pi(i,\cdot)).$$

Taking $\mathbf{u} > 0$ approaching $q$ proves the theorem. □

9.2. **Random walks in $\mathbb{R}^d$.**

In this subsection, we assume that $\mathcal{D}_\Lambda = \mathbb{R}^d$. Define

$$Z_n(t) = \frac{1}{n}\sum_{i=1}^{\lfloor nt\rfloor} X_i, \quad 0 \le t < 1,$$

and let $\mu_n$ denote the law of $Z_n(\cdot)$ (as a measure on $L^\infty([0,1])$).

**Theorem 9.5** (Mogulskii)**.** *The measures $\mu_n$ satisfy in $(L^\infty([0,1]), \|\cdot\|_\infty)$ the LDP with the good rate function*

$$I(\varphi) = \begin{cases} \int_0^1 \Lambda^*(\dot\varphi(t))dt & \text{if } \varphi \in \mathcal{AC},\ \varphi(0) = 0, \\ \infty & \text{otherwise,} \end{cases}$$

*where*

$$\mathcal{AC} = \{f \in C([0,1]) : f \text{ is absolutely continuous}\}.$$

**Remark 9.6.** *The derivative in Theorem 9.5 is well-defined and satisfies the fundamental theorem of calculus $\varphi(s) = \int_0^s \dot\varphi(t)dt$ since $\varphi$ is absolutely continuous.*

**Definition 9.7.** Let $(\mathcal{Y}, d)$ be a metric space. The probability measures $\{\mu_\varepsilon\}$ and $\{\tilde\mu_\varepsilon\}$ on $\mathcal{Y}$ are said to be exponentially equivalent if there exists a probability space $(\Omega, \mathcal{B}_\varepsilon, \mathbb{P}_\varepsilon)$ and two family of $\mathcal{Y}$-valued random variables $\{Z_\varepsilon\}$ and $\{\tilde Z_\varepsilon\}$ with joint laws $\mathbb{P}_\varepsilon$ and marginals $\{\mu_\varepsilon\}$ and $\{\tilde\mu_\varepsilon\}$, respectively, such that $\{\omega : (Z_\varepsilon, \tilde Z_\varepsilon) \in \Gamma_\delta\} \in \mathcal{B}_\varepsilon$ and

$$\limsup_{\varepsilon\to 0} \varepsilon \log \mathbb{P}_\varepsilon(\Gamma_\delta) = -\infty,$$

where

$$\Gamma_\delta = \{(y, \tilde y) : d(y, \tilde y) > \delta\}.$$

**Proposition 9.8.** *Suppose $\mu_\varepsilon$ and $\tilde\mu_\varepsilon$ are exponentially equivalent. If $\tilde\mu_\varepsilon$ satisfies the LDP with the good rate function $I$, then so does $\mu_\varepsilon$ with rate $I$.*

*Proof.* Since $I$ is good, it suffices to show the $\mu_\varepsilon$ satisfies the weak LDP with rate $I$. For the upper bound, we observe that for every compact set $K$ and every $x \in K$, there exists a finite open cover $\{A_{x_i}\}_{i=1}^N$ such that for every $\delta > 0$,

$$x_i \in K \text{ and } \inf_{x\in A_{x_i}} I(x) > I^\delta(x_i).$$

Hence,

$$\begin{aligned} &\limsup_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(K) \le \limsup_{\varepsilon\to 0} \varepsilon \log\big[\tilde\mu_\varepsilon\big(\cup_{i=1}^N A_i\big) + \mathbb{P}_\varepsilon(\Gamma_\delta)\big] \\ &\le -\min_{1\le i\le N} I^\delta(x_i) \le -\inf_{x\in K} I^\delta(x), \end{aligned}$$

which proves the upper bound when letting $\delta \to 0$. As for the lower bound, for every open set $G$, there exists $G^\delta := \{y \in G : d(y, G^c) \ge \delta\}$ such that

$$\mu_\varepsilon(G) \geq \tilde{\mu}_\varepsilon\big(G^\delta\big) + \mathbb{P}_\varepsilon\big(G^\delta\big),$$

amounting to

$$\liminf_{\varepsilon\to 0} \varepsilon \log \mu_\varepsilon(G) \geq -\inf_{x\in G^\delta} I(x).$$

The proof is hence complete by taking $\delta \to 0$. □

**Lemma 9.9.** *Let $\tilde{\mu}_n$ denote the law of $\tilde{Z}_n(\cdot)$ in $L^\infty([0,1])$, where*

$$\tilde{Z}_n(t) := Z_n(t) + \left(t - \frac{\lfloor nt \rfloor}{n}\right) X_{\lfloor nt \rfloor + 1} \tag{9.1}$$

*is the polygonal approximation of $Z_n(t)$. Then the probability measures $\mu_n$ and $\tilde{\mu}_n$ are exponentially equivalent in $L^\infty([0,1])$.*

*Proof.* The sets $\big\{\omega : \|\tilde{Z}_n - Z_n\| > \eta\big\}$ are obviously measurable. Note that

$$|\tilde{Z}_n(t) - Z_n(t)| \leq \frac{|X_{\lfloor nt \rfloor + 1}|}{n}.$$

Thus, for any $\eta > 0$ and any $\lambda > 0$,

$$\mathbb{P}\big(\|\tilde{Z}_n - Z_n\| > \eta\big) \leq n\mathbb{P}(|X_1| > n\eta) \leq n\mathbb{E}\big[e^{\lambda|X_1|}\big]e^{-\lambda n\eta}.$$

Since $\mathcal{D}_\Lambda = \mathbb{R}^d$, it follows, by considering first $n \to \infty$ and then $\lambda \to \infty$, that for any $\eta > 0$,

$$\limsup_{n\to\infty} \frac{1}{n} \log \mathbb{P}\big(\|\tilde{Z}_n - Z_n\| > \eta\big) = -\infty.$$

Therefore, the probability measures $\mu_n$ and $\tilde{\mu}_n$ are exponentially equivalent. □

**Lemma 9.10.** *Let $\mathcal{X}$ consist of all the maps from $[0,1]$ to $\mathbb{R}^d$ such that $t = 0$ is mapped to the origin, and equip $\mathcal{X}$ with the topology of pointwise convergence on $[0,1]$, namely, the product topology. Then the probability measures $\tilde{\mu}_n$ of Lemma 9.9 (defined on $\mathcal{X}$ by the natural embedding) satisfy the LDP in this Hausdorff topological space with the good rate function $I$ of (9.1).*

**Lemma 9.11.** *The probability measures $\tilde{\mu}_n$ are exponentially tight in the space $C_0([0,1])$ of all continuous functions $f : [0,1] \to \mathbb{R}^d$ such that $f(0) = 0$, equipped with the supremum norm topology.*

*Proof of Theorem 9.5.* Note that $C_0([0,1])$ is a measurable subset of $\big(\mathcal{X}, \tau_{\text{prod}}\big)$; explicitly, $\big\{f : [0,1] \to \mathbb{R}^d : f(0) = 0\big\}$ is closed and $C([0,1])$ is Borel since

$$C([0,1]) = \bigcap_{x\in\mathbb{Q}\cap[0,1]} \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \left\{ f : [0,1] \to \mathbb{R}^d : \sup_{y,z\in B_{n^{-1}}(x)} |f(y) - f(z)| \leq \frac{1}{m} \right\}.$$

In addition, since $\tilde{\mu}_n(C_0([0,1])) = 1$, we have that $\{\tilde{\mu}_n\}$ satisfies the LDP in $C_0([0,1])$ (in $\mathcal{X}$). Since (a) $\tilde{\mu}_n$ is exponentially tight in $\mathcal{X}$ by Lemma 9.11 and (b) the product topology is coarser than the norm-induced topology on $C_0([0,1])$, we infer from Theorem 5.2 that $\{\tilde{\mu}_n\}$ satisfies the LDP in $(C_0([0,1]), \|\cdot\|_\infty)$. Finally, since $C_0([0,1])$ is closed in $L^\infty([0,1])$, $\{\tilde{\mu}_n\}$ satisfies the LDP in $L^\infty([0,1])$. □

**Lemma 9.12.** *Let $J$ denote the collection of all ordered finite subsets of $(0,1]$. For any $j = \Big\{0 < t_1 < t_2 < \cdots < t_{|j|} \leq 1\Big\} \in J$ and any $f : [0,1] \to \mathbb{R}^d$, let $p_j(f)$*

*denote the vector* $\big(f(t_1), f(t_2), ..., f\big(t_{|j|}\big)\big) \in \big(\mathbb{R}^d\big)^{|j|}$ *. Then, the sequence of laws* $\big\{\mu_n \circ p_j^{-1}\big\}$ *satisfies the LDP in* $\big(\mathbb{R}^d\big)^{|j|}$ *with the good rate function*

$$I_j(\mathbf{z}) = \sum_{\ell=1}^{|j|} (t_\ell - t_{\ell-1}) \Lambda^* \left( \frac{z_\ell - z_{\ell-1}}{t_\ell - t_{\ell-1}} \right),$$

*where* $\mathbf{z} = \big(z_1, ..., z_{|j|}\big)$ *and* $t_0 = 0$*,* $z_0 = 0$*. In particular,* $\big\{\tilde{\mu}_n \circ p_j^{-1}\big\}$ *satisfies the LDP also with rate* $I_j$*.*

*Proof.* Note that $\mu_n \circ p_j^{-1}$ is the law of the vector

$$Z_n^j = \big(Z_n(t_1), Z_n(t_2), ..., Z_n\big(t_{|j|}\big)\big).$$

To study the LDP, we apply Gärtner–Ellis theorem (Theorem 3.5) and the contraction principle (Theorem 5.1) to the following transformed vectors

$$Y_n^j = \big(Z_n(t_1), Z_n(t_2) - Z_n(t_1), ..., Z_n\big(t_{|j|}\big) - Z_n\big(t_{|j|-1}\big)\big),$$

which satisfies the LDP with rate function (by writing $\mathbf{y} = \big(y_1, ..., y_{|j|}\big)$ and $\lambda = \big(\lambda_1, ..., \lambda_{|j|}\big)$):

$$\Lambda_j^*(\mathbf{y}) = \sum_{\ell=1}^{|j|} (t_\ell - t_{\ell-1}) \Lambda^* \left( \frac{y_\ell}{t_\ell - t_{\ell-1}} \right),$$

where

$$\Lambda_j(\lambda) = \sum_{\ell=1}^{|j|} (t_\ell - t_{\ell-1}) \Lambda(\lambda_\ell).$$

The LDP for $\big\{\tilde{\mu}_n \circ p_j^{-1}\big\}$ holds as a consequence of exponential equivalence. □

**Corollary 9.13.** *For any* $j \in J$*,* $\big\{\tilde{\mu}_n \circ p_j^{-1}\big\}$ *satisfies the LDP in* $\big(\mathbb{R}^d\big)^{|j|}$ *with the good rate function* $I_j$*.*

*Proof of Lemma 9.10.* By Dawson-Gärtner theorem (Theorem 5.10), the rate function governing the LDP is

$$I_{\mathcal{X}}(\varphi) = \sup_{0<t_0<t_1<\dots<t_k\le 1} \sum_{\ell=1}^{k} (t_\ell - t_{\ell-1}) \Lambda^* \left( \frac{\varphi(t_\ell) - \varphi(t_{\ell-1})}{t_\ell - t_{\ell-1}} \right).$$

Since $\Lambda^* \geq 0$, we may assume without loss of generality that $t_k = 1$.

We first demonstrate that $I_{\mathcal{X}} = I$. By convexity of $\Lambda^*$ and Jensen's inequality, it is clear that $I_{\mathcal{X}} \leq I$. For the other inequality, first assume $\varphi \in \mathcal{AC}$ and write

$$\psi_k(t) = \begin{cases} \int_{\frac{\lfloor kt \rfloor}{k}}^{\frac{\lfloor kt \rfloor + 1}{k}} \dot{\varphi}(s) ds \text{ if } t \in [0,1), \\ \int_{1-\frac{1}{k}}^{1} \dot{\varphi}(s) ds \quad \text{if } t = 1, \end{cases}$$

so that

$$I_{\mathcal{X}}(\varphi) \geq \liminf_{k\to\infty} \sum_{\ell=1}^{k} \frac{1}{k}\Lambda^*\left(k\left[\varphi\left(\frac{\ell}{k}\right) - \varphi\left(\frac{\ell-1}{k}\right)\right]\right) = \liminf_{k\to\infty} \int_0^1 \Lambda^*(\psi_k(t))dt$$
$$\geq \int_0^1 \liminf_{k\to\infty} \Lambda^*(\psi_k(t))dt \geq \int_0^1 \Lambda^*(\dot{\varphi}(t))dt,$$

where the second line follows from Fatou's lemma and lower semicontinuity of $\Lambda^*$. If $\varphi \notin \mathcal{AC}$, there exist $\delta > 0$ and $s_1^n < t_1^n \leq ... \leq s_{k_n}^n < t_{k_n}^n$ such that

$$\lim_{n\to 0} \sum_{\ell=1}^{k_n} t_\ell^n - s_\ell^n = 0 \text{ and } \sum_{\ell=1}^{k_n} \|\varphi(t_\ell^n) - \varphi(s_\ell^n)\| \geq \delta.$$

Observe that by non-negativity of $\Lambda^*$,

$$I_{\mathcal{X}}(\varphi) \geq \sup_{0\leq s_1<t_1\leq s_2<t_2\leq\ldots\leq s_k<t_k} \sum_{\ell=1}^{k} (t_\ell - s_\ell)\Lambda^*\left(\frac{\varphi(t_\ell)-\varphi(s_\ell)}{t_\ell - s_\ell}\right)$$
$$= \sup_{\substack{0\leq s_1<t_1\leq s_2<t_2\leq\ldots\leq s_k<t_k \\ \lambda_1,\ldots,\lambda_k\in\mathbb{R}^d}} \sum_{\ell=1}^{k} [\langle \lambda_\ell, \varphi(t_\ell)-\varphi(s_\ell)\rangle - (t_\ell - s_\ell)\Lambda(\lambda_\ell)].$$

If one chooses, for any $\rho > 0$,

$$\lambda_\ell = \rho \cdot \frac{\varphi(t_\ell)-\varphi(s_\ell)}{\|\varphi(t_\ell)-\varphi(s_\ell)\|} \text{ and } M = \sup_{\|\lambda\|=\rho} \Lambda(\lambda),$$

it follows that $I_{\mathcal{X}}(\varphi) \geq \rho\delta \to \infty$ as $\rho \to \infty$. □

*Proof of Lemma 9.11.* We first claim that given any real-valued random variable $X \sim \nu$, we have $E\left[e^{\delta\Lambda_\nu^*(X)}\right] < \infty$ for all $\delta < 1$. Indeed, by Markov's inequality and Lemma 3.4,

$$E\left[e^{\delta\Lambda_\nu^*(X)}\right] = \int_{-\infty}^{\overline{x}} e^{\delta\Lambda_\nu^*(x)}d\nu(x) + \int_{\overline{x}}^{\infty} e^{\delta\Lambda_\nu^*(x)}d\nu(x)$$
$$\leq \int_{-\infty}^{\overline{x}} \frac{d\nu(x)}{\nu((-\infty,x])^\delta} + \int_{\overline{x}}^{\infty} \frac{d\nu(x)}{\nu([x,\infty))^\delta},$$

By integration by part, the former integral can be evaluated as

$$\int_{-\infty}^{\overline{x}} \frac{d\nu(x)}{\nu((-\infty,x])^\delta} = \nu((-\infty,\overline{x}])^{1-\delta} + \delta\int_{-\infty}^{\overline{x}} \frac{d\nu(x)}{\nu((-\infty,x])^\delta},$$

and thus is bounded from above by $(1-\delta)^{-1}$. Since a similar estimates holds also for the latter integral, the claim is proved.

We now apply the inverse contraction principle to $\tilde{\mu}_n$ in $C_0([0,1])$ when equipped with the supremum norm topology. Denote by $X_1^j$ the $j$-th component of $X_1$, by $\Lambda_j$ the logarithmic moment generating function $X_1^j$, and by $\Lambda_j^*$ the Fenchel-Legendre transform of $\Lambda_j$. We claim that $\tilde{\mu}_n$ is exponentially tight via the closure of the sets

$$K_\alpha = \cap_{j=1}^d K_\alpha^j \text{ where } K_\alpha^j := \left\{\varphi \in \mathcal{AC} : \varphi(0) = 0, \int_0^1 \Lambda_j^*\big(\dot{\varphi}_j(\theta)\big)d\theta \leq \alpha\right\}.$$

To this end, note that $\frac{d\tilde{Z}_n}{dt}(t) = X_{\lfloor nt\rfloor+1}$ and hence

$$\tilde{\mu}_n(K_\alpha^c) \le d \max_{1\le j\le d} \mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n \Lambda_j^*\big(X_i^j\big) > \alpha\right).$$

By Markov's inequality and our claim,

$$\frac{1}{n}\log \tilde{\mu}_n(K_\alpha^c) \le -\delta\alpha + \log d + \max_{1\le j\le d} \log \mathbb{E}\Big[e^{\delta\Lambda_j^*\left(X_1^j\right)}\Big] \to \infty \text{ as } \alpha\to\infty.$$

It remains to show that $K_\alpha$ is precompact by the Arzelà–Ascoli theorem (uniformly bounded and equicontinuous). To this end, observe that by convexity of $\Lambda_j^*$ and Jensen's inequality, for any $\varphi \in \mathcal{AC}$ and any $0 \le s < t \le 1$,

$$\Lambda_j^*\left(\frac{\varphi_j(t)-\varphi_j(s)}{t-s}\right) \le \frac{1}{t-s}\int_s^t \Lambda_j^*\big(\dot{\varphi}_j(\theta)\big)d\theta \le \frac{\alpha}{t-s}$$

Since $\Lambda_j^*(x) \ge M\|x\| - \max\big\{\Lambda_j(M), \Lambda_j(-M)\big\}$ for all $M > 0$, we deduce from the inequality above that for all $0 < t-s < \delta$,

$$\|\varphi_j(t)-\varphi_j(s)\| \le \frac{1}{M}\big(\alpha + \delta\max\big\{\Lambda_j(M), \Lambda_j(-M)\big\}\big),$$

from which the equicontinuity follows naturally, and so does the uniform boundedness due to the fact $\varphi(0) = 0$. □

Mogulskii's theorem (Theorem 9.5) extends to the laws $\nu_\varepsilon$ of

$$Y_\varepsilon(t) = \varepsilon\sum_{i=1}^{\lfloor \frac{t}{\varepsilon}\rfloor} X_i, 0 \le t \le 1.$$

**Theorem 9.14.** *The probability measures $\nu_\varepsilon$ induced on $L^\infty([0,1])$ by $Y_\varepsilon$ satisfy the LDP with the good rate function $I$ in Theorem 9.5.*

*Proof.* It suffices to prove the LDP on arbitrary subsequences $\varepsilon_m \searrow 0$. To this end, let $\varepsilon_{n_m} = \left\lfloor \frac{1}{\varepsilon_m}\right\rfloor$ so that $\mu_{n_m}$ satisfies the LDP with rate $I$. It suffices to show the exponential equivalence, namely,

$$\frac{1}{n_m}\log \mathbb{P}\Big(\|Y_{\varepsilon_m} - Z_{n_m}\| \ge \delta\Big) \to -\infty.$$

Indeed,

$$\|Y_{\varepsilon_m} - Z_{n_m}\| \le 2\varepsilon_m \max_{1\le i\le n_m} |X_i|,$$

implying, due to the fact $0 \in \mathring{\mathcal{D}}_\Lambda$,

$$\frac{1}{n_m}\log \mathbb{P}\Big(\|Y_{\varepsilon_m} - Z_{n_m}\| \ge \delta\Big) \le \frac{1}{n_m}\log n_m + \frac{1}{n_m}\log \mathbb{P}\left(|X_1| \ge \frac{\delta}{2\varepsilon_m}\right) \to -\infty.$$

□

Now consider the process $w_\varepsilon(t) = w_t$ with $w_t$ the standard Brownian motion. We study the LDP of $w_\varepsilon$ in $C_0([0,1])$. To this end, consider the space $H_1 := \Big\{\int_0^t f(s)ds : f \in L^2([0,1])\Big\}$ equipped with the norm $\|\varphi\|_{H_1} = \|\dot{\varphi}\|_{L^2}$.

**Lemma 9.15.** *For any $d \in \mathbb{N}$ and any $\tau, \varepsilon, \delta > 0$,*

$$\mathbb{P}\left(\sup_{0\le t\le\tau}\|w_\varepsilon(t)\|\ge\delta\right)\le 4de^{-\frac{\delta^2}{2d\tau\varepsilon}}.$$

*Proof.* Observe that

$$\mathbb{P}\left(\sup_{0\le t\le\tau}\|w_\varepsilon(t)\|\ge\delta\right)=\mathbb{P}\left(\sup_{0\le t\le\tau}\|w_\varepsilon\|^2\ge\varepsilon^{-1}\delta^2\right)\le d\cdot\mathbb{P}\left(\sup_{0\le t\le\tau}(w_\varepsilon)_1^2\ge\frac{\delta^2}{d\varepsilon}\right)$$

By time rescaling,

$$P\left(\sup_{0\le t\le\tau}\|w_\varepsilon(t)\|\ge\delta\right)\le d\cdot\mathbb{P}\left(\sup_{0\le t\le 1}\left\|(w_\varepsilon)_1\right\|\right)\ge\frac{\delta}{\sqrt{d\tau\varepsilon}}.$$

Now for one-dimensional Brownian motion,

$$\mathbb{P}\left(\sup_{0\le t\le 1}\left\|(w_\varepsilon)_1\right\|\ge\eta\right)\le 2\mathbb{P}\left(\sup_{0\le t\le 1}(w_\varepsilon)_1\ge\eta\right)=4\mathbb{P}\left(\sup_{0\le t\le 1}(w_\varepsilon)_1\ge\eta\right)\le 4e^{-\frac{\eta^2}{2}},$$

where the equality follows from the Désiré André's reflection principle. □

**Theorem 9.16** (Schilder)**.** *The measure $\{\nu_\varepsilon\}$ satisfies, in $C_0([0,1])$, an LDP with good rate function*

$$I_w(\varphi)=\begin{cases}\frac{1}{2}\int_0^1|\dot\varphi(t)|^2\,dt & \text{if } \varphi\in H_1,\\ \infty & \text{otherwise.}\end{cases}$$

*Proof.* Observe that $\hat w_\varepsilon(t):=w_\varepsilon\big(\varepsilon\lfloor\frac{t}{\varepsilon}\rfloor\big)$ is the process of $Y_\varepsilon$ with $X_i$ the standard normal random variables in $\mathbb{R}^d$ with

$$\Lambda(\lambda)=\log\mathbb{E}\big[e^{\langle\lambda,X_1\rangle}\big]=\frac{\|\lambda\|^2}{2}\quad\text{and}\quad\Lambda^*(x)=\frac{\|x\|^2}{2}.$$

Hence, by Theorem 9.14, we have that the $\hat w_\varepsilon$ satisfies the LDP on $L^\infty([0,1])$ with rate function $I_w$. Now that $\hat w_\varepsilon$ and $w_\varepsilon$ are exponentially equivalent:

$$\mathbb{P}(\|\hat w_\varepsilon-w_\varepsilon\|\ge\delta)\le\left(\left\lfloor\frac{1}{\varepsilon}\right\rfloor+1\right)\mathbb{P}\left(\sup_{0\le t\le\varepsilon}|w_\varepsilon(t)|\ge\delta\right)\le 4d\varepsilon^{-1}(1+\varepsilon)e^{-\frac{\delta^2}{2d\varepsilon^2}}.$$

Hence, $w_\varepsilon$ satisfies the LDP with rate $I_w$ on $L^\infty([0,1])$, which carries naturally to $C_0([0,1])$. □

## 10. Concentration inequalities

In this section, we denote by $\Sigma$ to consistently mean a Polish space.

**Lemma 10.1** (Bennett)**.** *Suppose $X \le b$ is a real-valued random variable with $\overline{x} = \mathbb{E}(X)$ and $E\big[(X-\overline{x})^2\big] \le \sigma^2$ for some $\sigma > 0$. Then, for any $\lambda \ge 0$,*

$$\mathbb{E}\big(e^{\lambda X}\big) \le e^{\lambda \overline{x}} \mathbb{E}\left[\frac{(b-\overline{x})^2}{(b-\overline{x})^2+\sigma^2} e^{-\frac{\lambda \overline{x}}{b-\overline{x}}} + \frac{\sigma^2}{(b-\overline{x})^2+\sigma^2} e^{-\lambda(b-\overline{x})}\right]$$

*Proof.* If $b = \overline{x}$, then $X = \overline{x}$ almost surely and the inequality is trivial for all $\sigma > 0$. Likewise, when $\lambda = 0$, then the inequality is trivial. Hence, in the following we works with the assumption $b > \overline{x}$ and $\lambda > 0$. Let $Y := \lambda(X - \overline{x})$ so to reduce the inequality, by noting $Y \le \lambda(b-\overline{x}) =: m$ and $\mathbb{E}(Y^2) = \lambda^2\sigma^2 = v$, to

$$\mathbb{E}\big(e^Y\big) \le \frac{m^2}{m^2+v} e^{-\frac{v}{m}} + \frac{v}{m^2+v} e^m = \mathbb{E}\big(e^Z\big),$$

where $Z$ is a random variable taking values in $\big\{-\frac{v}{m}, m\big\}$ with $\mathbb{P}(Z = m) = v/m^{m^2+v}$. Consider a quadratic approximation $\varphi(y) = ay^2 + by + c$ of $e^y$ satisfying $f\big(-\frac{v}{m}\big) = f'\big(-\frac{v}{m}\big) = 0$ and $f(m) = 0$ with $f(y) := \varphi(y) - e^y$. By calculus and convexity of $f$, we know (a) $f(y) \ge 0$ for $y \in (-\infty, m]$, and (b) $\varphi''(y) > 0$ for all $y$. Thus, by noting that $\mathbb{E}(Y) = \mathbb{E}(Z) = 0$ and $\mathbb{E}(Y^2) \le \mathbb{E}(Z^2) = v$,

$$\begin{aligned} &\mathbb{E}(f(Y)) \ge 0 \\ \Rightarrow\ &\mathbb{E}\big(e^Y\big) \le \mathbb{E}(\varphi(Y))(= a\mathbb{E}(Y^2) + c) = \mathbb{E}(\varphi(Z)) = \mathbb{E}\big(e^Z\big) \end{aligned}$$

with equality holding if and only if $\mathbb{P}\big(Y \in \big\{-\frac{v}{m}, m\big\}\big) = 1$. □

**Corollary 10.2.** *Fix $a < b$. Suppose that $a \le X \le b$ is a real-valued random variable with $\overline{x} = \mathbb{E}(X)$. Then, for any $\lambda \in \mathbb{R}$,*

$$\mathbb{E}\big(e^{\lambda X}\big) \le \frac{\overline{x}-a}{b-a} e^{\lambda b} + \frac{b-\overline{x}}{b-a} e^{\lambda a}$$

*Proof.* Apply Lemma 10.1 with $\sigma^2 = (b-\overline{x})(\overline{x}-a)$. □

**Corollary 10.3.** *Suppose $v > 0$ and the real valued random variables $\{Y_n : n \in \mathbb{N}\}$ are such that both $Y_n \le 1$ almost surely, and $E[Y_n|S_{n-1}] = 0$, $E\big[Y_n^2|S_{n-1}\big] \le v$ for $S_n = \sum_{j=1}^n Y_j$, $S_0 = 0$. Then, for any $\lambda \ge 0$,*

$$\mathbb{E}\big(e^{\lambda S_n}\big) \le \left(\frac{e^{-v\lambda} + ve^{\lambda}}{1+v}\right)^n \tag{10.1}$$

*Moreover, for all $x \ge 0$,*

$$P(n^{-1}S_n \ge x) \le \exp\Big(-nH\Big(\frac{x+v}{1+v}\Big|\frac{v}{1+v}\Big)\Big), \tag{10.2}$$

*where $H(p|p_0) = p\log\frac{p}{p_0} + (1-p)\log\frac{1-p}{1-p_0}$. Finally, for all $y \ge 0$,*

$$\mathbb{P}\Big(n^{-\frac{1}{2}}S_n \ge y\Big) \le e^{-\frac{2y^2}{(1+v)^2}}. \tag{10.3}$$

*Proof.* Apply Lemma 10.1 to $Y_k|S_{k-1}$ with $x = 0$, $b = 1$ and $\sigma^2 = v$ to yield

$$\mathbb{E}\big(e^{\lambda Y_k}|S_{k-1}\big) \leq \frac{e^{-v\lambda}+ve^{\lambda}}{1+v}.$$

Taking the expectation of product of $e^{Y_k}$ then gives (10.1). Applying Markov's inequality to (10.1) then gives

$$P(n^{-1}S_n \geq x) \leq e^{-\lambda nx}\left(\frac{e^{-v\lambda}+ve^{\lambda}}{1+v}\right)^n,$$

from which (10.2) by taking $(1+v)^{-1}\log((x+v)/v(1-x))$. Note that $H(p|p_0) \geq 2(p-p_0)^2$ for any $p, p_0 \in [0,1]$. Hence, setting $x = n^{-\frac{1}{2}}y$ yields (10.3). □

**Corollary 10.4.** *Let $\Sigma$ be a polish space. Let $Z_n = g_n(X_1, ..., X_n)$ for independent $\Sigma$-valued random variables $\{X_i\}$ and real-valued, measurable $g_n$. Let $\left\{\hat{X}_i\right\}$ be an independent copy of $\{X_i\}$. Suppose that for $k = 1, ..., n$,*

$$|g_n(X_1, ..., X_n) - g_n\big(X_1, ..., X_{k-1}, \hat{X}_k, X_{k+1}, ..., X_n\big)| \leq 1,$$

*almost surely. Then, for all $x \geq 0$,*

$$P(n^{-1}(Z_n - \mathbb{E}(Z_n)) \geq x) \leq \exp\left(-nH\left(\frac{x+1}{2}\,\middle|\,\frac{1}{2}\right)\right)$$

*and for all $y \geq 0$,*

$$P\Big(n^{-\frac{1}{2}}(Z_n - \mathbb{E}(Z_n)) \geq y\Big) \leq e^{-\frac{y^2}{2}}.$$

10.1. **Talagrand's Concentration Inequalities.**

**Definition 10.5.** For any $n \in \mathbb{Z}_+$, let $(y, x) = (y_1, ..., y_n, x_1, ..., x_n)$ denote a generic point in $(\Sigma^n)^2$. Let $\mathcal{M}_n(\mathbf{Q}, \mathbf{P})$ denote the set of all probability measures on $(\Sigma^n)^2$ whose marginals are the prescribed probability measures $\mathbf{Q}$ and $\mathbf{P}$ on $\Sigma^n$.

- For any $\alpha > 0$ let $\varphi_\alpha : [0, \infty) \to [0, \infty)$ be such that

  $$\varphi_\alpha(x) = \alpha x \log x - (1+\alpha x)\log\left(\frac{1+\alpha x}{1+\alpha}\right).$$

  The *coupling distance* between $\mathbf{P}$ and $\mathbf{Q}$ is

  $$d_\alpha(\mathbf{Q}, \mathbf{P}) = \inf_{\pi \in \mathcal{M}_n(\mathbf{Q},\mathbf{P})} \int_{\Sigma^n} \sum_{k=1}^n \varphi_\alpha(\pi^{\mathbf{x}}\mathbf{x}(\{y_k = x_k\}))d\mathbf{P}(x).$$

- Define the relative entropy of the probability measure $\nu$ with respect to $\mu \in M_1(\Sigma^n)$ as

  $$H(\nu|\mu) = \begin{cases} \int_{\Sigma^n} \frac{d\mu}{d\nu}\log\frac{d\mu}{d\nu}\,d\mu & \text{if } \frac{d\mu}{d\nu} \text{ exists,} \\ \infty & \text{otherwise.} \end{cases}$$

**Theorem 10.6.** *Let $\Sigma$ be a Polish space. Suppose $\mathbf{R} = \prod_k R_k$ is a product probability measure on $\Sigma^n$. Then, for any $\alpha > 0$ and any probability measures $\mathbf{P}, \mathbf{Q}$ on $\Sigma_n$,*

$$d_\alpha(\mathbf{Q}, \mathbf{P}) \leq H(\mathbf{P}|\mathbf{R}) + \alpha H(\mathbf{Q}|\mathbf{R}). \tag{10.4}$$

**Lemma 10.7.** *For any $\alpha > 0$ and probability measures $P, Q$ on $\Sigma$, let*

$$\Delta_\alpha(Q,P) := \int_{\tilde\Sigma} \varphi_\alpha \min\left\{\frac{dQ}{dP}, 1\right\} dP,$$

*where* $\tilde\Sigma$ *is such that* $P\big(\tilde\Sigma^c\big) = 0$ *and* $\frac{dQ}{dP}$ *exists on* $\tilde\Sigma$.

*(1) There exists* $\pi \in \mathcal{M}_1(Q,P)$ *such that*

$$\int_{\sim} \varphi_\alpha\big(\pi^{x(\{y=x\})}\big)\, dP(x) = \Delta_\alpha(Q,P).$$

*(2) For any probability measure on* $R$ *on* $\Sigma$,

$$\Delta_\alpha(Q,P) \le H(P|R) + \alpha H(Q|R).$$

*Proof.* Let $(P-Q)_+$ denotes the positive part of the signed measure $P-Q$ and $Q \wedge P = P - (P-Q)_+ = Q - (Q-P)_+$. Let $q = (P-Q)_+(\Sigma) = (Q-P)_+(\Sigma)$. Define independent random variables $W_1 \sim (1-q)^{-1}(Q\wedge P)$, $W_2 \sim q^{-1}(P-Q)_+$, $Z \sim q^{-1}(Q-P)_+$. Let $I \in \{1,2\}$ be chosen independently of the random variables such that $\mathbb{P}(I=2) = q$, and set $(Y,X) = (W_1, W_1)$ when $I=1$ and $(Y,X) = (Z, W_2)$ otherwise. The coupling $(Y,X) \sim \pi \in \mathcal{M}_1(Q,P)$ is such that $\pi(\{(y,x) : y = x, x \in \cdot\}) = (Q\wedge P)(\cdot)$. Hence, for every $\Gamma \in \mathcal{B}_\Sigma$, $\Gamma \subset \tilde\Sigma$,

$$\begin{aligned}\int_\Gamma \pi^x(\{y=x\})\, dP(x) &= \int_\Gamma \pi^x(\{(y,x) : y = x\})\, dP(x)\\ &= \pi(\cup_{x\in\Gamma} \{(y,x) : y = x\})\\ &= Q\wedge P(\Gamma) = \int_\Gamma \frac{dQ\wedge P}{dP}(x)\, dP(x),\end{aligned}$$

implying $\pi^x(\{y=x\}) = \frac{dQ\wedge P}{dP} = \min\left\{\frac{dQ}{dP}, 1\right\}$ for $P$-almost every $x \in \tilde X$. With $P\big(\tilde\Sigma^c\big) = 0$, the equality follows.

It suffices to consider $R$ such that $f = \frac{dP}{dR}$ and $g = \frac{dQ}{dR}$ exists. Let $R_\alpha = \frac{P+\alpha Q}{1+\alpha}$ so that $h := \frac{dR_\alpha}{dR} = (f+\alpha g)(1+\alpha)$. Thus,

$$\begin{aligned}H(P|R) + \alpha H(Q|R) &= \int_\Sigma [f\log f + \alpha g \log g]\, dR\\ &\ge \int_\Sigma \left[f \log\frac{f}{h} + \alpha\alpha g \log\frac{g}{h}\right] dR\end{aligned}$$

since $\int_\Sigma h\log h\, dR = H(R_\alpha|R) \ge 0$. Let $\rho := \frac{dQ}{dP}$ on $\tilde\Sigma$. Then, on $\tilde\Sigma$, $\frac{f}{h} = \frac{1+\alpha}{1+\alpha\rho}$ and $\frac{g}{f} = \rho$, whereas $P\big(\tilde\Sigma^c\big) = 0$ implying that $\frac{g}{h} = \frac{1+\alpha}{\alpha}$ on $\tilde\Sigma^c$. Hence, the inequality above implies

$$H(P|R) + \alpha H(Q|R) \ge \int_{\tilde\Sigma} \varphi_\alpha(\rho)\, dP + Q\big(\tilde\Sigma^c\big) v_\alpha,$$

where $v_a = \alpha\log\big(\frac{1+\alpha}{\alpha}\big) > 0$. Given that $\varphi_\alpha(x) \ge \varphi_\alpha(1) = 1$ for all $x \ge 0$, the proof is complete. □

*Proof of Theorem 10.6.* Fix $\alpha > 0$, $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R} = \prod_{k=1}^n R_k$. For $k = 1, ..., n$, let $P_k^{\mathbf{x}^{k-1}}$ (respectively, $Q_k^{\mathbf{x}^{k-1}}$) denote the regular conditional probability distribution induced by $\mathbf{P}$ (respectively, $\mathbf{Q}$) on the $k$-th coordinate given the $\sigma$-algebra generated by the restriction of $\Sigma^n$ to the first $(k-1)$ coordinate. By Lemma 10.7, for any $k = 1, ..., n$, there exists a $\pi_k = \pi_k^{\mathbf{y}^{k-1},\mathbf{x}^{k-1}} \in \mathcal{M}_1\big(Q_k^{\mathbf{x}^{k-1}}, P_k^{\mathbf{x}^{k-1}}\big)$ such that

$$\int_{\Sigma} \varphi_\alpha(\pi_k^x)(\{y = x\})\, dP_k^{\mathbf{x}^{k-1}}(x) = \Delta_\alpha\Big(Q_k^{\mathbf{x}^{k-1}}, P_k^{\mathbf{x}^{k-1}}\Big)$$

Let $\pi \in M_1(\Sigma^{2n}) \cap \mathcal{M}_{n(\mathbf{Q},\mathbf{P})}$ be such that

$$\pi(\Gamma) = \int_{\Sigma^2} \cdots \int_{\Sigma^2} \mathbb{1}_\Gamma(\mathbf{y}, \mathbf{x})\, d\pi_1(y_1, x_1) d\pi_2^{\mathbf{y}^1, \mathbf{x}^1}(y_2, x_2) \cdots d\pi_2^{\mathbf{y}^{n-1}, \mathbf{x}^{n-1}}(y_n, x_n).$$

We then have, due to convexity of $\varphi_\alpha$, that

$$\begin{aligned}
\mathbb{E}(\varphi_\alpha(\pi^{\mathbf{X}}(\{y_k = x_k\}))) &\leq \mathbb{E}\Big(\varphi_\alpha\Big(\pi^{\mathbf{Y}^{k-1}, \mathbf{X}}(\{y_k = x_k\})\Big)\Big) \\
&= \mathbb{E}\Big(\varphi_\alpha\Big(\pi^{\mathbf{Y}^{k-1}, \mathbf{X}^k}(\{y_k = x_k\})\Big)\Big) \\
&= \mathbb{E}(\varphi_\alpha(\pi^{X_k}(\{y_k = x_k\}))) \\
&= \mathbb{E}\Big(\Delta_\alpha\Big(Q_k^{\mathbf{Y}^{k-1}}, P_k^{\mathbf{X}^{k-1}}\Big)\Big).
\end{aligned}$$

Hence, for any $n$, by Lemma 10.7,

$$\begin{aligned}
d_\alpha(\mathbf{Q}, \mathbf{P}) &\leq \sum_{i=1}^n \mathbb{E}\Big(\Delta_\alpha\Big(Q_k^{\mathbf{Y}^{k-1}}, P_k^{\mathbf{X}^{k-1}}\Big)\Big) \\
&\leq \mathbb{E}\left[\sum_{k=1}^n H\Big(P_k^{\mathbf{X}^{k-1}} | R_k\Big)\right] + \alpha \mathbb{E}\left[\sum_{k=1}^n H\Big(P_k^{\mathbf{Y}^{k-1}} | R_k\Big)\right].
\end{aligned}$$

Hence, by the chain rule for relative entropies,

$$H(\mathbf{P}|\mathbf{R}) = \sum_{k=1}^n \mathbb{E}\Big[H\Big(P_k^{\mathbf{Y}^{k-1}} | R_k\Big)\Big],$$

and a formula for $H(\mathbf{Q}|\mathbf{R})$ follows similarly. Based on the above, the theorem is established. □

10.1.1. *Applications.*

For $\alpha > 0$ and $A \in \mathcal{B}_{\Sigma^n}$ consider the function

$$f_\alpha(A, \mathbf{x}) := \inf_{\{\nu \in M_1(\Sigma^n) : \nu(A) = 1\}} \sum_{k=1}^n \varphi_\alpha(\nu(\{\mathbf{y} : y_k = x_k\}))$$

We then show the concentration inequality for $f_\alpha(\cdot, \cdot)$.

**Corollary 10.8.** *For any product probability measure* $\mathbf{R}$ *on* $\Sigma^n$*, any* $\alpha > 0$*,* $t > 0$ *and* $A \in \mathcal{B}_{\Sigma^n}$*,*

$$\mathbf{R}(\{\mathbf{x} : f_\alpha(A, \mathbf{x}) \geq t\}) e^t \leq \int_{\Sigma^n} e^{f_\alpha(A, \mathbf{x})}\, d\mathbf{R}(\mathbf{x}) \leq \mathbf{R}(A)^{-\alpha}. \tag{10.5}$$

*Proof.* Fix $\mathbf{R} \in M_1(\Sigma^n)$, $\alpha > 0$, and $A \in \mathcal{B}_{\Sigma^n}$. The inequalities in (10.5) trivially holds when $\mathbf{R}(A) = 0$. When $\mathbf{R}(A) > 0$, define $\cdot = \frac{\mathbf{R}(\cdot \cap A)}{\mathbf{R}(\cdot)}$ so that $H(\mathbf{Q}|\mathbf{R}) = -\log \mathbf{R}(A)$, and $\mathbf{P}$ such that

$$\frac{d\mathbf{P}}{d\mathbf{R}}(\mathbf{x}) = \frac{e^{f_\alpha(A, \mathbf{x})}\, d\mathbf{R}(\mathbf{x})}{\int_{\Sigma^n} e^{f_\alpha(A, \mathbf{y})}\, d\mathbf{R}(\mathbf{y})}$$

and that

$$H(\mathbf{P}|\mathbf{R}) = \int_{\Sigma^n} f_\alpha(A,\mathbf{x})\, d\mathbf{R}(\mathbf{x}) - \int_{\Sigma^n} e^{f_\alpha(A,\mathbf{x})}\, d\mathbf{R}(\mathbf{x}). \tag{10.6}$$

Since $\pi^{\mathbf{x}}(A \times \{\mathbf{x}\}) = 1$ for any $\pi \in \mathcal{M}_n(\mathbf{Q},\mathbf{P})$ and $\mathbf{P}$ almost every $\mathbf{x} \in \Sigma^n$. It follows that

$$\int_{\Sigma^n} f_\alpha(A,\mathbf{x})\, d\mathbf{R}(\mathbf{x}) \le d_\alpha(\mathbf{Q},\mathbf{P}). \tag{10.7}$$

Combining (10.4) with (10.6) and (10.7) gives the right inequality in (10.5). For any $t > 0$, the left inequality in (10.5) is a direct application of Markov's inequality. □

Define

$$g(A,\mathbf{x}) = \sup_{\{\beta:\, |\beta|\, \le 1\}} \inf_{\mathbf{y}\in A} \sum_{k=1}^n \beta_k \mathbb{1}_{y_k \ne x_k}$$

**Corollary 10.9.** *For any product probability measure* $\mathbf{R}$ *on* $\Sigma^n$, $A \in \mathcal{B}_{\Sigma^n}$, *and any* $u > 0$,

$$\mathbf{R}(\{\mathbf{x} : g(A,\mathbf{x}) \ge u\})\mathbf{R}(A) \le e^{-\frac{u^2}{4}}. \tag{10.8}$$

*If* $\mathbf{R}(A) > 0$ *and* $u > \sqrt{2\log \mathbf{R}(A)^{-1}}$, *then also*

$$\mathbf{R}(\{\mathbf{x} : g(A,\mathbf{x}) \ge u\}) \le \exp\Big(-\frac{1}{2}\Big(u - \sqrt{2\log \mathbf{R}(A)^{-1}}\Big)^2\Big). \tag{10.9}$$

Note

$$\inf_{\{\nu\in M_1(\Sigma^n):\nu(A)=1\}} \int_{\Sigma^n} \sum_{k=1}^n \beta_k \mathbb{1}_{y_k\ne x_k}\, d\nu(y) = \inf_{\mathbf{y}\in A} \sum_{k=1}^n \beta_k \mathbb{1}_{y_k \ne x_k}.$$

Hence,

$$g(A,\mathbf{x}) = \sup_{\{\beta:\, |\beta|\, \le 1\}} \inf_{\{\nu\in M_1(\Sigma^n):\nu(A)=1\}} \sum_{k=1}^n \beta_k \nu(\{\mathbf{y} : y_k \ne x_k\}).$$

By the Cauchy–Schwarz inequality,

$$g(A,\mathbf{x})^2 \le \inf_{\{\nu\in M_1(\Sigma^n):\nu(A)=1\}} \sum_{k=1}^n \nu(\{\mathbf{y} : y_k \ne x_k\})^2$$

Since $\varphi_\alpha(1) = 0$, $\varphi'(1) = 0$, and $\varphi''(x) = \frac{\alpha}{x(1+ax)} \ge \frac{\alpha}{1+\alpha}$ for $x \in [0,1]$. Hence, for any $\alpha > 0$, $A \in \mathcal{B}_{\Sigma^n}$, and $\mathbf{x} \in \Sigma^n$,

$$\frac{\alpha}{2(1+\alpha)} g(A,\mathbf{x}) \le f_\alpha(A,\mathbf{x}). \tag{10.10}$$

Taking $t = \frac{\alpha}{2(1+\alpha)} u^2$ in (10.5) and applying (10.10) proves (10.8). The inequality (10.9) follows similarly by choosing $\alpha = u/\sqrt{2\log \mathbf{R}(A)^{-1}} - 1 \in (0,\infty)$, where it is trivial when $\mathbf{R}(A) = 1$. Let

$$Z_n(x) := \max\Big\{m : x_{k_1} < \ldots < x_{k_m} \text{ for some } 1 \le k_1 < \ldots < k_m \le n\Big\}$$

denote the length of the longest increasing subsequence of $\mathbf{x} = \{x_k : 1 \le k \le n\}$ and $M_n :=$ Median $(Z_n(\mathbf{X}))$, where $\{X_k\}$ are i.i.d. with uniform distribution on $[0,1]$.

**Corollary 10.10.** *For any $v \in \mathbb{Z}_+$,*

$$\mathbb{P}(Z_n(X) \geq M_n + v) \leq 2\exp\left(-\frac{v^2}{4(M_n + v)}\right), \tag{10.11}$$

$$\mathbb{P}(Z_n(X) \leq M_n - v) \leq 2\exp\left(-\frac{v^2}{4M_n}\right), \tag{10.12}$$

*Proof.* Fix $v \in \mathbb{Z}_+$ and let $\Sigma = [0,1]$ with $A(j) := \{\mathbf{y} : Z_n(\mathbf{y}) \leq j\} \subset \Sigma^n$ for $j \in \{1, 2, ..., n\}$. Suppose $x \in \Sigma^n$ is such that $Z_n(\mathbf{x}) \geq j + v$, so there exist $1 \leq k_1 < k_2 \cdot \cdot \cdot < k_{j+v} \leq n$ with $x_{k_1} < x_{k_1} < ... < x_{k_{j+v}}$. Hence, $\sum_{i=1}^{j+v} \mathbb{1}_{y_{k_i} = x_{k_i}} \geq v$ for any $\mathbf{y} \in A(j)$. Thus, by setting $\beta_{k_i} = 1/\sqrt{j+v}$ for $i = 1, ..., j+v$ and $\beta_k = 0$ otherwise, it follows that $g(A(j), \mathbf{x}) \geq v/\sqrt{j+v}$. To establish (10.12), apply (10.8) with $\mathbf{R}$ the Lebesgue measure so that $\mathbf{R}(A) = \mathbb{P}(Z_n(\mathbf{X}) \leq M_n) \geq \frac{1}{2}$ and

$$\mathbb{P}(Z_n(\mathbf{X}) \geq M_n + v) \leq \mathbf{R}\Big(\Big\{\mathbf{x} : g(A, \mathbf{x}) \geq v/\sqrt{M_n + v}\Big\}\Big)$$

On the other hand, noting that $\mathbf{R}(A) = \mathbb{P}(Z_n(\mathbf{X}) \leq M_n - v)$ for $A = A(M_n - v)$ and that

$$\frac{1}{2} \leq \mathbb{P}(Z_n(\mathbf{X}) \geq M_n) \leq \mathbf{R}\Big(\Big\{\mathbf{x} : g(A, \mathbf{x}) \geq v/\sqrt{M_n}\Big\}\Big),$$

(10.12) holds by applying (10.8) with $A = A(M_n - v)$. $\square$

## A. Sion's minimax theorem

The following proofs are excerpted from [1].

**Definition A.1.** Let $V$ be a topological vector space. A function $f : V \to \mathbb{R}$ is said to be quasi-convex (respectively, quasi-concave) if $f(tx + (1-t)y) \leq \max\{f(x), f(y)\}$ (respectively, $f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$) for all $t \in [0, 1]$.

**Theorem A.2** (Sion's minimax theorem)**.** *Let $X$ be a compact convex set in a topological vector space and $Y$ is a convex set in a topological vector space. If $f : X \times Y \to \mathbb{R}$ is a function such that*

- *$f(x, \cdot)$ is upper semicontinuous and quasi-concave for all $x \in X$, and that*
- *$f(\cdot, y)$ is lower semicontinuous and quasi-convex for all $y \in Y$,*

*then*

$$\min_{x\in X} \sup_{y\in Y} f(x, y) = \sup_{y\in Y} \min_{x\in X} f(x, y)$$

**Lemma A.3.** *Suppose the assumptions of Theorem A.2 hold. If*

$$\min_{x\in X} \max\{f(x, y_1), f(x, y_2)\} > \alpha$$

*for some $\alpha \in \mathbb{R}$ and $y_1, y_2 \in Y$, then there exists $y_0 \in Y$ such that $\min_{x\in X} f(x, y_0) > \alpha$.*

*Proof.* Write $[y_1, y_2] = \{ty_1 + (1-t)y_2 : t \in [0, 1]\}$ for short.

Suppose toward a contradiction that

$$\min_{x\in X} f(x, y) \leq \alpha < \beta < \min_{x\in X} \max\{f(x, y_1), f(x, y_2)\} \quad \text{for all} \quad y \in Y \tag{A.1}$$

Define the superlevel set

$$C_y^\gamma := \{x \in X : f(x, y) \leq \gamma\}$$

and observe that

- For all $y \in Y$, $C_y^\alpha \subseteq C_y^\beta$ are nonempty (by (A.1)), convex (by quasi-convexity) and compact (by lower semicontinuity).
- $C_{y_1}^\beta \cap C_{y_2}^\beta = \emptyset$ by (A.1) and $C_y^\beta \subset C_{y_1}^\beta \cup C_{y_2}^\beta$ for all $y \in [y_1, y_2]$ by quasi-concavity.
- For all $y \in Y$, $C_y^\alpha \subset C_y^\beta$ are convex by quasi-convexity and thus connected.

Inferring from the observations, we have either $C_y^\beta \subset C_{y_1}^\beta$ or $C_y^\beta \subset C_{y_2}^\beta$ for all $y \in [y_1, y_2]$. Furthermore, the sets

$$I = \left\{y \in [y_1, y_2] : C_y^\alpha \subset C_{y_1}^\beta\right\} \quad \text{and} \quad J = \left\{y \in [y_1, y_2] : C_y^\alpha \subset C_{y_2}^\beta\right\}$$

form a partition of $[y_1, y_2]$. It remains to show that both $I$ and $J$ are closed sets to reach a contradiction, as $[y_1, y_2]$ is connected. Note that $[y_1, y_2]$ is metrizable with the usual Euclidean distance. Hence, to demonstrate closedness one might examine whether the sequential limit $z$ of a convergent sequence $(z_n)_{n=1}^\infty \subset I$ lies in $I$. Indeed, given any $x \in C_y^\alpha$

$$\alpha \geq f(x, z) \geq \limsup_{n\to\infty} f(x, z_n)$$

implying that

$$x \in C^{\beta}_{z_n} \subset C^{\beta}_{y_1}$$

for all sufficiently large $n$ as desired. □

**Lemma A.4.** *Suppose the assumptions of Theorem A.2 hold. If*

$$\min_{x \in X} \max_{1 \le i \le n} f(x, y_i) > \alpha$$

*for some* $\alpha \in \mathbb{R}$ *and* $(y_i)_{i=1}^n \subset Y$*, then there exists* $y_0 \in Y$ *such that* $\min_{x \in X} f(x, y_0) > \alpha$*.*

*Proof.* We prove the lemma by induction. The case $n = 1$ is trivial. For induction step, we assume the hypothesis holds for $n - 1$ and define

$$X' = \{x \in X : f(x, y_n) \le \alpha\},$$

which is a convex set due to quasi-convexity. To proceed, we assume $X'$ is nonempty, for the proof is complete otherwise. Under the circumstances,

$$\min_{x \in X'} \max_{1 \le i < n} f(x, y_i) > \alpha$$

and the induction hypothesis applies to yield some $y' \in Y$ satisfying

$$\min_{x \in X'} f(x, y') > \alpha.$$

Now that

$$\min_{x \in X} \max\{f(x, y'), f(x, y_n)\} > \alpha,$$

we once again apply Lemma A.3 to obtain the desired $y_0$. □

*Proof of Theorem A.2.* Let $\alpha < \min_{x \in X} \max_{y \in Y} f(x, y)$ and $C^{\alpha}_y$ as in Lemma A.3. Then, $C^{\alpha}_y$ is compact and $\cap_{y \in Y} C^{\beta}_y = \emptyset$. By the finite intersection property, there exists finite points $(y_i)_{i=1}^n$ such that $\cap_{i=1}^n C^{\beta}_{y_i} = \emptyset$ and therefore

$$\alpha < \min_{x \in X} \max_{1 \le i \le n} f(x, y_i).$$

By Lemma A.4, there exists $y_0 \in Y$ such that

$$\alpha < \min_{x \in X} f(x, y_0) \le \sup_{y \in Y} \min_{x \in X} f(x, y).$$

Immediately, this gives

$$\min_{x \in X} \sup_{y \in Y} f(x, y) \le \sup_{y \in Y} \min_{x \in X} f(x, y).$$

The remaining inequality simply follows from definition. □

## B. Probability theory

### B.1. Basic inequalities.

**Proposition B.1** (Borel-Cantelli). *Let $(S_n)_{n=1}^{\infty}$ be a sequence of events in a probability space $(\mathcal{X}, \mathcal{B}, \mu)$. Then,*

$$\mu\Big(\limsup_{n\to\infty} S_n\Big) \le \lim_{N\to\infty} \sum_{n=N}^{\infty} \mu(S_n).$$

**Proposition B.2** (Markov). *Let $X$ be a nonnegative random variable. Then, for every $\alpha > 0$,*

$$\mu(X \ge \alpha) \le \alpha^{-1} \cdot \mathbb{E}[X].$$

**Proposition B.3** (Hölder). *Let $X, Y$ be non-negative random variables. Then, for $p, q \in [1, \infty]$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$,*

$$\mathbb{E}(X^p)^{1/p}\mathbb{E}(Y^q)^{1/q} \ge \mathbb{E}(XY),$$

*where the equality holds if and only if $X = cY$ for some $c > 0$.*

*Proof.* Without loss of generality, assume $\mathbb{E}(X^p) = \mathbb{E}(Y^q) = 1$ to paraphrase the proposition as $1 \ge \mathbb{E}(XY)$ with equality holding if and only if $X = cY$ for some $c > 0$. Under the circumstances, it is not hard to verify the Young's inequality

$$XY \le \frac{X^p}{p} + \frac{Y^q}{q}$$

holds if and only if $X^p = Y^q$. The paraphrased proposition then follows by integrating both sides. □

**Proposition B.4** (Jensen). *Let $X$ be a real-valued random variable and $\varphi : \mathbb{R} \to \mathbb{R}$ is convex. If $\mathbb{E}(X)$ and $\mathbb{E}(\varphi(X))$ are defined, then $\mathbb{E}(\varphi(X)) \ge \varphi(\mathbb{E}(X))$ with the convention $\varphi(+\infty) = \lim_{x\to\infty} \varphi(x)$ and $\varphi(-\infty) = \lim_{x\to-\infty} \varphi(x)$.*

*Proof.* We make use the property of convex functions that

$$\varphi(x) = \sup\{ax + b : \varphi(x) \ge ax + b \text{ for all } x \in \mathbb{R}\}. \tag{B.1}$$

The left-hand side is by definition larger than the other, and thus it remains to show the remaining. To this end, it suffices to show that for all $(x_0, y_0)$ satisfying $\varphi(x_0) \ge y_0$, there exists a linear function $\psi(x) = a(x - x_0) + y_0$ such that $\varphi(x) \le \psi(x)$ for all $x$. Essentially, this is achieved by choosing

$$\begin{aligned} a &\in \left[\sup_{x<x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0}, \inf_{x>x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0}\right] \\ &= \left[\liminf_{x\to x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0}, \limsup_{x\to x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0}\right]. \end{aligned}$$

Now that (B.1) coincides with

$$\varphi(x) = \sup\{ax + b : \varphi(x) \ge ax + b \text{ for all } x \in \mathbb{R}, a, b \in \mathbb{Q}\}, \tag{B.2}$$

one can enumerate the linear functions on the right-hand side by $(\psi_i)_{i=1}^{\infty}$ and obtain

$$\mathbb{E}(\varphi(X)) \geq \mathbb{E}\Big(\max_{1\leq i\leq n} \psi_i(X)\Big) \geq \max_{1\leq i\leq n} \psi_i(\mathbb{E}(X)).$$

If $\mathbb{E}(X) \in \mathbb{R}$, then the proposition holds by letting $n \to \infty$. If $\mathbb{E}(X) = \infty$ (similar for $\mathbb{E}(X) = -\infty$) and $\varphi(\infty) > -\infty$, then right-hand side of the above still converges to $\varphi(\infty)$, while the proposition is trivial when $\mathbb{E}(X) = \infty$ and $\varphi(\infty) = -\infty$. □

B.2. **Radon-Nikodym theorem.**

**Definition B.5** (absolute continuity)**.** Let $\mu$ and $\nu$ be defined on a common measurable space $(\mathcal{X}, \mathcal{B})$. We say $\nu$ is *absolutely continuous* with respect to $\mu$, denoted by $\nu \ll \mu$, if $\nu(A) = 0$ for every $A \in \mathcal{B}$ satisfying $\mu(A) = 0$.

**Theorem B.6** (Radon-Nikodym)**.** *Suppose $\nu, \mu$ are two $\sigma$-finite measures on a common measurable space $(\mathcal{X}, \mathcal{B})$ and $\nu \ll \mu$, then there exists a $\mathcal{B}$-measurable function $f : \mathcal{X} \to [0, \infty)$ such that for every $A \in \mathcal{B}$,*

$$\nu(A) = \int_A f d\mu.$$

B.3. **Laws of large numbers.**

**Theorem B.7** (Weak Law of Large Numbers)**.** *Suppose $(X_i)_{i=1}^\infty$ are i.i.d. and $\mathbb{E}(X_1) = 0$. Then, $n^{-1}\sum_{i=1}^n X_i \to 0$ in probability.*

*Proof.* Equivalently, we can assume $X_i$ is nonnegative and $\mathbb{E}(X_1) < \infty$ and prove $n^{-1}\sum_{i=1}^n X_i \to \mathbb{E}(X_1)$. Let $M > 0$ and $X_i = Y_{i,1} + Y_{i,2}$ with $Y_{i,1} = \max\{X_i, M\}$. Immediately,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n Y_{i,1} > \varepsilon\right) \leq \frac{nM}{n^2\varepsilon^2} \to 0 \text{ as } n \to \infty.$$

On the other hand,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n Y_{i,2} > \varepsilon\right) \leq \frac{n\mathbb{E}\big(X_1 \mathbb{1}_{X_1 \geq M}\big)}{n\varepsilon} \to 0 \text{ as } M \to \infty.$$

The theorem is proved by combining the above. □

**Lemma B.8** (Kronecker)**.** *Suppose $a_n > 0$ and $a_n \nearrow \infty$. Then $\sum_{i=1}^\infty a_n^{-1} x_n < \infty$ implies $a_n^{-1}\sum_{i=1}^n x_i \to 0$.*

*Proof.* Writing $b_n = (a_n^{-1} - a_{n-1}^{-1})$ with $a_0^{-1} := 0$, one may use summation by part to deduce

$$a_n^{-1}\sum_{i=1}^n x_i = \sum_{i=1}^n a_i(a_i^{-1}x_i) = \sum_{i=1}^n a_i^{-1}x_i + \sum_{i=1}^{n-1} \frac{a_i - a_{i+1}}{a_n} \sum_{j=1}^i a_j^{-1}x_j.$$

The last expression converges to 0 as $n \to \infty$. □

**Proposition B.9** (Kolmogorov's Criterion of SLLN)**.** *Suppose $(X_i)_{i=1}^\infty$ are independent such that $\mathbb{E}(X_n) = 0$ and $\sum_{i=1}^\infty i^{-2}\operatorname{Var}(X_i) < \infty$. Then, $n^{-1}\sum_{i=1}^n X_i \to 0$ a.s.*

*Proof.* By virtue of the independence, one observes that

$$\mathbb{E}\left(\left|\sum_{i=1}^{\infty}\frac{X_i}{i}\right|^2\right) \leq \sum_{i=1}^{\infty}\frac{\mathrm{Var}(X_i)}{i^2} < \infty,$$

which, in particular, implies $\sum_{i=1}^{\infty}\frac{X_i}{i}$ is finite almost surely and in turn implies the conclusion by Lemma B.8. □

**Proposition B.10** (Strong Law of Large Numbers)**.** *Suppose* $(X_i)_{i=1}^{\infty}$ *are i.i.d. and* $\mathbb{E}(X_1) = 0$*. Then,* $n^{-1}\sum_{i=1}^{n} X_i \to \mathbb{E}(X_1)$ *a.s.*

*Proof.* It suffices to prove the case $\mathbb{E}(|X_1|) < \infty$, for if otherwise, one may still apply the result to $Y_n = \min\{M, X_n\}$ for every $M > 0$ if $\mathbb{E}(X_1^+) < \infty$ (resp., $Y_n = \max\{-M, X_n\}X_n$ if $\mathbb{E}(X_1^-) > -\infty$) so that $\mathbb{E}(|Y_n|) < \infty$ and that $Y_n \leq X_n$ (resp., $Y_n \geq X_n$), leading to the conclusion by letting $M \to \infty$.

To begin, truncate $X_n$ to define

$$Y_n = \mathbb{1}_{\{|X_n| \leq n\}}X_n - \mathbb{E}\Big(X_n\mathbb{1}_{\{|X_n| \leq n\}}\Big).$$

Then, verify the assumption of Proposition B.9

$$\begin{aligned}\sum_{i=1}^{\infty}\frac{\mathrm{Var}(Y_i)}{i^2} &\leq \sum_{i=1}^{\infty}\frac{\mathbb{E}\Big(|X_1|^2\ \mathbb{1}_{\{|X_1|\leq i\}}\Big)}{i^2} = \left(\mathbb{E}\left(|X_1|^2 \sum_{i=\max\{1,|X_1|\}}^{n}\frac{1}{i^2}\right)\right)\\ &\leq \mathbb{P}(|X_1| < 1) + \mathbb{E}\left(|X_1|^2\sum_{i=|X_1|}^{n}\frac{2}{i(i+1)}\right)\\ &\leq \mathbb{P}(|X_1| < 1) + 2\mathbb{E}(|X_1|).\end{aligned}$$

to deduce

$$\frac{1}{n}\sum_{i=1}^{\infty} Y_i \to 0. \tag{B.3}$$

On the other hand, by the Borel-Cantelli lemma,

$$\mathbb{P}(|X_n| > n \text{ infinitely often}) \leq \limsup_{n\to\infty}\sum_{i=n}^{\infty}\mathbb{P}(|X_i| \geq i) \leq \limsup_{n\to\infty}\mathbb{E}(|X_1|). \tag{B.4}$$

Combining (B.3) and (B.4) gives that

$$\lim_{n\to\infty} n^{-1}\sum_{i=1}^{n} X_i = \lim_{n\to\infty} n^{-1}\sum_{i=1}^{n}\Big(Y_i + \mathbb{E}\Big(|X_1|\ \mathbb{1}_{\{|X_1|\leq i\}}\Big)\Big) = \mathbb{E}(X_1).$$

□

## C. Functional analysis

**Theorem C.1** (Hahn-Banach Extension Theorem)**.** *Let $\mathcal{X}$ be a real vector space. Suppose that*

- *$\mathcal{Y}$ is a linear subspace of $\mathcal{X}$,*
- *$p : \mathcal{X} \to \mathbb{R}$ is a convex functional,*
- *$f : \mathcal{Y} \to \mathbb{R}$ is linear and $f \le p|_{\mathcal{Y}}$.*

*Then, there exists a linear functional $g : \mathcal{X} \to \mathbb{R}$ such that $g|_{\mathcal{Y}} = f$* and *$g \le p$.*

*Proof.* The theorem is proved by transfinite induction.

If $\mathcal{Y} \neq \mathcal{X}$, choose $y \in \mathcal{Y} \setminus \mathcal{X}$, $\mathcal{Y}' = (\mathrm{span}\{y\} \cup \mathcal{X})$ and extend $f$ to $f' : \mathcal{Y}' \to \mathbb{R}$ by

$$f'(x + ty) = f(x) + t\alpha \text{ for all } x \in \mathcal{Y}, t \in \mathbb{R},$$

where one may choose, if possible,

$$\alpha(= f'(x')) \in \left[\sup_{t<0} \sup_{x \in \mathcal{Y}} \frac{p(x + tx') - f(x)}{t}, \inf_{t>0} \inf_{x \in \mathcal{Y}} \frac{p(x + tx') - f(x)}{t}\right]$$

so that $f'(x) \le p(x)$ on $\mathcal{Y}'$. To see the above interval is indeed nonempty, note that the convexity of $p$ together with $p \ge f$ implies that for each $t > 0$,

$$\frac{p(z + tx') - f(z)}{t} \ge \frac{p(z' - tx') - f(z')}{-t} \text{ for all } z, z' \in \mathcal{Y}, t > 0,$$

and hence the non-emptiness follows from the finite intersection property.

To conclude the proof, consider the set $\mathcal{C}$ of all pairs $(\mathcal{Y}, f)$ such that $\mathcal{Y}$ is a subspace of $\mathcal{X}$ and that $f : \mathcal{Y} \to \mathbb{R}$ is linear on $\mathcal{Y}$ with $f \le p|_{\mathcal{Y}}$. Define further a partial order $\le$ by

$$(\mathcal{Y}, f) \le (\mathcal{Y}', f') \text{ if and only if } \mathcal{Y} \subseteq \mathcal{Y}' \text{ and } f'|_{\mathcal{Y}} = f.$$

Hence, for any chain $\{(\mathcal{Y}_i, f_i)\} \subset \mathcal{C}$, the space $\mathcal{Y} = \cup_i \mathcal{Y}_i$ is a vector space and the function $f(x) := f_i(x)$ whenever $x \in \mathcal{Y}_i$, which form an maximal element $(\mathcal{Y}, f) \in \mathcal{C}$ of the chain. Hence, by Zorn's lemma, there exists a maximal element $(\mathcal{Y}, f) \in \mathcal{C}$, and $\mathcal{Y} = \mathcal{X}$ by the first part of the proof. □

In the context of topological vector space, an equivalent of Theorem C.1 is phrased in terms of separation as in the following. Recall that a *core point* $x$ of a subset $A$ of a vector space $\mathcal{X}$ is the point satisfying that for every $y \subset \mathcal{X}$ there exists $\varepsilon = \varepsilon(y)$ such that $y + \delta y \in A$ for all $|\delta| \le \varepsilon$, for which we denote $\mathrm{cor}A = \{x \in A : x \text{ is a core point}\}$.

**Theorem C.2** (Hahn-Banach Separation Theorem)**.** *Let $A$ and $B$ be disjoint, nonempty, convex subsets of a real topological vector space $\mathcal{X}$. Assume that $\mathrm{cor}A \neq \emptyset$. Then there is a linear functional $f : \mathcal{X} \to \mathbb{R}$ satisfying $\sup_{a \in A} f(a) \le \inf_{b \in B} f(b)$ and $A \cup B \not\subset \{f = \alpha\}$ for all $\alpha$. Moreover, if in addition $\mathrm{int}A \neq \emptyset$, then $f$ can be chosen so that $f \in \mathcal{X}^*$ and $f(a) < f(b)$ for all $a \in A$ and $b \in B$.*

*Proof of Theorem C.2 by Theorem C.1.* We first prove the general case, after which we point out the necessary modifications for the case where $\mathrm{int}A \neq \emptyset$.

Let $C = A - B$ and $z \in \mathrm{cor}(A - B)$. Define $C = A - B - z$ so that $-z \notin C$ and its associated *gauge*

$$p_C(x) = \inf\{t > 0 : x \in tC\}.$$

Then, $p_C : \mathcal{X} \to \mathbb{R}$ is a sublinear functional. Indeed, since $0 \in \mathrm{cor}C$, $p_c$ is real-valued. Moreover, $p_C(tx) = tp_C(x)$ for all $t > 0$ by definition and $p_C(x+y) \leq p_C(x) + p_C(y)$ by convexity of $C$. Consider the subspace $\mathcal{Y} = \mathrm{span}\{-z\}$ and define the linear functional $g(-tz) = t$. To apply the extension theorem, observe that $p_C(-z) \geq 1$ and thus $g \leq p|_y$, yielding a linear extension $f : \mathcal{X} \to \mathbb{R}$ of $g$ satisfying $f \leq p_C$ by Theorem C.1. To see $f$ separates $A$ and $B$, observe that for any $a \in A$ and $b \in B$,

$$\begin{aligned} f(a) &= f(a-b-z) + f(z) + f(b) \\ &\leq p_C(a-b-z) + f(z) + f(b) \leq 1 - 1 + f(b) = f(b). \end{aligned} \tag{C.1}$$

The number $\alpha$ in question can be arbitrary in $[\sup_{a\in A} f(a), \inf_{b\in B} f(b)]$. To see that that $A \cup B \not\subset \{f = a\}$, note that $z = a - b$ for some $a \in A$ and $b \in B$ and $f(z) = -1 = f(a) - f(b)$ as desired.

Now if $\mathrm{int}A \neq \emptyset$, pick $z \in \mathrm{int}(A - B)$ and proceed as before. Under the circumstances, $C$ is a neighborhood of 0. Hence, $|f(x)| \leq \max\{|p_C(x)|, |p_C(-x)|\} \leq 1$ for all $x$ in the neighborhood $C \cap -C$ of 0, proving the continuity. Furthermore, $p_C(a - b - z) < 1$ due to the fact $z \in \mathrm{int}(A - B)$, rendering (C.1) a strict inequality. □

*Proof of Theorem C.1 by Theorem C.2.* Let

$$A = \{(x, \alpha) \in \mathcal{X} \times \mathbb{R} : p(x) < \alpha\} \text{ and } B = \{(y, \beta) \in \mathcal{Y} \times \mathbb{R} : f(y) \geq \beta\}.$$

Then, $A$ and $B$ are convex and every point in $A$ is a core point. Thus, by Theorem C.2, there exists a linear functional $h : \mathcal{X} \to \mathbb{R}$ and numbers $\rho \in \mathbb{R}$ such that

$$\sup_{(x,\alpha)\in A} h(x) + \rho\alpha \leq \gamma := \inf_{(y,\beta)\in B} h(y) + \rho\beta \quad (\gamma \in \mathbb{R}). \tag{C.2}$$

Under the circumstances, we observe, by letting $\alpha \to \infty$, that $\rho \leq 0$. It is clear that $\rho \neq 0$, for if otherwise, the inequality (C.1) above holds if and only if $h(x) = 0$ for all $x \in \mathcal{X}$, contradicting Theorem C.2 as $h(A) = h(B) = \{0\}$. Now that $\rho < 0$, the inequality (C.2) implies that

$$\begin{aligned} \rho^{-1}h(y) + f(y) \leq \rho^{-1}\gamma \text{ for all } y \in \mathcal{Y} \quad &\Rightarrow \quad f = \rho^{-1}(\gamma - h|_y), \\ \rho^{-1}g(x) + p(x) \geq \rho^{-1}\gamma \text{ for all } x \in \mathcal{X} \quad &\Rightarrow \quad p \geq \rho^{-1}(\gamma - h). \end{aligned}$$

The proof is finished by choosing $g = \rho^{-1}(\gamma - h)$. □

**Theorem C.3.** *Suppose $\mathcal{X}$ is a locally convex real topological vector space. Suppose $A$ and $B$ are two disjoint, nonempty, convex sets in the locally convex topological vector space $\mathcal{X}$. If $A$ is compact and $B$ is closed, then there exists an $f \in \mathcal{X}^*$ such that* $\sup_{a\in A} f(a) < \inf_{b\in B} f(b)$.

*Proof.* Suppose $V$ is a convex neighborhood of 0 such that $(A + V) \cap B = \emptyset$. Apply Theorem C.2 to $A + V$ and $B$ to obtain $f \in \mathcal{X}^*$ such that $\sup_{a\in A} f(a) < \sup_{a\in A+V} f(a)$ since $f(A)$ is compact and $f(A + V)$ is open. □

## D. Basics in convex analysis

**Lemma D.1** (duality)**.** *Let $\mathcal{X}$ be a locally convex topological vector space. If $f : \mathcal{X} \to (-\infty, \infty]$ is convex and lower semicontinuous, then $f^{**}(x) = f(x)$.*

*Proof.* We should assume, without loss of generality, that $f$ is not identically $\infty$, for the lemma holds obviously otherwise. Define

$$\mathcal{E} = \{(x, \alpha) \in \mathcal{X} \times \mathbb{R} : f(x) \le \alpha\},$$
$$\mathcal{E}^* = \{(\lambda, \alpha) \in \mathcal{X}^* \times \mathbb{R} : f^*(\lambda) \le \alpha\},$$

which are convex subsets in the locally convex topological vector spaces $\mathcal{X} \times \mathbb{R}$ and $\mathcal{X}^* \times \mathbb{R}$, respectively.

It is not hard to verify by definition that $f^{**} \le f$. Hence, it suffices to prove the other inequality. Equivalently, it asserts that if $f(x) > \alpha$, then there is $(\lambda, \beta) \in \mathcal{X}^*$ such that $\langle \lambda, x\rangle - \beta > \alpha$. Then, observe that $A = \{(x, \alpha)\}$ is compact and $B = \mathcal{E}$ is closed by lower semicontinuity of $f$ and nonempty by the fact $f$ is not identically $\infty$. Moreover, under the assumption $f(x) > \alpha$, $A \cap B = \emptyset$. These altogether allow us to apply Hahn-Banach separation theorem (specifically, Theorem C.3) to yield some $\eta \in \mathcal{X}^*$ and $\rho \in \mathbb{R}$ satisfying

$$\gamma := \sup_{(y,\beta)\in\mathcal{E}} \langle \eta, y\rangle - \rho\beta < \langle \eta, x\rangle - \rho\alpha \tag{D.1}$$

where $\rho \ge 0$ must hold as there exists $y \in \mathcal{X}$ with $f(y) < \infty$.

If $\rho > 0$, then (D.1) implies that $\alpha < \langle \rho^{-1}\eta, x\rangle - \rho^{-1}\gamma$. Moreover, $(\rho^{-1}\eta, \rho^{-1}\gamma) \in \mathcal{E}^*$ as desired, for if otherwise, $f^*(\rho^{-1}\eta) > \rho^{-1}\gamma$ contradicting the definition of $\gamma$.

If $\rho = 0$, then for all $(\mu, \beta) \in \mathcal{E}^*$, define $(\mu_\delta, \beta_\delta) := \left(\frac{\eta}{\delta} + \mu, \frac{\gamma}{\delta} + \beta\right)$ for $\delta > 0$. In fact, $(\mu_\delta, \beta_\delta) \in \mathcal{E}^*$ since

$$\begin{aligned} f^*(\mu_\delta) - \beta_\delta &\le \sup_{y\in\mathcal{X}} [\langle \mu_\delta, y\rangle - f(y) - \beta_\delta] \\ &\le \sup_{y\in\mathcal{X}} \left[\frac{1}{\delta}(\langle \eta, y\rangle - \gamma) + (\langle \mu, y\rangle - f^*(\mu)) - f(y)\right] \le 0. \end{aligned}$$

Moreover,

$$[\langle \eta_\delta, x\rangle - f^*(\eta_\delta)] \ge \lim_{\delta\to 0} \left[\frac{1}{\delta}(\langle \eta, x\rangle - \gamma) + (\langle \mu, x\rangle - f^*(\mu))\right] = \infty,$$

yielding the desired $\lambda = \eta_\delta$ when $\delta$ is sufficiently small. □

**Lemma D.2.** *Suppose $f : \mathbb{R}^d \to [0, \infty]$ is a convex, lower semicontinuous function with $\inf_{\lambda\in\mathbb{R}^d} f(\lambda) = 0$ and $0 \in \operatorname{ri} \mathcal{D}_{f^*}$, then $f(\eta) = 0$ for some $\eta \in \mathbb{R}^d$*

*Proof.* Note that $f^*(0) = 0$, and hence the lemma, by duality (Lemma D.1), is equivalent to the existence of $\eta \in \mathbb{R}^d$ such that $\langle \eta, x\rangle \le f^*(x)$ for all $x \in \mathbb{R}^d$. This latter claim, by convexity of $f^*$, is equivalent to

$$\langle \eta, x\rangle \le \lim_{\delta\to 0+} \frac{f^*(\delta x) - f^*(0)}{\delta} = \inf_{\delta>0} \frac{f^*(\delta x) - f^*(0)}{\delta} =: g(x) \text{ for all } x \in \mathbb{R}^d.$$

To prove the claim, define

$$A = \overline{\{(x, \alpha) \in \mathbb{R}^d \times \mathbb{R} : g(x) \le \alpha\}} \text{ and } B = \{(0, -1)\},$$

Obviously, $B$ is convex, compact, and nonempty. On the other hand, $A$ is nonempty, convex, and $A \cap B = \emptyset$. Obviously, $(0,0) \in A$. To prove convexity, note that $g$ is a convex function and closure of a convex set is convex. Indeed,

$$g(tx + (1-t)y) = \lim_{\delta \to 0+} \frac{f^*(t\delta x + (1-t)\delta y)}{\delta}$$
$$\leq \lim_{\delta \to 0+} t \cdot \frac{f^*(\delta x)}{\delta} + (1-t) \cdot \frac{f^*(\delta y)}{\delta} = tg(x) + (1-t)g(y).$$

Finally, to prove $A \cap B = \emptyset$, we first show that $0 \in \text{ri } \mathcal{D}_g = \mathcal{D}_g$. Under the circumstances, we deduce that $g|_{\mathcal{D}_g}$ is continuous at $0 \in \text{ri } \mathcal{D}_g = \mathcal{D}_g$ and thus $A \cap B = \emptyset$ follows naturally. To this end, note that $0 \in \text{ri } \mathcal{D}_{f^*}$ implies $0 \in \text{ri } \mathcal{D}_g$. Indeed, $g(0) = 0$ and if $x \in \mathcal{D}_g$, then $f^*(x) < \infty$ and for all small $\varepsilon > 0$,

$$g(-\varepsilon x) = \inf_{\delta > 0} \frac{f^*(\delta(-\varepsilon)x)}{\delta} \leq f^*(-\varepsilon x) < \infty.$$

In particular, the above implies $0 \in \text{ri } \mathcal{D}_g$. Moreover, $0 \in \text{ri } \mathcal{D}_g$ implies also $\text{ri } \mathcal{D}_g = \mathcal{D}_g$ since $x \in \mathcal{D}_g$ if and only if $tx \in \mathcal{D}_g$ for all $t > 0$.

We now may apply Theorem C.3 to $A$ and $B$ to yield some $\mu \in \mathcal{X}$ such that

$$\langle \mu, 0 \rangle + \rho = \rho > \sup_{(x,\alpha) \in A} \langle \mu, x \rangle - \rho\alpha,$$

where $\rho > 0$ since $(0,0) \in A$. Hence, for every $x \in \mathcal{X}$ with $f^*(x) < \infty$, we have that

$$\langle \rho^{-1}\mu, \varepsilon x \rangle \leq g(\varepsilon x) + 1 \text{ for all } \varepsilon > 0 \Rightarrow \langle \rho^{-1}\mu, x \rangle \leq g(x).$$

The proof is concluded by choosing $\eta = \rho^{-1}\mu$. □

**Lemma D.3.** *Let $f$ be an essentially smooth, convex function. If $f(0) = 0$ and $f^*(x) = 0$ for some $x \in \mathbb{R}^d$, then $0 \in \mathring{\mathcal{D}}_f$.*

*Proof.* Since $f(0) = 0$, it follows by convexity of $f$ that

$$f(t\lambda) \leq tf(\lambda), \text{for all } t \in [0,1], \lambda \in \mathbb{R}^d.$$

Moreover, since $f^*(x) = 0$,

$$f(t\lambda) \geq \langle t\lambda, x \rangle - f^*(x) \geq -t|\lambda||x|.$$

Because $f$ is essentially smooth, there exists a closed ball $\overline{B}_r(z) \subset \mathring{\mathcal{D}}_f$ in which $f$ is differentiable. Hence,

$$M = \sup_{\lambda \in \overline{B}_r(z)} \{f(\lambda) \vee |\lambda||x|\} < \infty$$

Hence, for any $t \in (0,1]$ and any $\theta \in \overline{B}_{tr}(tz)$ different from $tz$, by the convexity of $f$,

$$f(\theta) - f(tz) \leq \frac{|\theta - tz|}{tr}(f(y) - f(tz)) \leq \frac{2tM}{tr}|\theta - tz|,$$

where $y = tz + \frac{tr}{|\theta - tz|}(\theta - tz) \in \overline{B}_{tr}(tz)$. Similarly, $f(\theta) - f(tz) \geq -\frac{2tM}{tr}|\theta - tz|$. Hence,

$$|f(\theta) - f(tz)| \leq \frac{2tM}{tr}|\theta - tz| \text{ for all } \theta \in \overline{B}_{tr}(tz).$$

Observe that $tz \in \mathring{\mathcal{D}}_f$ because of the convexity of $\mathcal{D}_f$. Hence, by assumption, $\nabla f(tz)$ exists, and by the preceding inequality, $|\nabla f(tz)| \leq 2\frac{M}{r}$. Since $f$ is steep, it follows by considering $t \to 0$, in which case $tz \to 0$, that $0 \in \mathring{\mathcal{D}}_f$. □

We define the *approximate subgradient* as follows:

$$\begin{aligned}\partial_\varepsilon f(x) &= \{x^* \in \mathcal{X}^* : f(z) \geq (f(x) - \varepsilon) + \langle x^*, z - x\rangle \text{ for all } z \in \mathcal{X}\} \\ &= \{x^* \in \mathcal{X}^* : f(z) + f^*(x^*) - \langle x^*, x\rangle \leq \varepsilon \text{ for all } z \in \mathcal{X}\}\end{aligned}$$

By Hahn-Banach separation theorem (Theorem C.2), such set is always non-empty as long as $x \in \mathcal{D}_f$.

**Theorem D.4** (Brønsted–Rockafellar)**.** *Suppose $\mathcal{X}$ is a Banach space. Then, every lower semicontinuous convex function taking value in $(-\infty, \infty]$ which is not identically $+\infty$ satisfies the following:*

$$f(y) = \liminf_{x \to y} \overline{f}(x) \text{ for all } y, \text{where } \overline{f}(x) = \begin{cases} f(x) \text{ if } x \in \mathcal{D}_{\partial f}, \\ +\infty \ \text{ otherwise,} \end{cases}$$

$$f(y) = \sup\big\{f(\overline{x}) + (y - \overline{x}, x^*) \mid \in \overline{x} \in \mathcal{D}_f, x^* \in \mathcal{D}_{\partial f}\big\} \text{ for all } y \in E.$$

*Moreover, the conjugates $f^*$ of such functions actually satisfy the stronger conditions derived from the above by restricting attention to the existence of subgradients of $f^*$ belonging to $\mathcal{X}$ instead of $\mathcal{X}^{**}$.*

**Lemma D.5.** *Assume that $\mathcal{X}$ is a Banach space and that $x^* \in \partial_\varepsilon f(x)$. Then, for any $\lambda > 0$, there exist vectors $\overline{x}$ and $\overline{x}^*$ such that $\|\overline{x} - x\| \leq \lambda$, $\|\overline{x}^* - x^*\| \leq \frac{\varepsilon}{\lambda}$ and $\overline{x}^* \in \partial f(x)$.*

*Proof.* Define a partial ordering $y \prec z$ on $\mathcal{D}_f$ if

$$\frac{\varepsilon}{\lambda}\ \|y - z\| \leq [f(y) - \langle x^*, y\rangle] - [f(z) - \langle x^*, z\rangle].$$

By Zorn's lemma, there exists a maximal totally ordered subset $M$ of $\big\{z \in \mathcal{D}_f : x\big\}$. Therefore,

$$f(z) - \langle x^*, z\rangle \searrow \rho \geq f(x) - \langle x^*, x\rangle - \varepsilon > -\infty \text{ as } z \nearrow \text{in } M.$$

implying that $z \to \overline{x}$ as $\overline{x} \nearrow$ by completeness, that $\overline{x} \in \mathcal{D}_f$, and that

$$\frac{\varepsilon}{\lambda}\ \|\overline{x} - z\| > [f(\overline{x}) - \langle x^*, \overline{x}\rangle] - [f(z) - \langle x^*, z\rangle] \text{ if } z \neq \overline{x}.$$

We now apply Hahn-Banach separation theorem (Theorem C.2) to

$$A = \{(y, \mu) : \mu \geq f(\overline{x} + y) - f(\overline{x}) - \langle x^*, y\rangle\} \text{ and } B = \Big\{(y, \mu) : \mu < -\frac{\varepsilon}{\lambda}\|y\|\Big\},$$

we find some $z^* \in \mathcal{X}^*$ such that

$$-\frac{\varepsilon}{\lambda}\ \|y\| \leq \langle z^*, y\rangle \leq f(\overline{x} + y) - f(\overline{x}) - \langle x^*, y\rangle,$$

which completes the proof if we take $\overline{x}^* = x^* + z^*$, assuring $\|\overline{x}^* - z^*\| \leq \frac{\varepsilon}{\lambda}$ and $\overline{x}^* \in \mathcal{D}_{\partial f}$. □

*Proof of Theorem D.4.* It suffices to show that $\liminf_{y \to x} \overline{f}(y) \leq f(x)$ whenever $x \in \mathcal{D}_f$. Given any $\delta > 0$, choose any $x^* \in \partial_\varepsilon f$, where $\varepsilon = \frac{\delta}{2}$. Choose $\lambda > 0$ so small

that $\lambda < \delta$ and $\lambda \|x^*\| < \frac{\delta}{2}$. Now let $\overline{x}$ and $\overline{x}^*$ be the vectors whose existence is guaranteed by Lemma D.5, meaning

$$f(\overline{x}) - f(x) \leq -(x^*, x - \overline{x}) \leq \|\overline{x} - x\| \; \|x^*\| \leq \lambda \left( \|x^*\| + \frac{\varepsilon}{\lambda} \right) < \delta.$$

This implies $\overline{x} \in \mathcal{D}_f$, $\|\overline{x} - x\| < \delta$ and $f(\overline{x}) < f(x) + \delta$. These proves the first part. The modified condition is proved using Lemma D.5 similarly.

The second equality is proved as follows. □

## References

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Department of Mathematics, University of Exampleville, Tennessee, TN 59341
*Email address:* jdoe@math.ue.edu
*URL:* math.ue.edu/~jdoe